MishMash The Meal Prep Planner Developer Guide

MishMash is a command-line meal prep planner for managing recipes, generating meal plans, and preparing grocery lists. This guide describes the current design, project practices, requirements, use cases, and manual tests.

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Feature navigation:

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Acknowledgements

- The project originates from the NUS CS2113 team-project template. The team's shared repository is AY2627S1-CS2113-W13-4/tp.
- The guide structure follows the AddressBook-Level3 project's User Guide and Developer Guide, adapted to the CS2113 deliverable requirements. This reuse concerns documentation structure; the feature descriptions document MishMash's implementation.
- Gradle provides build automation and dependency management.
- JUnit Jupiter provides the automated testing framework.
- Checkstyle checks Java coding conventions.
- Shadow packages the application JAR.
- JLine provides interactive line editing, completion, and session history in the development build. The JNI terminal provider supports native consoles; unsupported environments retain the existing Scanner-based input. JLine is distributed under the BSD 3-Clause license.
- Jackson reads and writes the versioned JSON snapshots used for local persistence.
- PlantUML generates the UML diagrams from the editable sources under `docs/diagrams`, organised by contributor and feature. The diagram index records locations, verification, and PDF export instructions.

Setting up, getting started

Refer to Setting Up and Getting Started for Java 25 prerequisites, importing the project, running tests, packaging the JAR, and troubleshooting.

Interactive Input in the Development Build

`ConsoleInputs` uses JLine on supported interactive consoles and falls back to Scanner if terminal setup fails. The launcher owns its input adapter; injected test streams remain caller-owned.

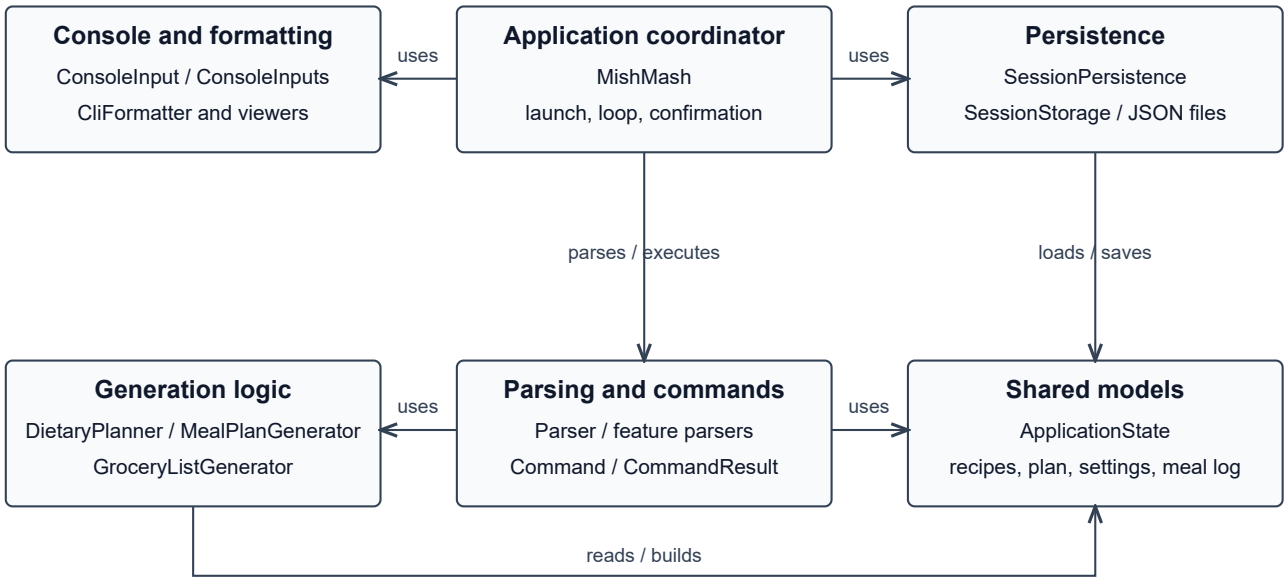
Completion uses `Parser.COMMAND_USAGE`; history stays in memory. Names and confirmation answers use neither. Ctrl+C clears the command or supplies a blank confirmation, while EOF ends input. JLine preserves the existing grammar by disabling history expansion, quote interpretation, and escape interpretation.

Gradle packages the reader and JNI provider, merges service registrations, and enables native access on Java 25. Tests use in-memory terminals and injected factories; check native behaviour with the packaged JAR.

This guide documents current `master` for the upcoming submission release. The published v2.0 release predates persistence and some meal-editing features described here. Build the current checkout when testing this guide; when the submission release is published, verify its JAR against these instructions and update this release reference. User Stories show milestone assignments and current source support.

Design

Architecture



MishMash architecture and dependency directions

Arrows show dependency or data-access direction, not the timing of calls. The editable SVG groups classes by responsibility; it does not imply AB3-style manager interfaces. `MishMash` connects the console, parser, shared state, and persistence at startup. Commands use model data and generation services; persistence restores and saves the same shared roots.

The application separates terminal interaction, command handling, application data, and output formatting. The main application coordinates these components throughout a single command session.

Area	Current responsibility
Application and user interaction	Display the greeting, read commands, obtain confirmation when required, and show results or errors.
Command handling	Recognise commands, validate their arguments, and perform the requested action.
Model	Hold recipes, nutrition information, meal plans, generated groceries, and recorded meals.
Generation logic	Produce meal plans and consolidate grocery quantities.
Display formatting	Present recipes, meal plans, grocery lists, and recorded meals consistently as text.

Command names and prefixes are case-sensitive. Invalid commands normally produce an error and allow the user to continue. The application accepts one command per line and ends when the user enters `bye` or console input ends.

UI Component

`MishMash` owns the terminal input loop, greeting, name prompt, command execution, confirmation prompts, and recoverable error display. Classes in `mishmash.ui` format recipes, plans, groceries, and recorded meals as text. Commands return `CommandResult` values to the loop rather than reading console input themselves. Each result explicitly selects `DONE`, `NOTE`, `WARNING`, `INFO`, or `HELP`; `CliFormatter` never guesses from a command class or message prefix. The first three render as `[DONE]`, `[NOTE]`, and `[WARN]`; `INFO` remains unlabelled. The default is `INFO`, so a new command cannot accidentally claim success. This keeps command execution and presentation testable without an interactive terminal.

The command loop passes original input to `Parser`, which rejects unsupported control characters before trimming. Exit matching accepts only `bye` and its ordinary surrounding whitespace; NUL-padded text cannot silently become an exit request. `CliFormatter` escapes untrusted terminal controls before adding its own ANSI styles, including text restored from saved data. Printable Unicode, tabs and multiline feedback remain intact. `HostileCommandTest`, formatter tests, session tests and `StorageJarTest` cover malformed numbers, literal symbols, Unicode, recovery after rejected commands, and retained data after JAR restarts.

Logic Component

The logic covers command parsing, application actions, and generation. A normal command follows this flow:

1. `MishMash` reads a line through `ConsoleInput`.
2. `Parser.parseCommand` validates syntax and constructs a command; feature parsers handle specialised fields.
3. `MishMash` obtains the command's optional confirmation prompt. Cancellation skips execution.
4. `Command.executeWithResult` reads or updates the shared models and returns a `CommandResult`.
5. `SessionPersistence.afterCommand` saves mutation outcomes before `CliFormatter` displays feedback. A save failure retains live state and produces a warning.
6. Recoverable `MishMashException` failures display error/usage feedback and return to the loop; `bye` and EOF flush pending saves and exit.

The logic supports the following commands:

Command	Purpose
`add-recipe`	Add a recipe with nutrition information and ingredients.
`list-recipes`	List stored recipes or display one recipe in full.
`edit-recipe`	Replace selected fields of a recipe at its displayed index.

Command	Purpose
`delete-recipe`	Remove a recipe by its displayed index after confirmation.
`set-diet`	Replace the session dietary profile and repair or clear conflicting planned days.
`view-diet`	Display the session dietary profile.
`clear-diet`	Reset the dietary profile while preserving all other session data.
`set-targets`	Save, replace, or remove daily calorie, protein, carbohydrate, and fat targets for later plans.
`view-targets`	Display the saved daily targets.
`generate-plan`	Generate one to seven days of meals, optionally using daily nutrition targets. Saved targets fill in nutrients that the command leaves out.
`reset-plan`	Clear the current plan, its generation request, and derived groceries; preserve other session data.
`view-plan`	Display the active meal plan or one selected day.
`adjust-portion`	Change one planned meal’s positive whole-recipe portion count.
`customize-meal`	Replace one planned meal occurrence’s ingredients without editing its recipe.
`replace-meal`	Replace one planned meal with a recipe chosen by its displayed index, subject to dietary preferences.
`check-shortfall`	Compare each planned day’s nutrition with explicit daily targets and report deficits.
`generate-list`	Generate groceries for the active meal plan.
`view-list`	Display the generated grocery list.
`record-meal`	Record the recipe at a displayed index as a meal eaten.
`view-meals`	Display the recorded meals, or one date’s meals, with their nutrition and totals.
`edit-meal`	Correct the date or servings of a recorded meal.
`delete-meal`	Remove a recorded meal after confirmation.
`help`	Display the supported commands and their formats.
`bye`	End the session.

Commands validate their arguments and relevant application state. Requests that fail validation leave the existing data unchanged. Generation and viewing are separate actions: viewing does not create a new plan or grocery list.

Model Component

Model group	Responsibility
`ApplicationState`	Shares the recipe book, active plan, dietary settings, target settings, and meal log.
`RecipeBook`, `Recipe`, `GroceryItem`	Ordered recipes, their nutrition/classification, and decimal ingredient quantities with units.
`ActiveMealPlan`, `MealPlan`, `DayPlan`, `PlannedMeal`	One active plan, its generation request and groceries; days contain recipe snapshots with whole portions and optional ingredient overrides.
`DietarySettings`, `DietaryPreferences`, `Diet`, `FoodTag`, `FoodInfo`	Current restrictions and declared recipe classifications.
`TargetSettings`, `NutritionTargets`, `NutritionTotals`	Saved targets, applied constraints, and totals held as `long` values.
`MealLog`, `MealEntry`	Recorded recipe snapshots with dates and whole-serving counts, separate from planned meals.

Validated request/change objects and their interactions are described under the corresponding Implementation feature.

Recipe indexes start at 1 and change after deletion; duplicate names are allowed. Editing replaces an entry without shifting indexes. Existing plans and recorded meals retain independent recipe snapshots after recipe edits or deletion.

Saved targets apply during generation; changing them later does not change an existing plan's applied targets. Successful replacement generation clears old groceries; failed generation preserves both plan and groceries. Ingredients and groceries use `GroceryItem`; there are no separate `Ingredient` or `GroceryList` classes.

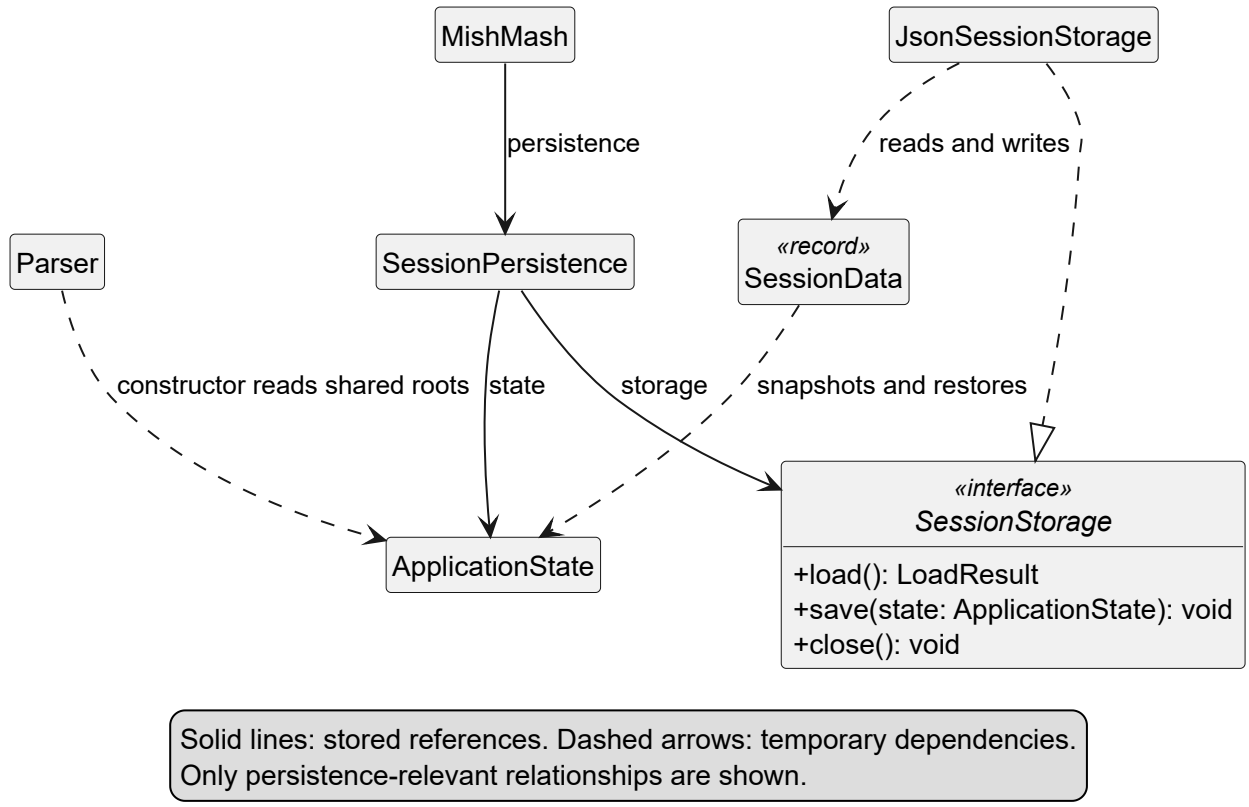
Storage Component

`ApplicationState` owns all five shared model roots. The production launcher restores them through `SessionStorage`, builds `Parser(ApplicationState)`, and uses `SessionPersistence` to autosave completed changes. See Persistence and recovery.

Diagnostic logs remain separate from the versioned JSON user-data snapshot.

Persistence and recovery

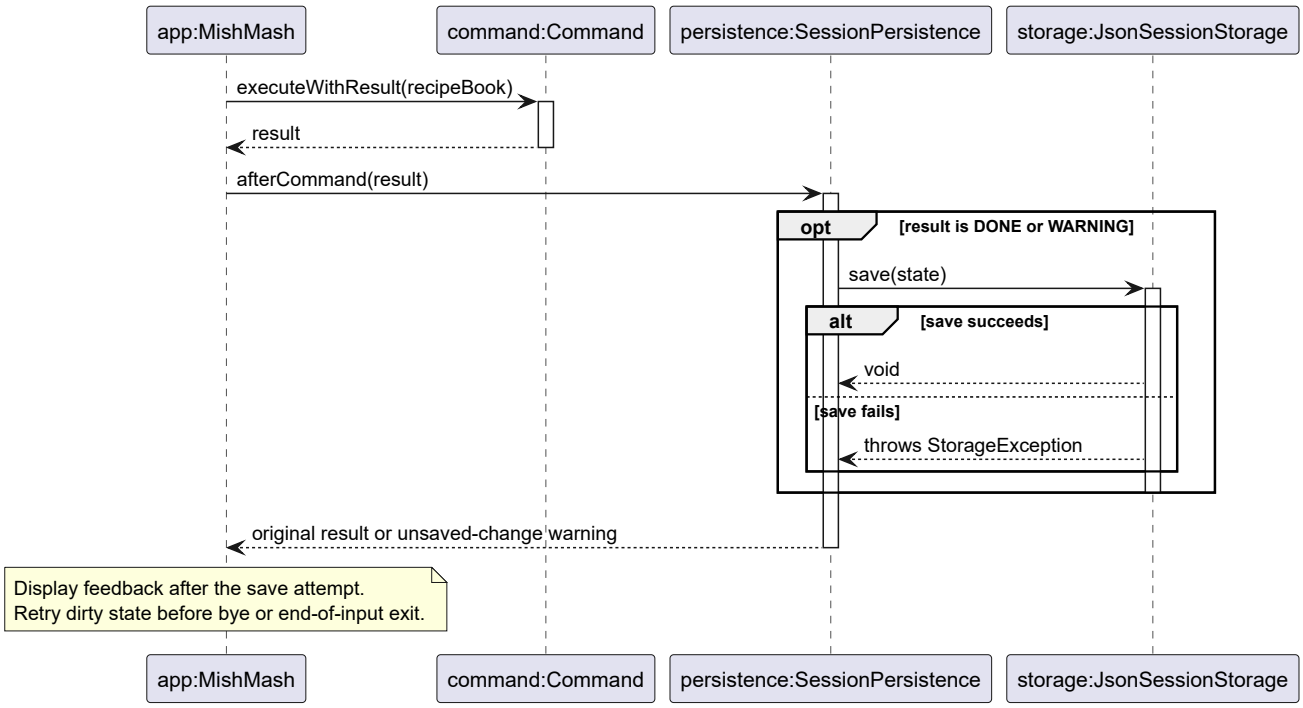
`ApplicationState` shares recipes, `ActiveMealPlan`, dietary settings, target settings, and `MealLog` between the parser and storage. Existing parser/session constructors still create isolated in-memory sessions; the public production entry point restores state before opening the command loop. This keeps tests from reading or overwriting a developer's real data. `StorageJarTest` covers the actual production launcher.



Persistence ownership and dependencies

`SessionData` is a storage-only record hierarchy with schema version 2. Recipe values are embedded independently in the recipe book, planned occurrences, and recorded meals. Orders and duplicates are preserved; sets are sorted for deterministic output. Recorded entries retain their ISO calendar date and whole-serving count. Decimal quantities are plain strings and nullable targets use JSON null. Null groceries mean not generated; an empty array means a generated empty snapshot. Restoration uses domain constructors, including exact ordered ingredient equality from the defensive PR fixes.

Jackson rejects duplicate/unknown/missing fields, scalar coercion, fractional integer values, trailing JSON, and null primitives. Parsing limits document size to 10 MiB, depth to 100, strings to 1,000,000 characters, and numeric tokens to 1,000 characters. No polymorphic typing or native Java serialisation is enabled. Only validated snapshots are published; malformed data cannot partially update application state.



Autosave sequence

`SessionPersistence` marks `DONE` and `WARNING` outcomes dirty and saves before presenting feedback. Notes, read-only output, cancelled commands, and parse/execution errors do not trigger autosave. A failed save retains the live state, produces warning feedback, and retries after the next mutation and on normal exit. Future state-changing commands must report a mutation outcome; presentation status currently supplies that contract. Unexpected application failures are still logged and rethrown; they do not trigger an unsafe final save.

`JsonSessionStorage` holds an exclusive `FileChannel` lock for the process lifetime. Writes validate the JSON, flush a same-directory temporary file, preserve the previous valid backup, and replace the primary with an atomic move where supported. Filesystems without atomic replacement fall back to replacement plus backup recovery. Temporary-file cleanup is best effort. An unreferenced `.tmp` is never loaded automatically. Operating-system locks and file flushing do not provide a guarantee against every power or hardware failure.

Startup distinguishes invalid JSON from filesystem access failure. Only invalid files are quarantined; an inaccessible file or a failed backup restoration stops startup. Recovery never quarantines a valid backup merely because writing the primary failed. Unsupported versions are preserved like malformed snapshots; version 1 snapshots are upgraded automatically to version 2. Legacy undated records receive the local upgrade date and one serving, with warning feedback. The original file is retained as the backup. The upgrade is saved immediately so repeated restarts do not change those assigned dates. Failed upgrades stop startup and preserve the original primary; they never classify a valid legacy file as corrupt.

Run Java 25 `./gradlew check shadowJar`. Storage tests use temporary directories and cover round trips, validation, locks, backups, quarantine, write failures, and retry behaviour. The check task also builds and runs the shadow JAR through multiple processes to verify retention from a working directory outside the repository. JAR tests also cover forced termination after autosave, lock release, corrupt-primary recovery, unsaved-change warnings and shutdown retries, failed dietary repair, and unchanged save files after no-ops or cancellation.

Common classes

Shared types remain in the packages that define their responsibilities:

Type	Shared responsibility
`command.CommandResult`	Carries feedback and an explicit outcome from commands to the UI and persistence coordinator.
`exception.MishMashException`	Carries recoverable errors, optional command-specific usage, and the original cause from parsing or execution to the application.
`parser.IngredientParser`	Validates ingredient syntax for adding recipes, editing recipes, and customising planned meals.
`model.GroceryItem`, `model.NutritionTotals`	Provide validated ingredient quantities and nutrition values used by planning, viewing, and storage.

Implementation

This section explains the design and implementation of MishMash's completed features, including their key components, workflows, and error handling.

Adding a recipe

Section author: peilingggg

`add-recipe` creates a recipe from its name, nutrition, ingredients, and optional food categories. This supports the user story: **As a first-time user, I want to add a recipe with its ingredients and quantities so that I can create a personalised meal.** The User Guide gives the command rules and examples.

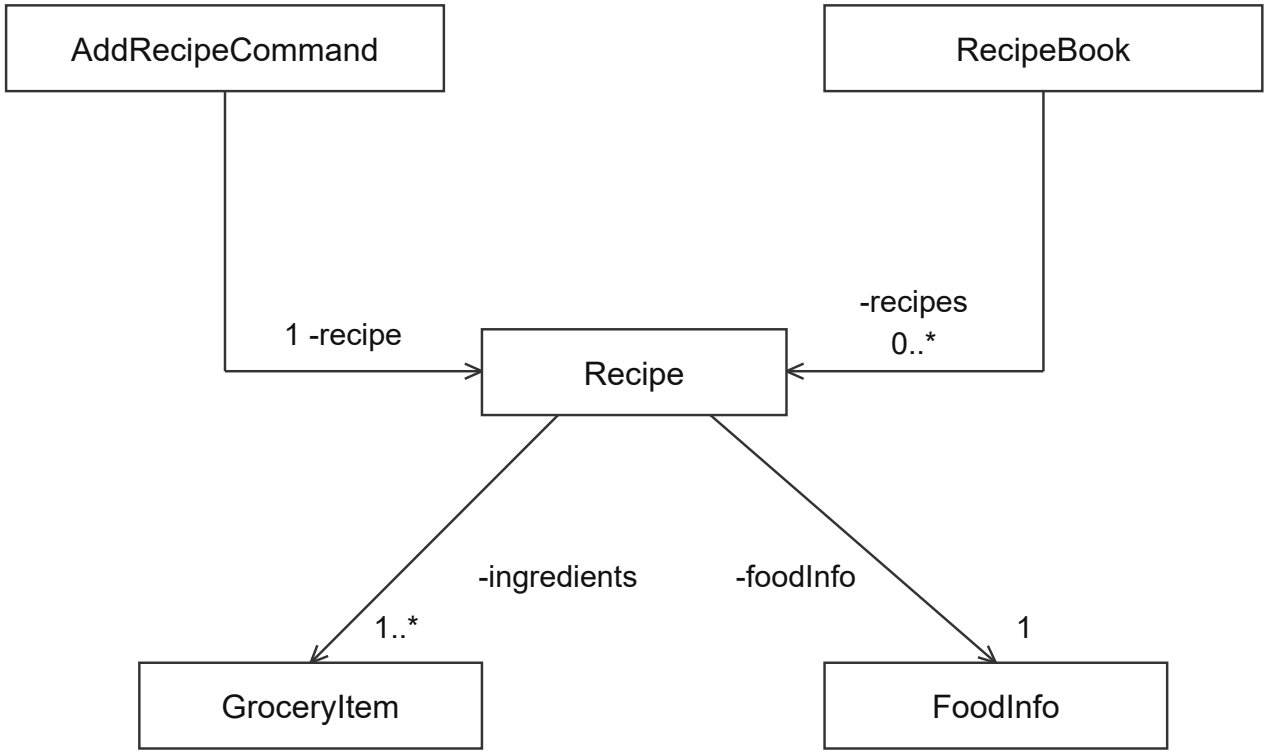
```
add-recipe n/DISHNAME cal/CALORIES p/PROTEIN c/CARBS f/FATS i/INGREDIENT:QUANTITYUNIT
[i/INGREDIENT:QUANTITYUNIT ...] [food/TAG[,TAG...]]
```

All five recipe fields and at least one ingredient are required. Fields may appear in any order; only `i/` may repeat. Recipes are autosaved for later sessions. Adding a recipe leaves existing recipes, meal plans, groceries, and recorded meals unchanged.

Responsibilities

Component	Responsibility
`Parser`	Read prefixed fields, validate values, and create a recipe and command without changing the book.
`Recipe`, `GroceryItem`	Hold validated recipe data and precise decimal ingredient quantities.
`FoodInfo`, `FoodTag`	Represent the optional food-category declaration and validate supported tags.

Component	Responsibility
`AddRecipeCommand`	Prepare the confirmation, then append its validated recipe to the supplied book.
`RecipeBook`	Store recipes in insertion order, including duplicate names.
`RecipeViewer`	Format recipe details consistently with listing and editing.
`MishMash`	Parse and execute the command, print its result or error, and continue the input loop.



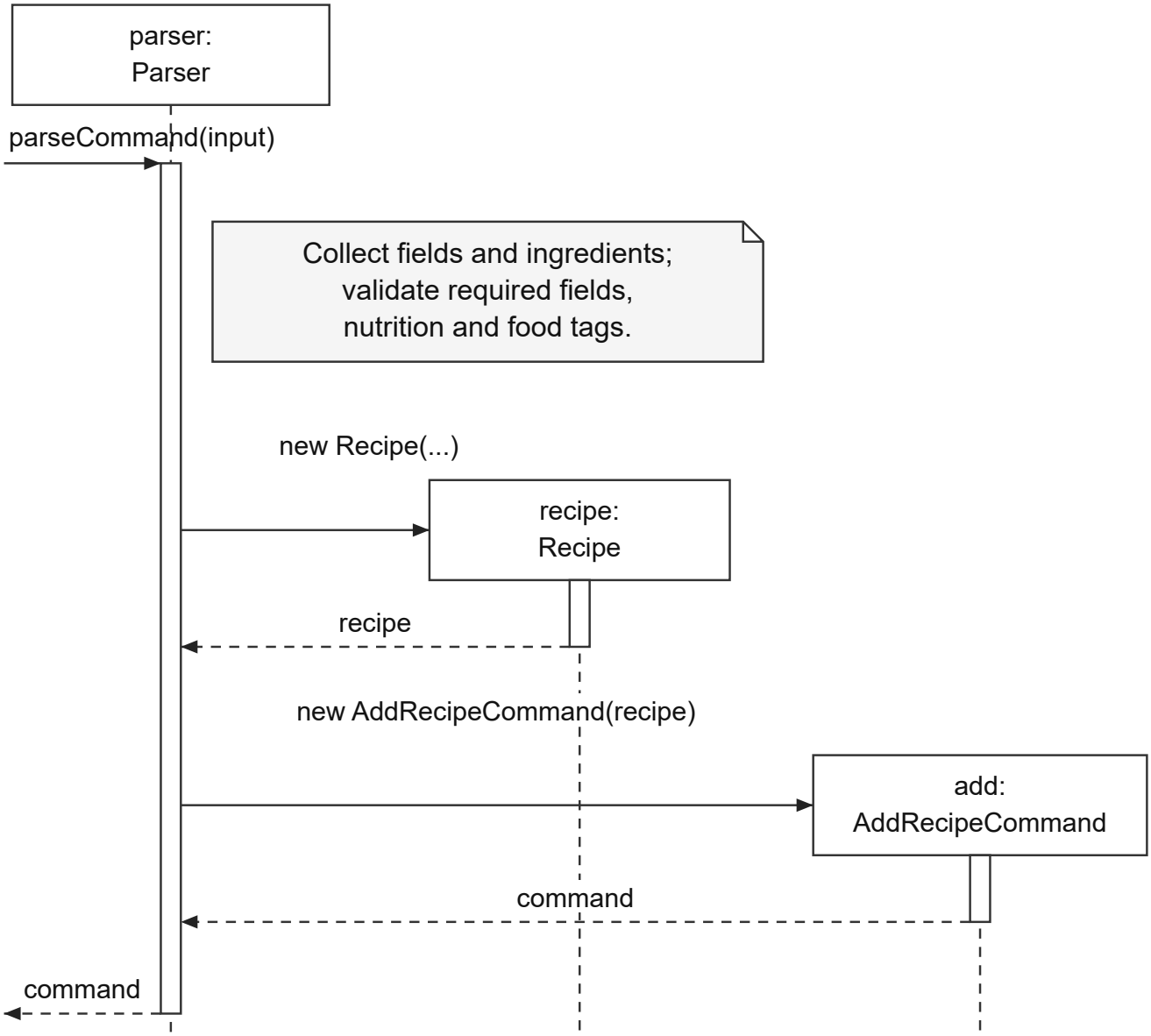
Class diagram of the stored recipe references

Selected stored references. Open arrowheads show navigability, with target multiplicities and private role names. Members, temporary dependencies, and `Command` realization are omitted; `AddRecipeCommand` implements `Command`. PlantUML source

The command stores one recipe; the book and viewer are supplied or created during execution. The book and ingredient lists preserve order and allow repeated entries. `Recipe` copies the ingredient list with `List.copyOf`, protecting its structure without deep-copying the `GroceryItem` objects. Plain associations allow references to be shared with other features.

Parsing and object creation

For example, `add-recipe n/Rice cal/200 p/4 c/45 f/1 i/Rice:150g food/none` creates one recipe containing a 150 g ingredient and an explicit empty food-category declaration.



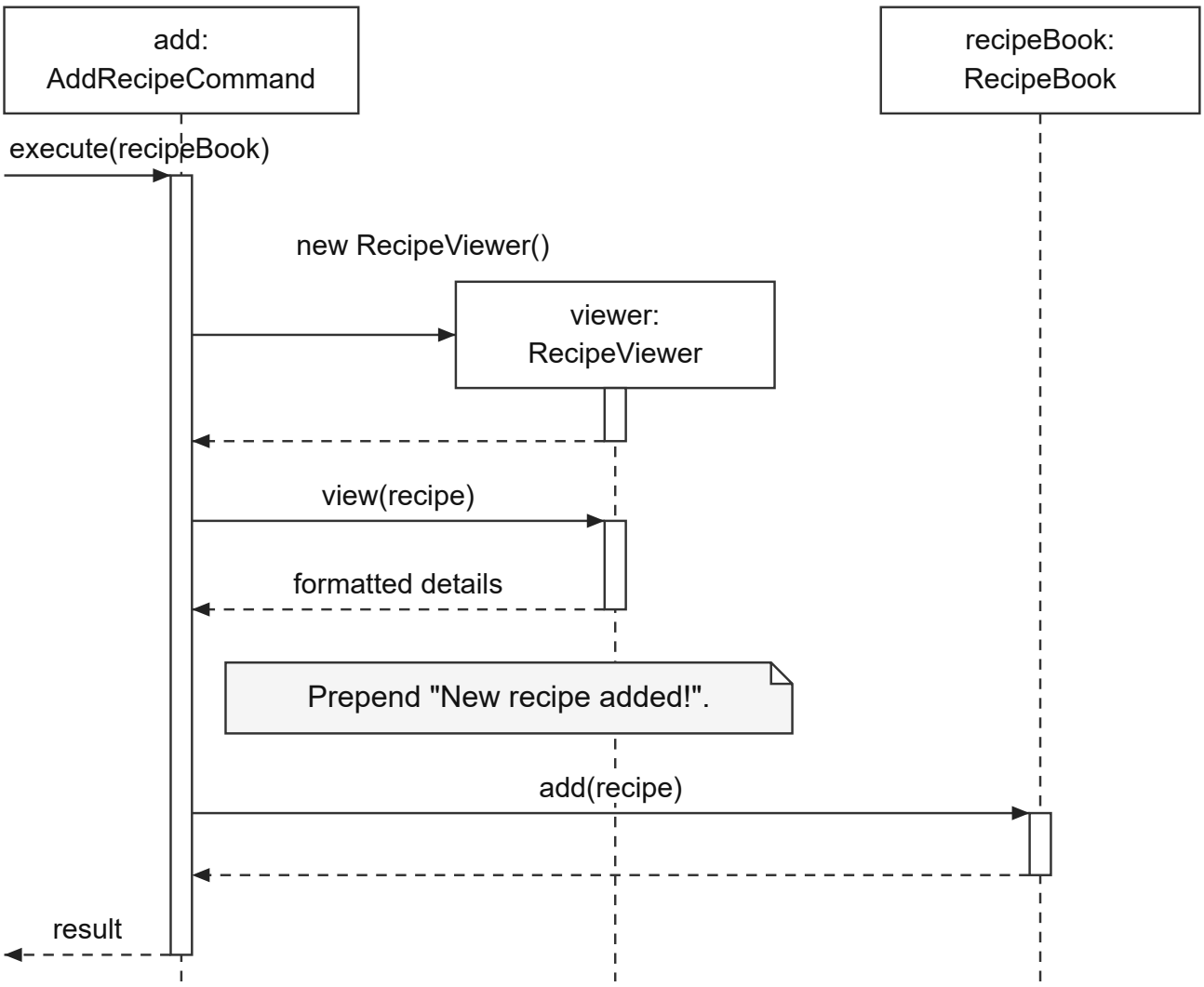
Sequence diagram of validating and creating an addition command

Successful parsing. Dispatch helpers, ingredient creation, field validation, and library operations are condensed in the note. Only the recipe and command constructors are expanded. Parameters abbreviated with `...` are described below; parsing never updates the book. PlantUML source

1. `collectRecipeArguments` identifies prefixed fields, including unsupported prefixes, so an unknown field cannot silently become part of a name. `readArgument` stores single-value fields and appends each parsed `i/` item to a local list.
2. `validateRequiredFields` checks `n/`, `cal/`, `p/`, `c/`, `f/`, and at least one ingredient. `parseNutrition` then reads calories, protein, carbohydrates, and fats, in that order, as non-negative integers. Ingredient quantities use `BigDecimal`.
3. `FoodInfo.parseField` reads the optional declaration. Missing `food/` means unclassified; `food/none` means no supported categories apply. Other values are validated tags, never inferred from ingredient names.
4. Only after validation succeeds does the parser construct `Recipe` and `AddRecipeCommand`. The recipe constructor independently checks its constraints to protect direct Java callers.

Execution and output

The inherited `Command.confirmationPrompt` returns an empty `Optional`, so addition does not ask for confirmation. The execution diagram starts when `MishMash` calls `execute(recipeBook)`.



Sequence diagram of formatting and storing a recipe

Successful execution. Calls have filled arrowheads and returns are dashed. Constructor activations start below the newly created object boxes. Formatting internals, logging, the confirmation check, and application printing are omitted. PlantUML source

1. `AddRecipeCommand.execute` rejects a null book, creates `RecipeViewer`, and prepares the full confirmation before inserting anything. A formatting failure therefore cannot leave an addition without a prepared confirmation.
2. `RecipeBook.add` checks the recipe is non-null and appends its reference. Duplicate names and repeated ingredients remain separate entries; grocery generation handles aggregation.
3. The command returns the confirmation. `MishMash` prints it and prompts for another command. For the Rice example:

```
New recipe added!
Rice
200 kcal | Protein: 4g | Carbs: 45g | Fats: 1g
```

```
Ingredients:
- Rice: 150g
Food tags: None declared
```

Ingredient quantities display as plain decimals without unnecessary trailing zeros. `RecipeViewer` also displays sorted food tags, or `Unclassified` when `food/` was omitted.

Error handling and defensive checks

Input or condition	Result
Missing required field, no ingredient, or unprefixed leading text	Reject before creating an addition command.
Blank value, unknown prefix, embedded `/`, or repeated single-value field	Reject; repeated `i/` fields remain valid.
Nutrition outside `0` to `Integer.MAX_VALUE`, including signs and decimals	Reject; zero and leading zeros are valid.
Non-positive quantity, missing unit, malformed decimal, or extra colon	Reject; quantities require digits before and after a decimal point when present.
Unit containing characters other than ASCII letters	Reject; accepted units are not converted or checked against a fixed vocabulary.
Unsupported, blank, or repeated food tag	Reject; `food/none` cannot be combined with other tags.
Null Java dependency or required model argument	Fail with `NullPointerException`, indicating a caller contract violation.

For example, `i/Rice:0g` is rejected by `parseIngredient` before a `GroceryItem` is created. `parseAddRecipeCommand` wraps parsing failures in `MishMashException`, preserves the cause, and adds `ADD_RECIPE_USAGE`. The application displays the error and accepts another command; existing recipes remain unchanged. Although `GroceryItem` permits zero, recipe parsing and the `Recipe` constructor enforce a strictly positive ingredient quantity.

Alternatives and limitations

Decision	Alternative	Reason
Validate before insertion	Add a partial recipe first	Invalid input cannot leave an incomplete book entry.
Validate in both parser and model	Trust the parser alone	Syntax checks and model contracts protect different callers.
Use `BigDecimal` quantities	Use floating-point quantities	Preserve entered decimal precision.
Reuse `RecipeViewer`	Format separately in each command	Keep addition, listing, and editing output consistent.

Names identify display text rather than unique keys, and repeated names are accepted. Food categories depend on user declarations. No unit conversion is provided by addition. To use a newly added recipe in generated data, generate a new plan, then generate its grocery list.

Automated verification

- `AddRecipeBoundaryTest` covers nutrition limits, quantities, whitespace, ordering, and repeated ingredients.
- `AddRecipeEdgeCaseTest` uses `add-recipe-invalid.tsv` to check rejection, unchanged data, and recovery.
- `AddRecipeExceptionTest` checks command usage and preserved causes.
- `AddRecipeCommandTest`, `RecipeTest`, and `RecipeBookTest` cover insertion, output, null contracts, and collection protection.
- `MishMashTest` covers console interaction and continued input after errors.

Run `./gradlew test checkstyleMain checkstyleTest` with Java 25 (Windows: `.\gradlew.bat test checkstyleMain checkstyleTest`). Sources and SVG maintenance instructions are in diagrams/add-recipe.

Listing recipes

Section author: neil0618

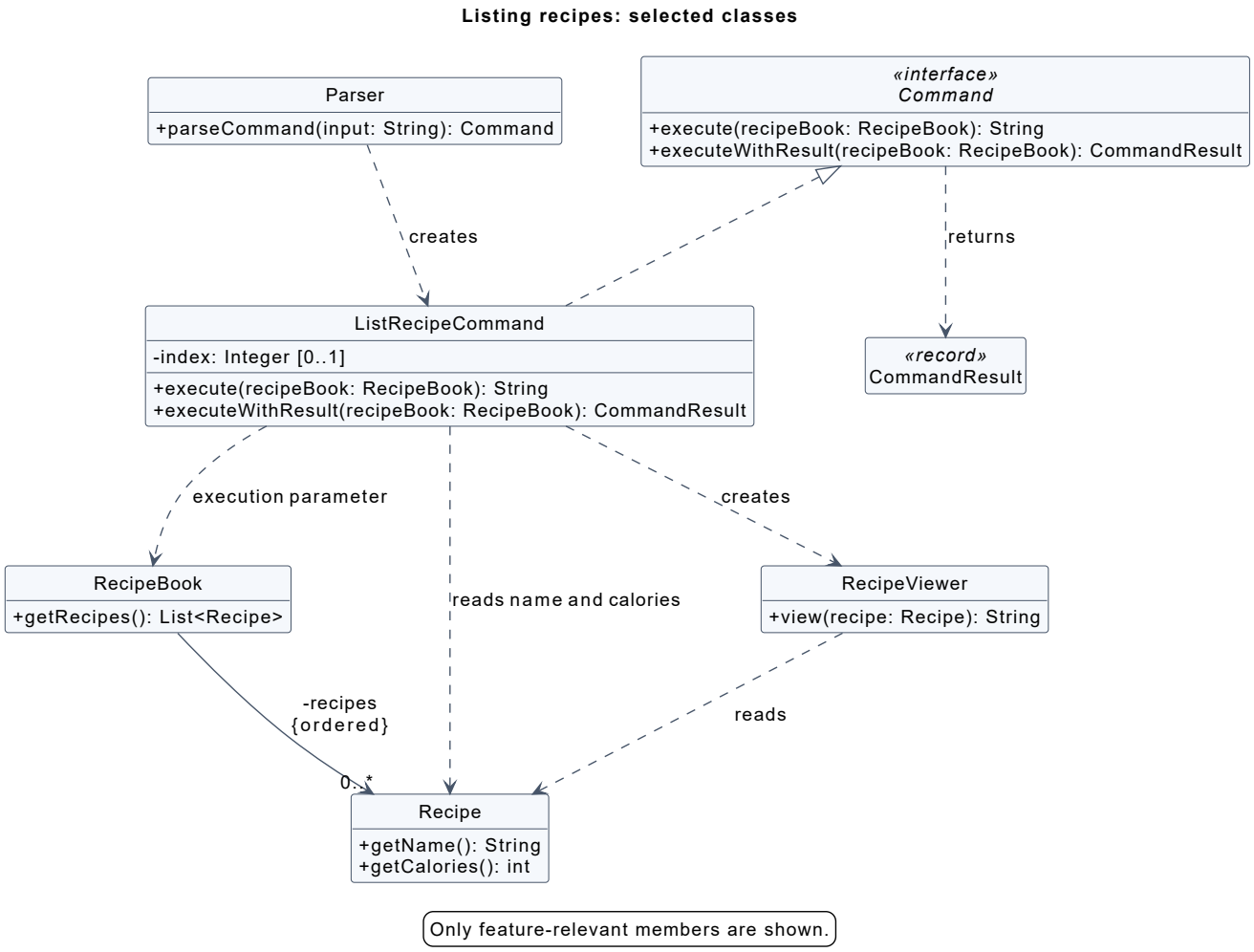
Purpose and command contract

This feature implements the story: **As a user managing recipes, I want to list my recipes and inspect a selected recipe so that I can find and check its stored details.**

`list-recipes [INDEX]` has two forms. Without an index it shows every recipe as a numbered line with its name and calories. With an index it shows that one recipe in full: nutrition, ingredients, and food tags. The index is one-based and equals the number shown in the list, which is also the number that `edit-recipe` and `delete-recipe` expect. For example, `list-recipes 2` shows the second recipe. Listing is read-only: it never changes the recipe book, the active plan, or the grocery list.

Class responsibilities and relationships

`Parser` validates the optional index and creates a `ListRecipeCommand`. The command keeps only a nullable `Integer index`, where `null` means “list every recipe”. The recipe book is supplied only when `executeWithResult` runs, so the command reads the recipes as they are at that moment rather than a copy taken during parsing. Recipe details are formatted by `RecipeViewer`, the class that also formats the `add-recipe` and `edit-recipe` confirmations, so a recipe looks the same wherever it is shown.

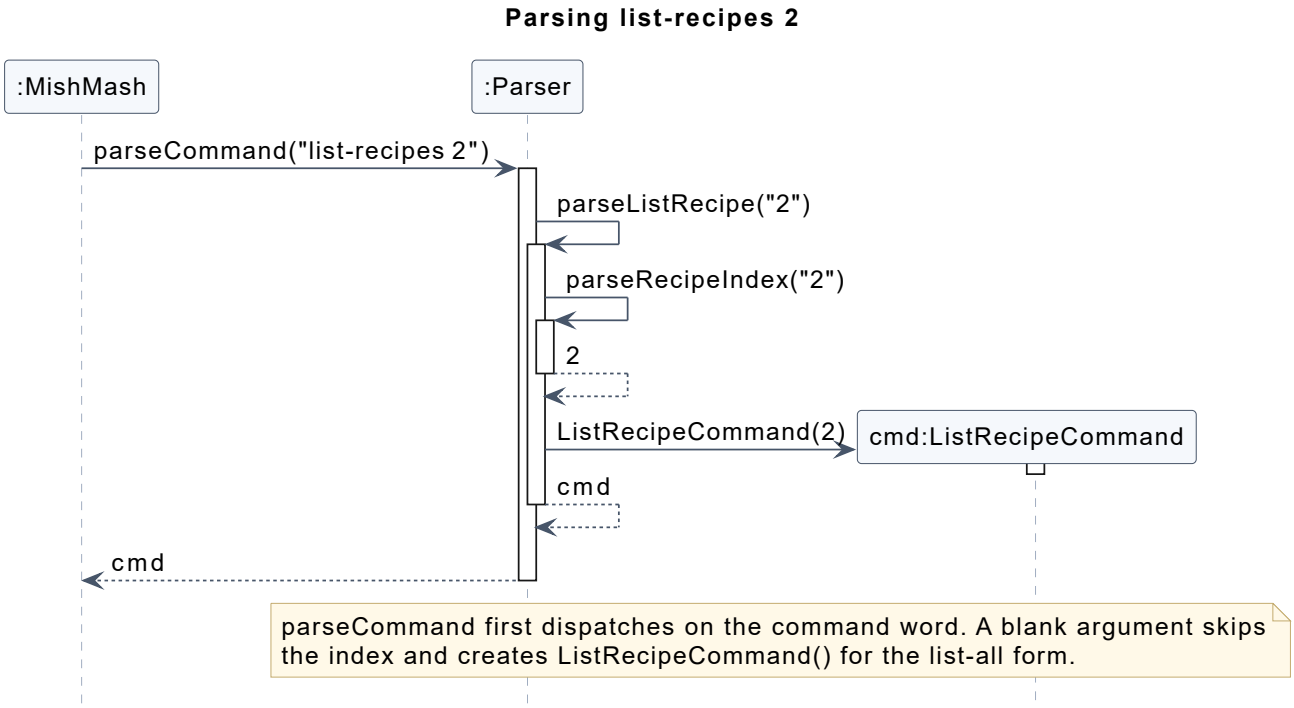


Selected classes and relationships for recipe listing.

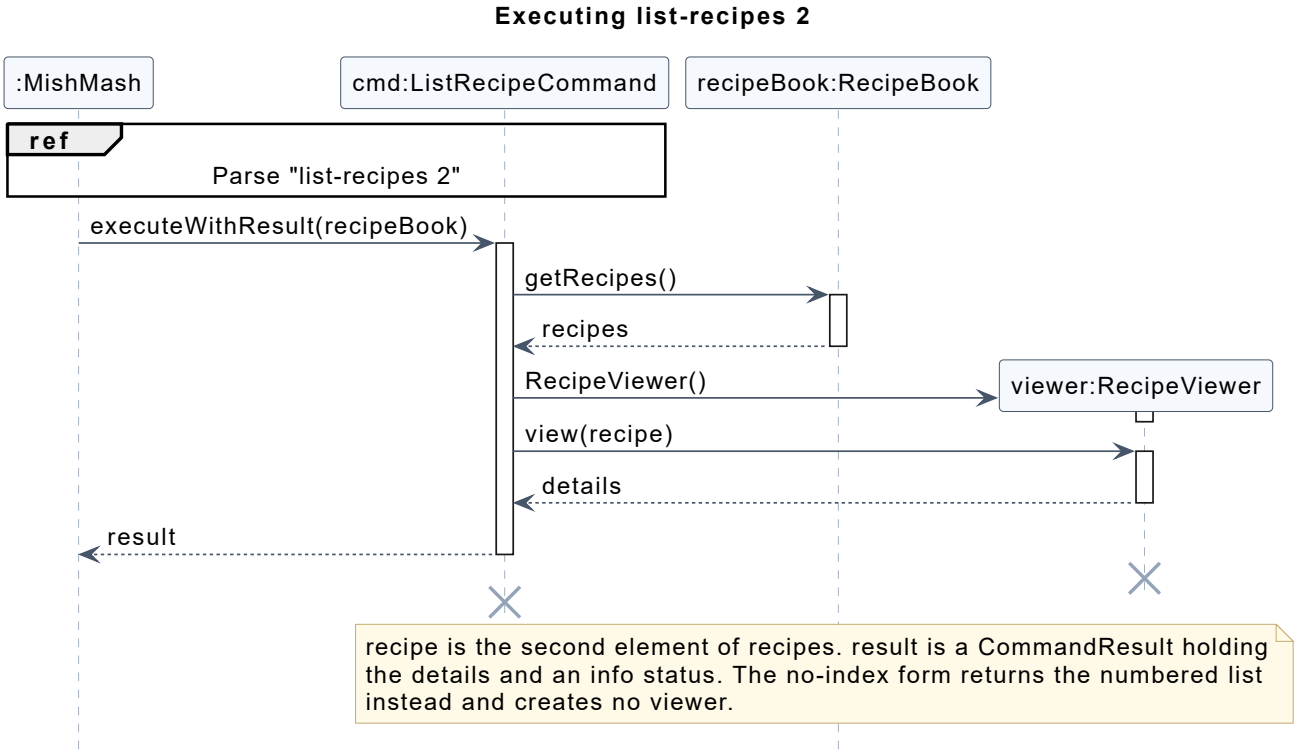
Only feature-relevant members are shown. Dashed arrows with an open head are dependencies; the dashed arrow with a hollow triangle is interface realization; the solid arrow is an association with a role name and multiplicity. `RecipeBook.getRecipes` returns a read-only copy, so listing cannot change the book.

Parsing and execution

The successful `list-recipes 2` scenario is separated into parsing and execution. Invalid and no-index outcomes are described afterwards so both interactions remain readable at guide width.



Parsing list-recipes 2.



Executing list-recipes 2.

The execution diagram refers to the parsing diagram with a `ref` frame. The no-index form skips index parsing and the viewer. The x marks objects that are no longer used after the command returns.

`Parser.parseListRecipe` treats blank arguments as the list-all form. Otherwise it calls `parseRecipeIndex`, the helper that `delete-recipe` also uses. The helper accepts digits only and rejects zero and values above `Integer.MAX_VALUE`. Leading zeros and surrounding spaces are accepted, so `list-recipes 01` selects recipe 1.

The parser does not receive the recipe book, so whether the index exists is decided at execution. `MishMash` calls `executeWithResult`, which reads the book, then either returns a `CommandResult` holding the details of `recipes.get(index - 1)` or throws a `MishMashException`. `execute` is kept for callers that only need the text and returns the `feedback` of that `CommandResult`. The list-all form returns `Recipes:` followed by one `N. name (calories kcal)` line per recipe, or `No recipes yet. Use 'add-recipe' to add one.` for an empty book. An empty book is therefore an error only when an index was requested.

Invalid input and recovery

Input or state	Where it is caught	Result
Index that is not digits only (`abc`, `-1`, `1.5`, `1 2`), is `0`, or exceeds `Integer.MAX_VALUE`	`Parser.parseRecipeIndex`	Error ending with `Format: list-recipes [INDEX]`.
Valid index, empty book	`ListRecipeCommand.execute`	Error telling the user to add a recipe.
Valid index above the recipe count	`ListRecipeCommand.execute`	Error stating the valid range.
No index, empty book	`ListRecipeCommand.execute`	Normal message, not an error.
Command word with a different case or spelling (`List-Recipes`, `list-recipe`)	`Parser.parseNamedCommand`	`Unknown command` error.

Syntax errors start as an `IllegalArgumentException`. `parseListRecipe` wraps it in a `MishMashException` that keeps the original cause and ends with the list-recipes usage, so `MishMash.displayError` does not print the general command list. The two execution-time errors pass `ListRecipeCommand.USAGE` as well, so they also show only the list-recipes format. In every case the session continues and the recipe book is unchanged.

Design choices and alternatives

Decision	Alternative	Reason for the current design
One command with an optional index	A separate `show-recipe INDEX` command	There is one command word to learn, and the numbered list shows exactly the numbers that `edit-recipe` and `delete-recipe` need.
Look the recipe up when the command executes	Look it up while parsing	The parser stays independent of application data, as for the other recipe commands, and the lookup sees the book as it is when the command runs.
A nullable `index` means “list all”	Two command classes, or an `Optional<Integer>` field	It is the simplest option that is sufficient, and both forms share the book access. Two classes would be worth it if the forms diverged.

Decision	Alternative	Reason for the current design
The list shows only names and calories	Show all nutrition for every recipe	A compact list stays readable for a long book, and full details are one command away.
Reuse `RecipeViewer` for the details	Format the text inside the command	A recipe is displayed identically after `add-recipe`, `edit-recipe`, and `list-recipes`.

Verification and maintenance

`ListRecipeCommandTest` checks the numbered list, empty and duplicate-name books, the full details of the first, last, and a duplicate-named recipe, out-of-range and empty-book errors with the book left unchanged, repeated calls, and null arguments. `ParserTest` checks both forms, accepted variations such as leading zeros and extra whitespace, invalid arguments showing only the list-recipes usage, the largest index (which parses and fails only at execution), and near-miss command words. `MishMashTest` checks the console flow: the list followed by one recipe in full, an empty book, a bad index followed by recovery, and the numbers matching those used by `delete-recipe`.

For manual verification, follow List Recipes in the manual-testing appendix, or the User Guide's Listing Recipes example. The class diagram and sequence diagram have editable sources and rendering instructions in `docs/diagrams/neil0618/list-recipes`. When the implementation changes, check method names, message order, and the error paths against the code, then regenerate and inspect both SVGs.

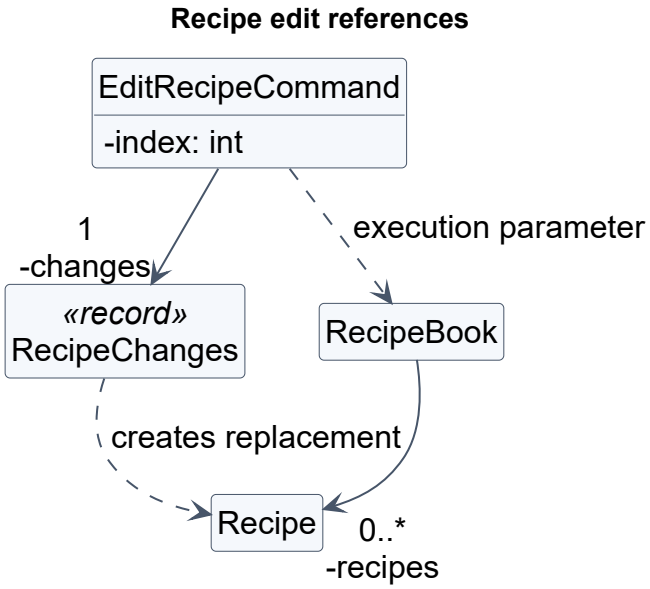
Editing a recipe

Section author: Rixte

`edit-recipe INDEX` changes selected fields of the recipe at a one-based displayed index. For example, `edit-recipe 1 p/55 i/Chicken:220g i/Rice:150g` replaces its protein and entire ingredient list. Omitted fields stay unchanged. The User Guide gives the complete syntax.

Responsibilities

Component	Responsibility
`Parser.parseEditRecipe`	Validate the index and supplied fields using the shared recipe helpers.
`RecipeChanges`	Store immutable optional edits and build a validated replacement.
`EditRecipeCommand`	Look up the current recipe, format feedback, and then publish the edit.
`RecipeBook`	Replace one entry without changing the recipe count or indexes.
`RecipeViewer`	Format every field of the edited recipe.



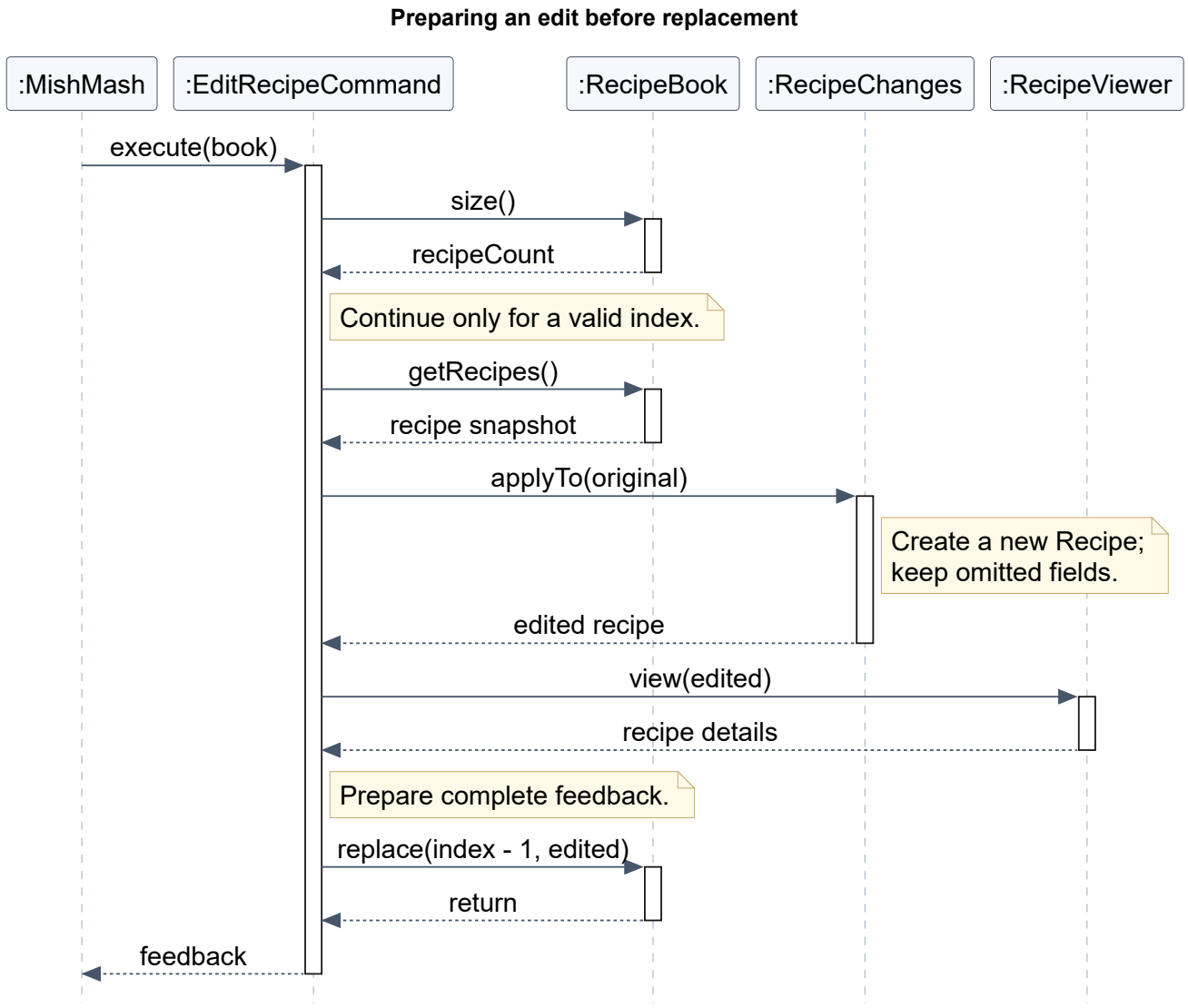
Class diagram of recipe editing references

Selected stored references and temporary dependencies. The book's ordered list permits repeated references; existing plans may retain the original recipes. Parser creation, viewer dependencies, and the command interface are omitted. PlantUML source

Nullable scalar fields in `RecipeChanges` mean unchanged; an empty ingredient list also means unchanged. Nullable nutrition distinguishes an omitted field from an explicit zero. Supplied ingredients replace the complete list. Omitted `food/` preserves the previous declaration, and changing ingredients does not infer new food categories.

Parsing and execution

Parsing reuses `collectRecipeArguments`, `parseOptionalNutrition`, and `parseIngredient`. Single-value prefixes cannot repeat; ingredient prefixes can. Validation finishes before a command is created, without looking up or changing any stored recipe.



Sequence diagram of formatting and publishing a recipe replacement

Successful execution for a valid index. `original` is the snapshot entry at `index - 1`. Field getters, replacement construction, logging, and assertions are condensed; viewer construction is omitted. Feedback is complete before replacement. PlantUML source

1. `execute` checks the current book size and index, then reads the recipe at `index - 1`.
2. `RecipeChanges.applyTo(original)` creates a new `Recipe` from supplied edits and retained fields.
3. `RecipeViewer.view(edited)` formats the replacement; the command prepares the complete feedback.
4. Only then does `RecipeBook.replace(index - 1, edited)` update the book entry.

Errors and snapshot behaviour

Condition	Result
Missing fields, malformed index, unknown or repeated single-value prefix, or invalid field	Parsing raises `MishMashException` with edit usage and the validation cause; no edit is applied.
Empty book or index beyond the current size	Reject during execution before creating a replacement.

Condition	Result
Replacement validation or formatting fails	The book entry has not yet been replaced.
Edit succeeds	Recipe count and indexes stay the same; existing plans, groceries, and recorded meals retain their old snapshots.

Commands resolve indexes during execution, including any renumbering caused by earlier deletions. To incorporate an edit into planned meals and groceries, generate a new plan and then a new grocery list. Regenerating groceries alone still uses the existing plan.

Design choice and verification

Immutable replacement keeps the change local to the recipe book. Mutating the original would also change references held by planned or recorded meals and could invalidate groceries. Replacing the whole ingredient list supports quantity changes and removals without introducing ingredient identifiers or extra subcommands.

`ParserEditRecipeTest`, `RecipeChangesTest`, and `EditRecipeCommandTest` cover validation, partial edits, current-index lookup, food declarations, and snapshot preservation. `EditRecipeIntegrationTest` and `EditRecipeDiagnosticsTest` check command integration and diagnostics. Manually edit a recipe after generating meals and groceries: both views should keep the old values until their data is regenerated. A rejected `p/-5` must leave the last valid recipe unchanged.

Deleting a recipe

Section author: neil0618

Purpose and command contract

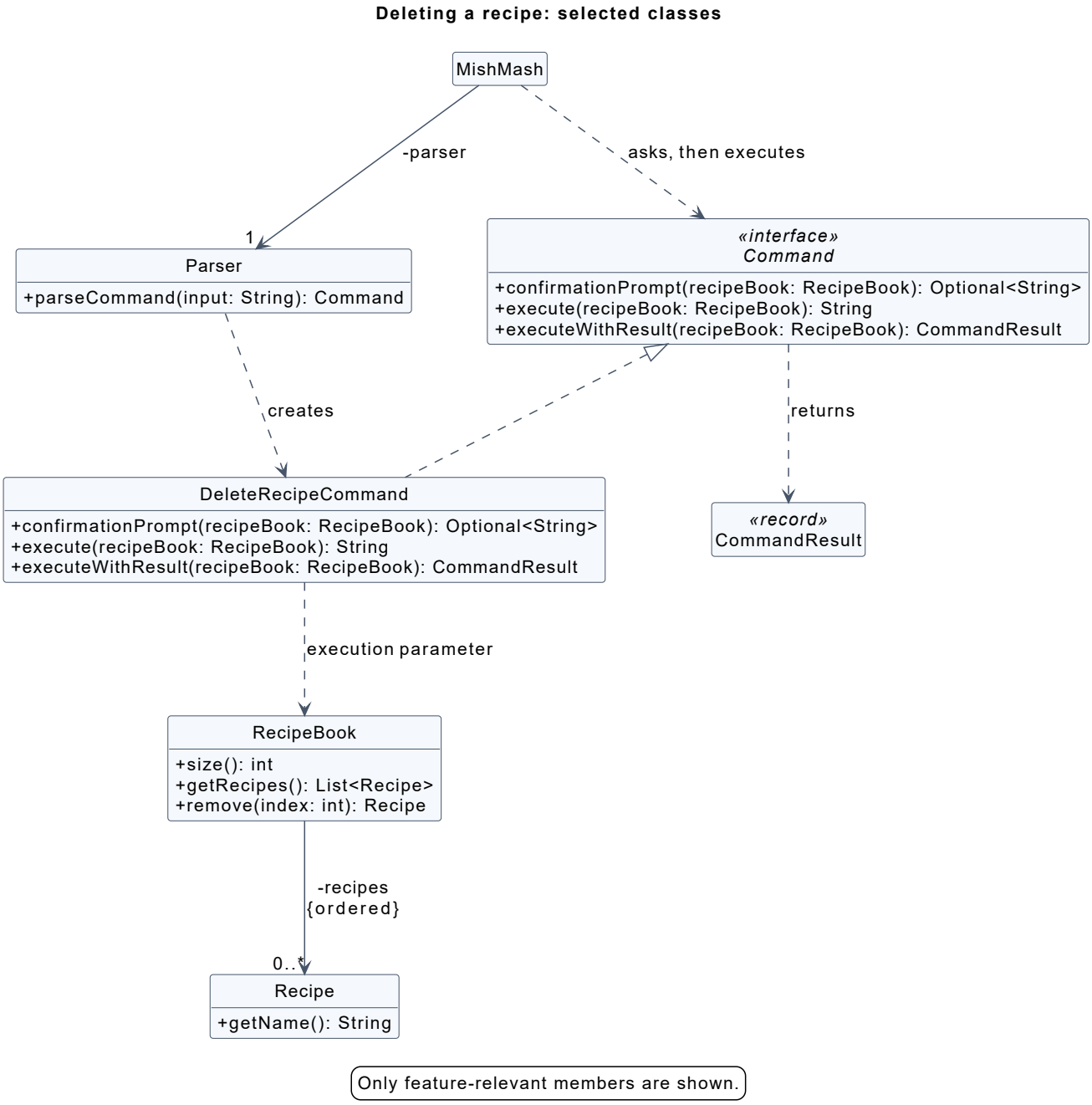
This feature implements the story: **As a user managing recipes, I want to confirm the deletion of a selected recipe so that I can check that I am removing the intended entry.**

`delete-recipe INDEX` removes the recipe at the displayed, one-based position. Deletion cannot be undone, so the application first names the recipe and `CliFormatter` presents the question as `[CONFIRM] ... [y/N]:`. Only `y` or `yes`, ignoring letter case and surrounding whitespace, deletes the recipe. Any other answer, including a blank line or the end of input, cancels. For example, `delete-recipe 2` followed by `y` removes the second recipe, and every later recipe moves up one position.

The command changes only the `RecipeBook`. It does not use `ActiveMealPlan`, so an existing meal plan and its grocery list keep the recipes they were generated with. No undo is provided.

Where the confirmation lives

`Command` declares `confirmationPrompt(RecipeBook)`, a default method that returns an empty `Optional`. A command that needs confirmation overrides it and returns its question. `MishMash.executeCommand` calls it for every command. If it returns a question, `userConfirms` displays it and reads the next input line, and `execute` runs only if the answer agrees.



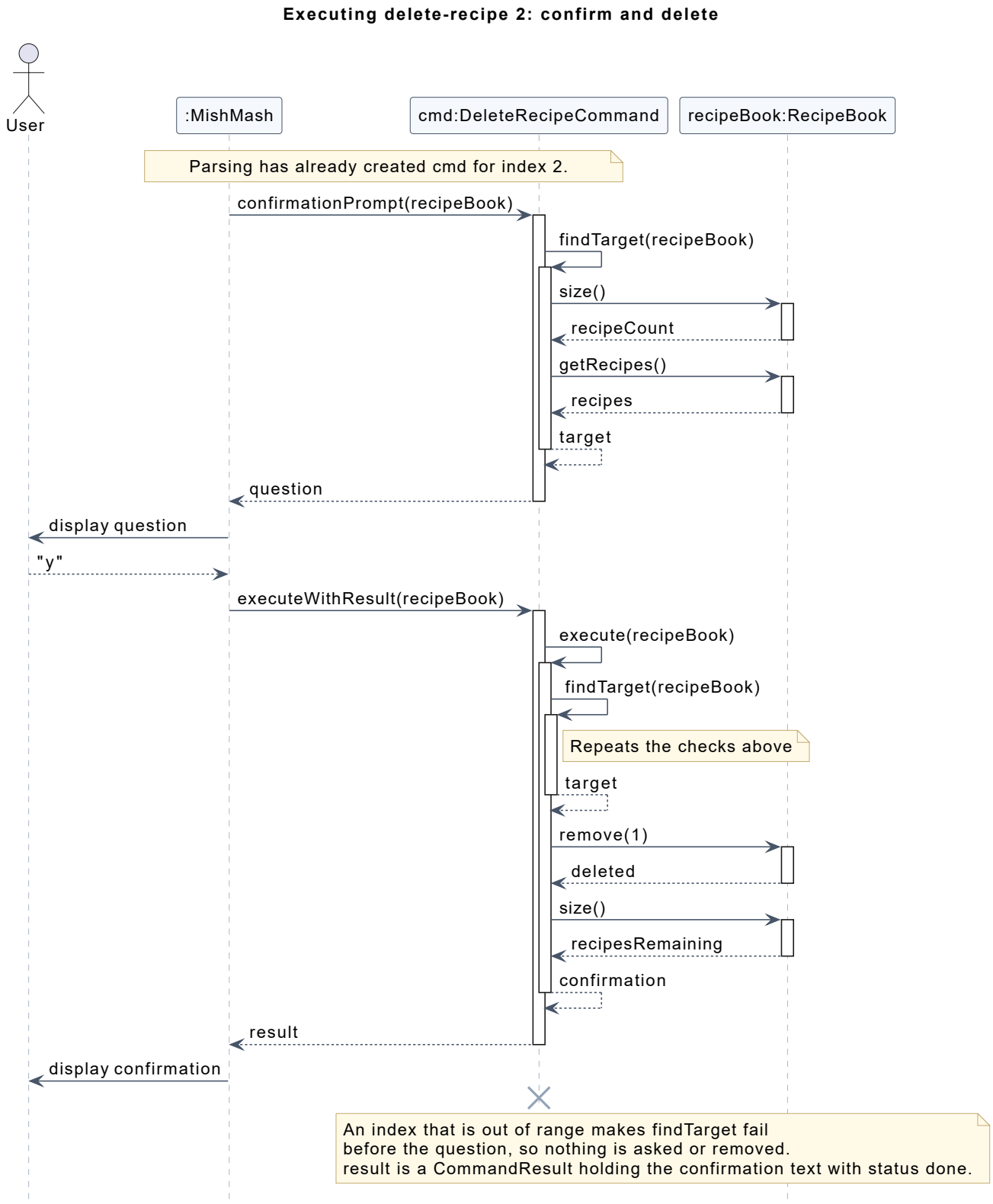
Selected classes and relationships for recipe deletion.

Only feature-relevant members are shown. Dashed arrows with an open head are dependencies; the dashed arrow with a hollow triangle is interface realization; solid arrows are associations with a role name and multiplicity. `RecipeBook.remove` takes a zero-based index, while `DeleteRecipeCommand` stores the one-based number the user typed.

This keeps all console input and output in `MishMash`. `DeleteRecipeCommand` only builds the question and performs the deletion, so tests can call both methods directly without a `Scanner`. The question contains the index as well as the name, so recipes with the same name can still be told apart.

Execution and confirmation

The first sequence starts after parsing and follows a confirmed `delete-recipe 2`.



Execution of recipe deletion, with confirmation.

The command validates the index once while building the prompt and again immediately before removal. The x marks the command object, which is no longer used once it has returned. An invalid index is not drawn here because it is covered by the activity diagram below.

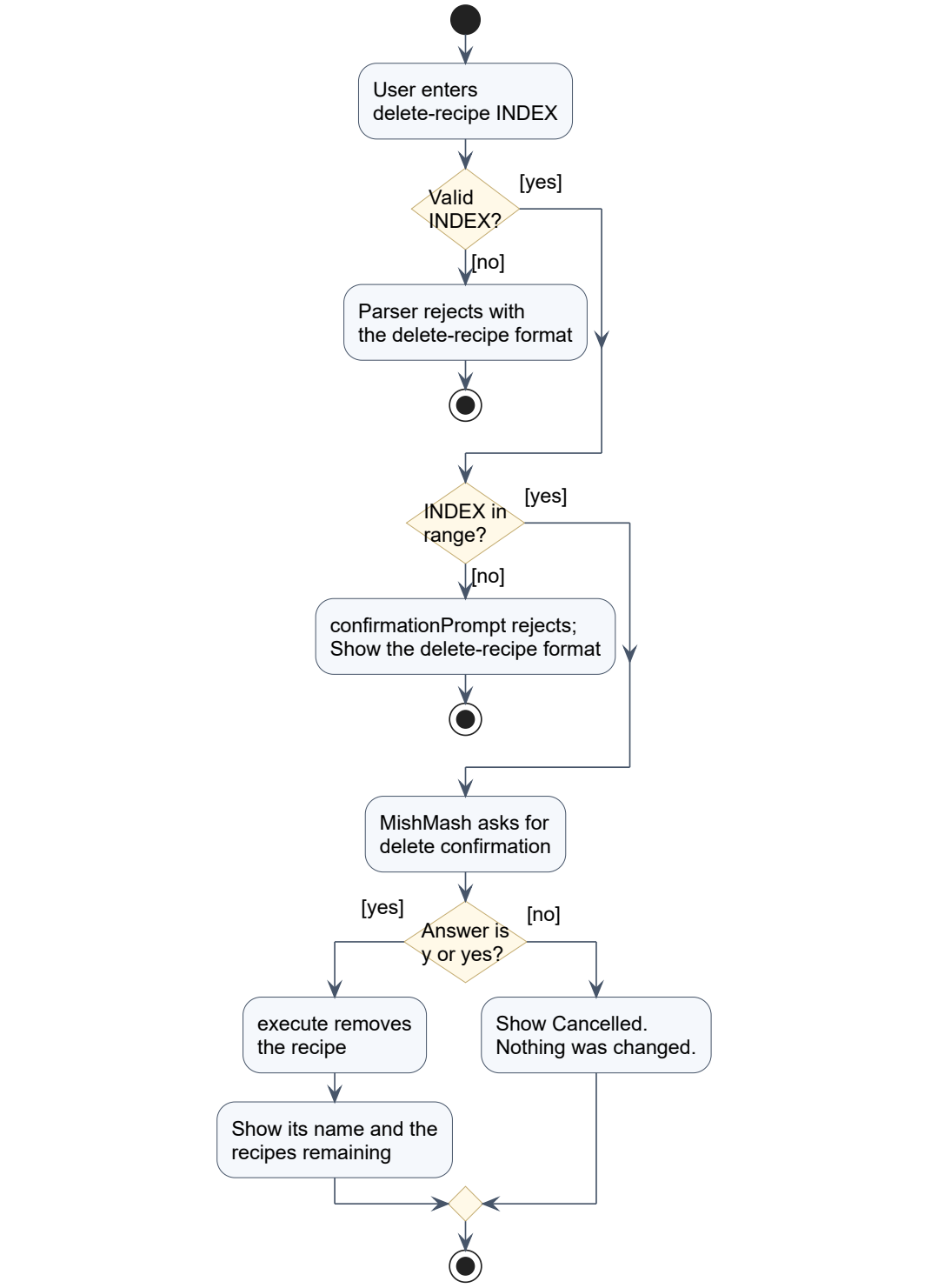
`confirmationPrompt` finds the target through `findTarget`, which throws if the book is empty or the index is too large. An impossible request therefore fails before any question is asked. `execute` calls `findTarget` again instead of trusting an earlier prompt, so it is also safe when called on its own. It then

calls `RecipeBook.remove(index - 1)` and returns `Recipe deleted!`, the recipe's name, and `Recipes remaining: N`. `RecipeBook.remove` checks the index itself with `Objects.checkIndex`.

Outcomes and invalid input

The activity diagram shows which step decides each outcome.

Workflow of delete-recipe: where each outcome is decided



Valid INDEX: one positive whole number that fits in an int. In range: the book is not empty and INDEX does not exceed the recipe count. Only the confirmed path changes the recipe book.

Where each outcome of delete-recipe is decided.

Diamonds are decisions, with their guards in square brackets. Both error paths end before the question is asked. Only the confirmed path changes the recipe book.

Input or state	Where it is decided	Result
Missing, non-numeric, zero, or oversized index	`Parser.parseRecipeIndex`	Error ending with `Format: delete-recipe INDEX`; no question.
Empty book, or index above the recipe count	`DeleteRecipeCommand.confirmationPrompt`	Error followed by the delete-recipe usage; no question.
Valid index, answered `y` or `yes`	`MishMash.userConfirms`, then `execute`	Recipe removed and confirmation shown.
Valid index, any other answer or end of input	`MishMash.userConfirms`	`Cancelled. Nothing was changed.`

Syntax errors start as an `IllegalArgumentException`, which `parseDeleteRecipe` wraps in a `MishMashException` that keeps the cause and ends with the delete-recipe usage. After any error or cancellation the session continues with the next command.

Design choices and alternatives

Decision	Alternative	Reason for the current design
Ask through a default `confirmationPrompt` method on `Command`	Let the delete command read the `Scanner` itself, or pass a context object to `execute`	Commands stay free of console input and output and are easy to test, and every other command is unchanged because of the default. A context object would be worth it if many commands needed shared state.
Check the index in `confirmationPrompt`, before asking	Ask first and report the error afterwards	The user is never asked to confirm something that cannot happen.
`execute` repeats the checks	Assume the prompt already ran	`execute` is public and must be safe on its own, and the check is only two comparisons.
Anything except `y` or `yes` cancels	Ask again until the answer is `y` or `n`, or default to yes	A destructive action must never run by accident, including when input ends.
Identify the recipe by position and name it in the question	Delete by name	Recipe names may repeat, so the position is the only unambiguous identifier, and naming the recipe lets the user catch a wrong number.
Leave existing plans and groceries unchanged	Remove the recipe from plans or clear them	Plans are snapshots, as for `edit-recipe`, so deleting never silently changes a plan the user is relying on. Automatic repair could be added if it were specified.

Verification and maintenance

`DeleteRecipeCommandTest` checks the confirmation text and removal, the first and last positions, repeated deletion at one index, duplicate names, emptying the book, out-of-range and empty-book errors with the book left unchanged, prompts that never modify the book, and null arguments. `ParserTest` checks a valid index, the question naming the recipe, leading zeros and extra whitespace, invalid arguments showing only the delete-recipe usage, and near-miss command words. `MishMashTest` checks the console flow: a confirmed transcript, a cancellation, end of input cancelling instead of deleting, a missing recipe never producing a question, and a later `edit-recipe` using the renumbered index.

For manual verification, follow Delete a Recipe in the manual-testing appendix, or the User Guide's Deleting a Recipe example. The editable diagram sources and rendering instructions are in `docs/diagrams/neil0618/delete-recipe`. When the implementation changes, check method names, message order, and the decision branches against the code, then regenerate and inspect every SVG.

Applying dietary restrictions

Section author: peilingggg

`set-diet` replaces the session's dietary profile and applies it to future meal generation. If an existing plan conflicts, the command can replace its conflicting days and refresh its groceries. The User Guide describes the syntax, supported diets, food tags, and examples.

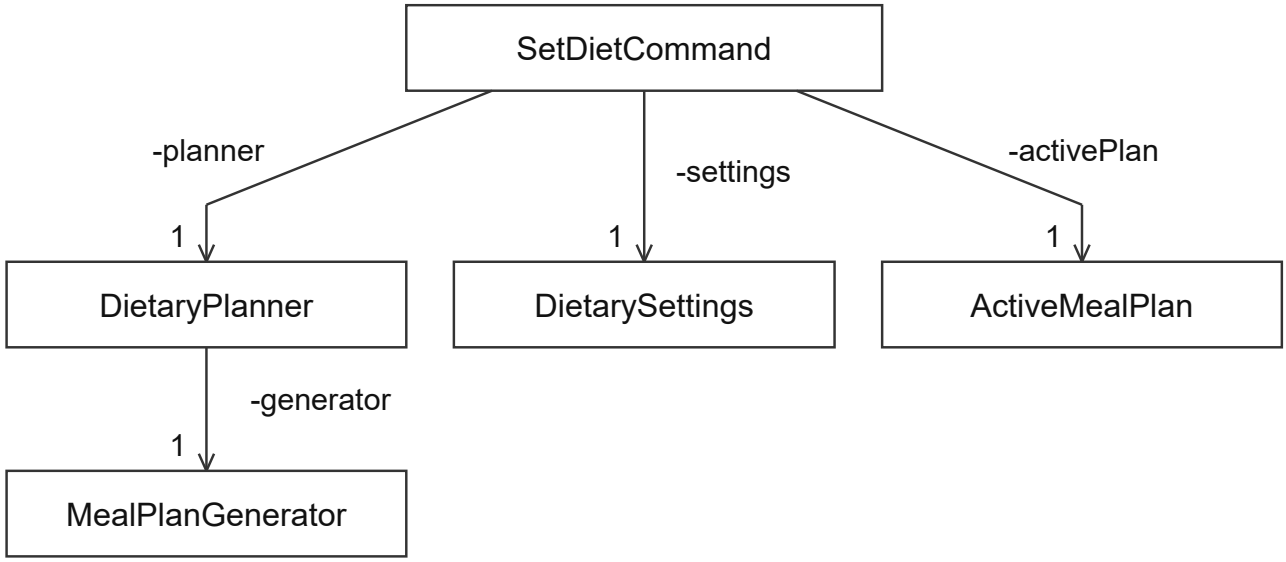
```
set-diet [diet/DIET] [avoid/TAG[,TAG...]] [exclude/INGREDIENT[,INGREDIENT...]]
```

Omitted fields return to unrestricted defaults. For example, `set-diet avoid/peanut` clears a previously selected vegan diet. Profiles and recipe classifications are held in shared holders and retained across sessions.

Responsibilities

Component	Responsibility
`Parser`, `DietaryPreferencesParser`	Validate arguments, build an immutable replacement profile, and create `SetDietCommand` without changing session data.
`Diet`, `FoodTag`, `FoodInfo`	Define diet rules, supported categories, and a recipe's declared classification.
`DietaryPreferences`	Check recipes against the named diet, avoided categories, and normalised ingredient exclusions.
`DietarySettings`	Hold the current profile shared by setting, viewing, and generation commands.
`SetDietCommand`	Request confirmation when needed, coordinate repair, and save the resulting session state.
`DietaryPlanner`	Filter eligible recipes and replace incompatible days using `MealPlanGenerator`.

Component	Responsibility
`ActiveMealPlan`	Hold the current plan, original generation request, and optional grocery snapshot.
`GroceryListGenerator`	Rebuild existing groceries from the repaired plan.



Class diagram of the dietary update collaborators

Stored references used to apply a dietary update. Each arrow points to one retained object; unrelated members and references are omitted. `SetDietCommand` also implements `Command` and stores the proposed `DietaryPreferences`. PlantUML source

The shared holders expose current session data. `DietaryPreferences` copies its exclusion sets and normalises ingredient names by trimming, lowercasing with `Locale.ROOT`, and collapsing whitespace. Collection copies do not deep-copy shared recipes or days.

Classifying existing recipes

Recipes added without `food/` are **unclassified**. In contrast, `food/none` explicitly declares that no supported food categories apply. Any active dietary restriction excludes unclassified recipes; declarations are not inferred from ingredient names.

Users can classify a recipe they have already added through Editing a Recipe. For example, if recipe 1 contains soy and legumes and no other supported categories, enter:

```
edit-recipe 1 food/soy,legume
```

Use the recipe's current index from `list-recipes`. The `food/` field replaces its complete category declaration, so include every applicable tag. Omitting other fields preserves the recipe's name, nutrition, and ingredients. Existing plans and groceries keep their previous recipe snapshots; run `generate-plan`, then `generate-list` to use the updated classification in a new plan and grocery list.

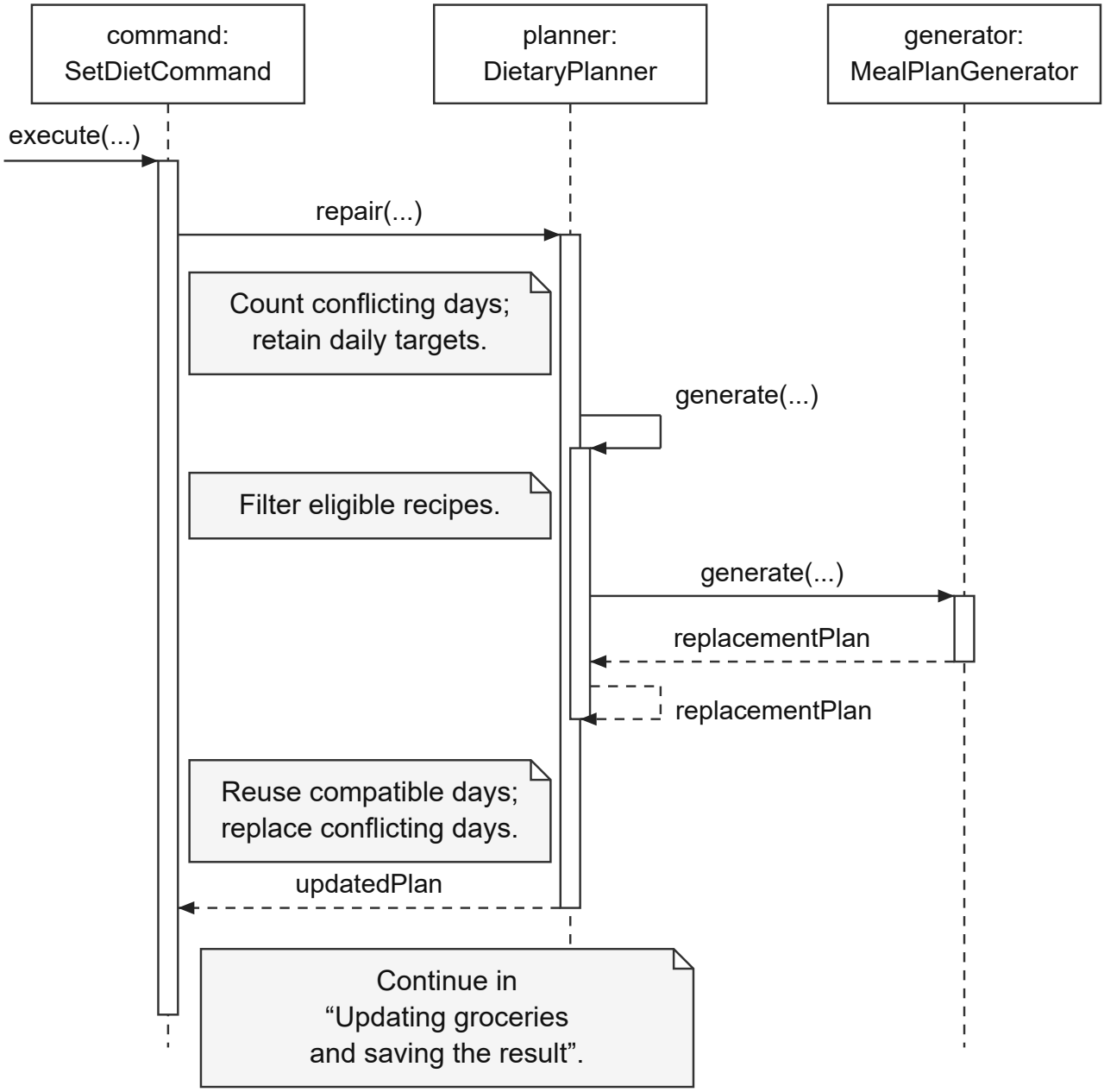
Parsing and confirmation

`Parser.parseSetDiet` delegates validation to `DietaryPreferencesParser.parse`, then creates `SetDietCommand` without changing session data. At least one supported field is required; each prefix may appear once in any order. Unsupported prefixes, unprefixed text, blank values, embedded slashes, and duplicates are rejected. Diet and tag values ignore case; aliases and ingredient names are normalised before duplicate detection.

Before execution, `MishMash` calls `confirmationPrompt`. Only a conflicting active plan requires agreement. Declining skips execution and preserves all state.

Repairing conflicting days

The repair and publication sequence diagrams split one confirmed `SetDietCommand.execute` into planning and publication. This example has conflicting days, a retained generation request, and existing groceries.



Sequence diagram of repairing incompatible days

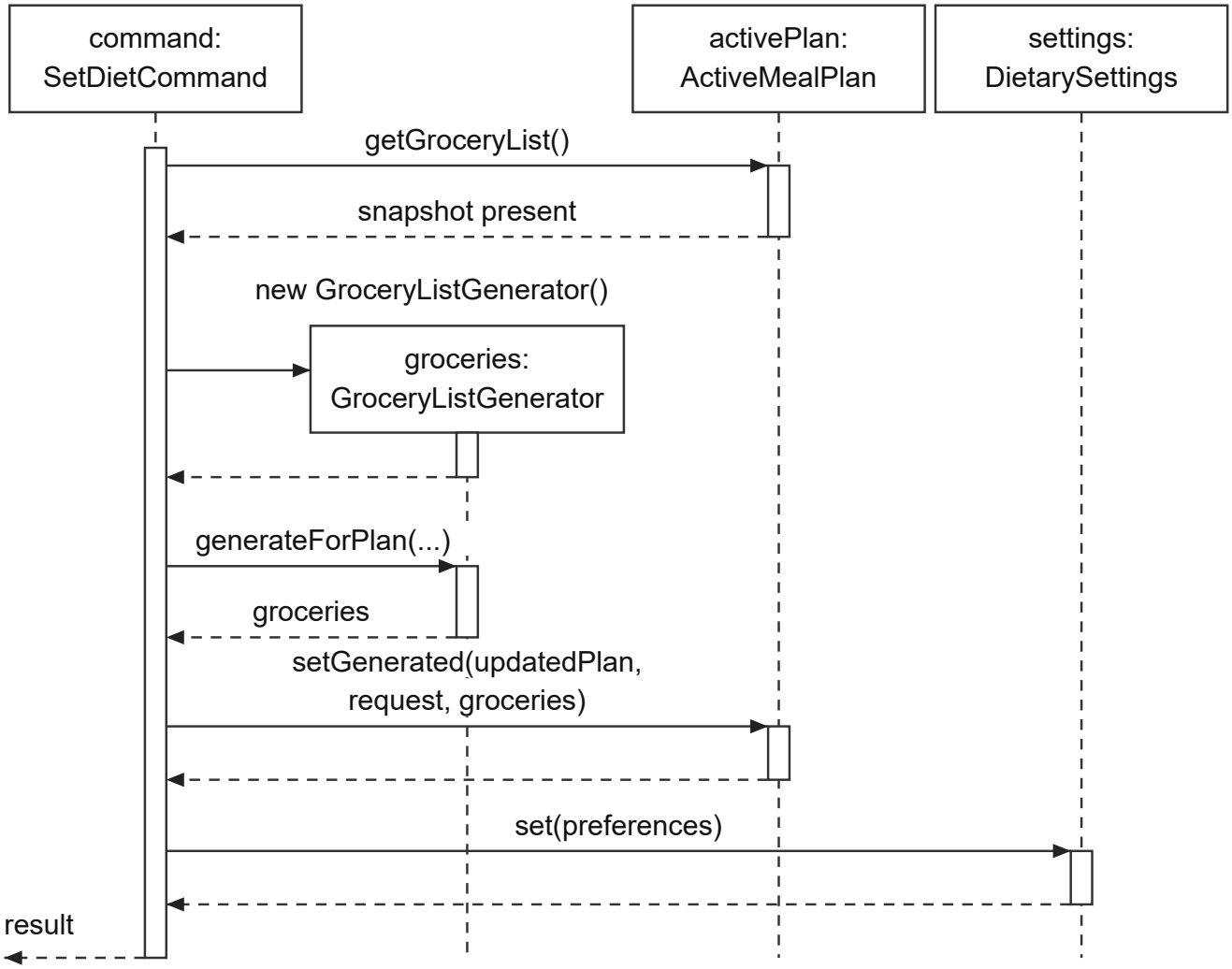
Building a replacement plan. Parameters are abbreviated with `...`; compatibility checks, filtering, and day assembly are summarized in notes. The original plan is unchanged during this sequence. The publication diagram continues the same execution. PlantUML source

1. The command reads the active plan and its original `GeneratePlanRequest`, then calls `DietaryPlanner.repair` with the current recipe pool and proposed preferences.
2. The planner counts incompatible days and requests that many replacements, retaining the original daily nutrition targets.
3. `generate` filters recipes through `DietaryPreferences.isAllowed`, then delegates to `MealPlanGenerator`. Any active restriction requires classified recipes. A recipe must satisfy the named diet, explicit category exclusions, and complete ingredient-name exclusions together. Matching does not infer synonyms, substrings, or undeclared tags.

4. The planner reuses compatible `DayPlan` objects at their original positions and replaces every conflicting day in full. For example, if only Day 2 conflicts in a three-day plan, Days 1 and 3 remain the same objects; all three meals in Day 2 may change.

Generation retains three distinct recipes per day, the +/-15% nutrition tolerance, and bounded candidate search. Three eligible recipes alone do not guarantee a feasible plan.

Updating groceries and saving the result



Sequence diagram of publishing a repaired plan and groceries

Continuing the repair sequence when groceries already exist. The grocery generator is created at its constructor message. Optional handling, validation, and formatting are omitted and arguments abbreviated with `...`; dashed arrows show returns. PlantUML source

1. If groceries already exist, `GroceryListGenerator.generateForPlan` rebuilds them from every meal occurrence in the updated plan. Otherwise groceries remain absent, and the confirmation advises the user to run `generate-list` later.
2. `ActiveMealPlan.setGenerated` validates the plan length and copies the grocery snapshot before assigning the new plan, original request, and groceries.
3. `DietarySettings.set` saves the proposed profile, and the command returns its confirmation.

Other outcomes are summarized below:

Situation	Result
No active plan	Clear stale generated state and save the profile for future generation.
Entire plan is compatible	Save the profile; preserve the plan and groceries.
Conflicting plan; user declines confirmation	Skip execution; preserve the profile, plan, and groceries.
Repair succeeds	Save the profile and repaired plan; rebuild groceries only if they already existed.
No feasible replacement or original request unavailable	Save the profile, clear the plan, request, and groceries, and explain how to generate again.

`ActiveMealPlan.set(MealPlan)`, used by legacy callers, discards the original request and groceries. A compatible legacy plan can be preserved, but a conflicting one cannot be repaired without its original generation settings.

Error handling and defensive checks

Parsing failures raise `IllegalArgumentException`. `Parser.parseSetDiet` wraps the error in `MishMashException`, preserves its cause, and adds usage. For example, `diet/keto` reports supported diets and aliases. The main loop displays the error and continues without changing session data.

During repair, `IllegalArgumentException` instead triggers the save-and-clear outcome above. Null Java dependencies fail with `NullPointerException`. Assertions check plan length, dietary compatibility, nutrition targets, and exact use of replacement days; `AssertionError` denotes an internal programming error.

Alternatives and limitations

Decision	Alternative	Reason
Use complete, explicit food declarations	Infer categories from ingredient text	Avoid unreliable guesses about synonyms and compound ingredients.
Replace the whole profile	Merge supplied fields	Each command defines a complete profile with clear defaults for omitted fields.
Filter before invoking the existing generator	Add dietary rules throughout the generator	Reuse its nutrition and variety rules.
Replace whole conflicting days	Replace individual meals	Reuse daily nutrition checks while preserving compatible days.
Clear generated data after failed repair	Keep the incompatible plan	Keep generated data consistent with the saved profile.

Classification depends on the user’s declarations. The application does not detect undeclared allergens, cross-contact, or certification.

Automated verification

- `DietaryPreferencesParserTest` covers syntax, values, aliases, and malformed lists.
- `DietaryPreferencesTest` and `FoodInfoTest` cover restrictions, classification, normalised exclusions, and defensive collection ownership.
- `DietaryPlannerTest` covers filtering, original targets, preserved days, infeasible requests, and generator invariant failures.
- `DietaryCommandsTest` covers shared state, profile replacement, grocery rebuilding, failed-repair invalidation, and recovery. `ActiveMealPlanGeneratedTest` checks retained requests and snapshots. `PlanConfirmationTest` covers dietary-update confirmation and cancellation.

Run `./gradlew test checkstyleMain checkstyleTest` with Java 25 (Windows: `.\gradlew.bat test checkstyleMain checkstyleTest`). Editable sources and SVG maintenance instructions are in diagrams/dietary.

Viewing dietary preferences

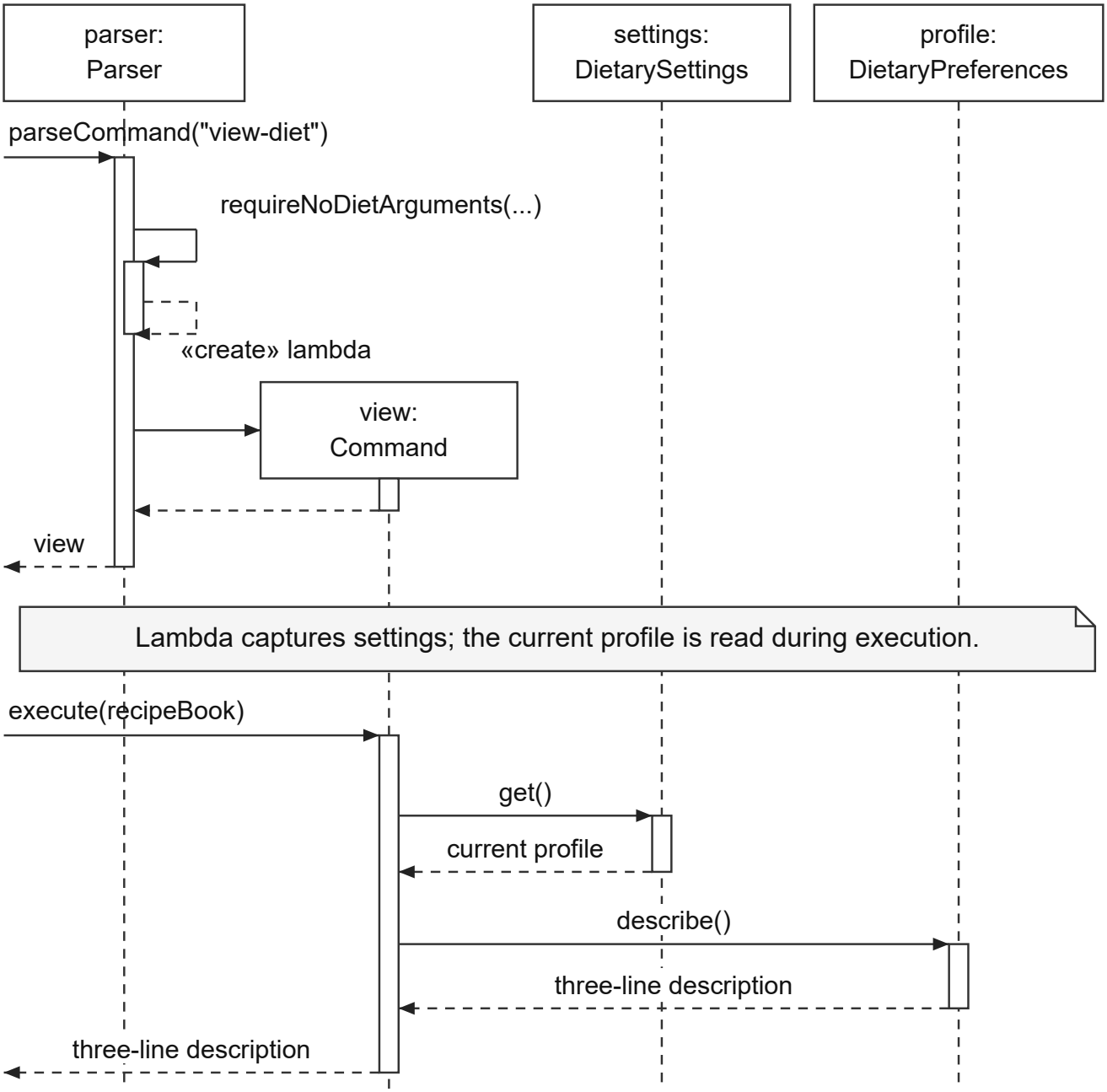
Section author: peilingggg #### Purpose and implemented scope

`view-diet` displays `DietarySettings.get().describe()` and accepts no arguments. It works without recipes or an active plan, including an unrestricted fresh session.

Parsing and execution

`Parser.parseNamedCommand` calls `requireNoDietArguments` and returns a `Command` lambda. The lambda captures the shared holder, rather than a profile snapshot, so a command parsed before a later update displays the settings current at execution time. It does not read or change the recipe book, active plan, original generation request, or grocery snapshot. `DietaryPreferences.describe()` prints the named diet and sorted explicit category and ingredient exclusions on three lines, using `none` for empty lists.

The viewing sequence diagram (source) shows parsing and execution-time profile lookup without changing session data.



Sequence for viewing current dietary preferences.

The lambda captures the shared settings holder and reads its current profile during execution. Parsing helpers and formatting internals are condensed; no session data is changed.

Invalid input and verification

Extra arguments raise `MishMashException` with the no-arguments message and `Format: view-diet`. No command executes and all state is preserved. `ViewDietCommandTest` covers unrestricted and populated profiles, execution-time reads, and argument rejection without profile changes. Parser tests cover argument rejection. Run the existing Java 25 test suite when changing this behaviour.

Clearing dietary preferences and restrictions

Section author: oilybikechain

`clear-diet` accepts no arguments. `Parser` constructs `ClearDietCommand` without changing state, and execution calls `DietarySettings.clear` to install a fresh unrestricted profile. The command deliberately retains only a `DietarySettings` reference, so it cannot mutate the recipe book, active plan, customised ingredients, portion counts, groceries, saved targets, or meal log. It also validates the command execution context before clearing, which keeps a failed invocation atomic.

Clearing an already unrestricted profile is idempotent and reports that nothing changed. Unlike a restrictive `set-diet` update, clearing cannot make a planned recipe incompatible, so it does not request confirmation or repair the existing plan. Future generation reads the new unrestricted profile.

`DietarySettingsTest` and `ClearDietCommandTest` cover complete reset, repeated clearing, and null boundaries. `ClearDietIntegrationTest` covers deferred parser execution, invalid arguments, help text, future unrestricted generation, and identity-preservation of unrelated planning state.

Setting and viewing daily targets

Section author: neil0618

Purpose and command contract

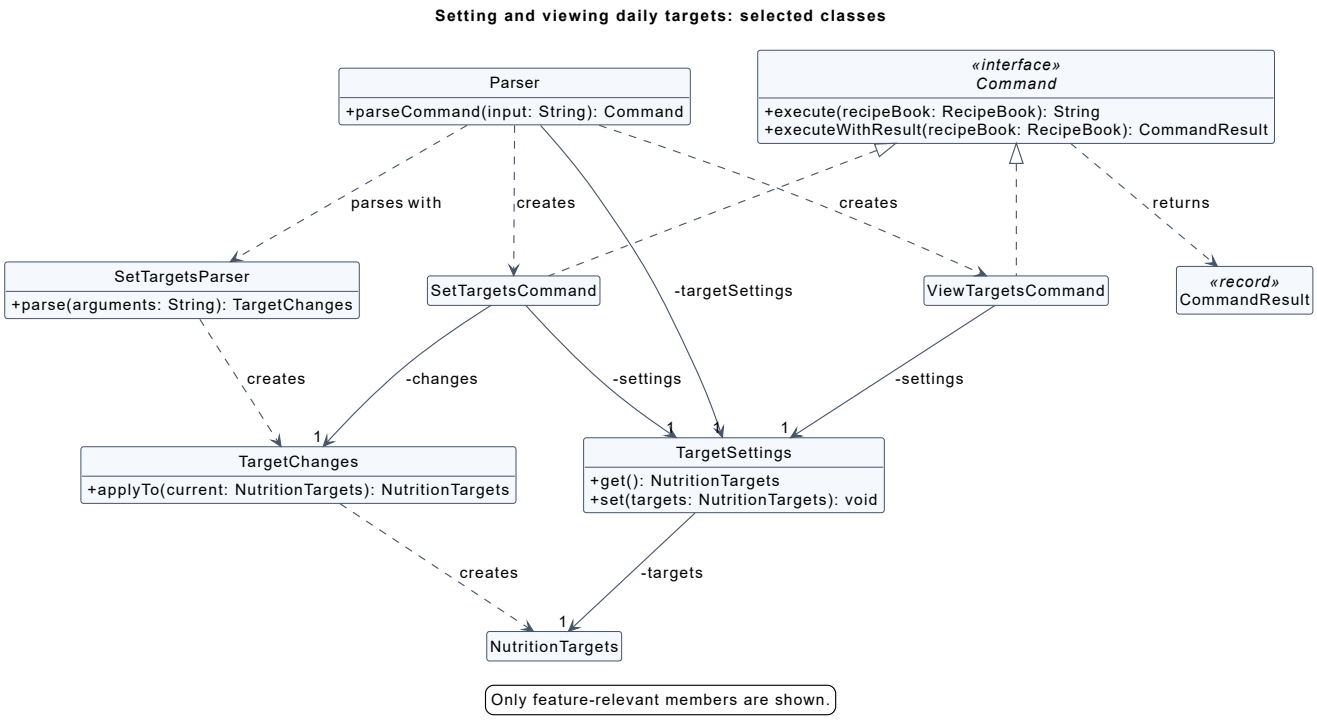
This feature implements two stories: **As a user, I want to set a daily calorie target so that my meal plan can meet my desired calorie intake**, and **As a user, I want to set my daily protein, carbohydrate, and fat targets so that my meal plan can meet my desired macronutrient intake.** One command covers both, because calories and macronutrients are the four fields of `NutritionTargets` and are used together.

`set-targets [cal/CALORIES|none] [p/PROTEIN|none] [c/CARBS|none] [f/FATS|none]` saves daily targets across application sessions. At least one field is required. A field with a positive whole number replaces that target, `none` removes it, and a field that is not mentioned keeps its saved value. `view-targets` shows the saved targets. For example, `set-targets cal/2000 p/150` followed by `set-targets p/none` leaves only the 2000 kcal target.

The saved targets are used by `generate-plan`. For every nutrient that the `generate-plan` command leaves out, the saved target is used; a target typed in `generate-plan` replaces the saved one for that plan only. Neither command changes an existing meal plan, and `check-shortfall` still requires explicit targets. Targets are included in the autosaved application snapshot.

Class responsibilities and relationships

`Parser` owns one `TargetSettings` object and passes it to `SetTargetsCommand`, `ViewTargetsCommand`, and `GeneratePlanCommand`, in the same way that it shares `DietarySettings`. `TargetSettings` holds one immutable `NutritionTargets`, which is empty at the start. `SetTargetsParser` turns the arguments into a `TargetChanges`, and `Nutrient` names the four nutrients with their prefix, label, and unit, so the parser, the merge, and the display loop over one list instead of repeating four near-identical branches.



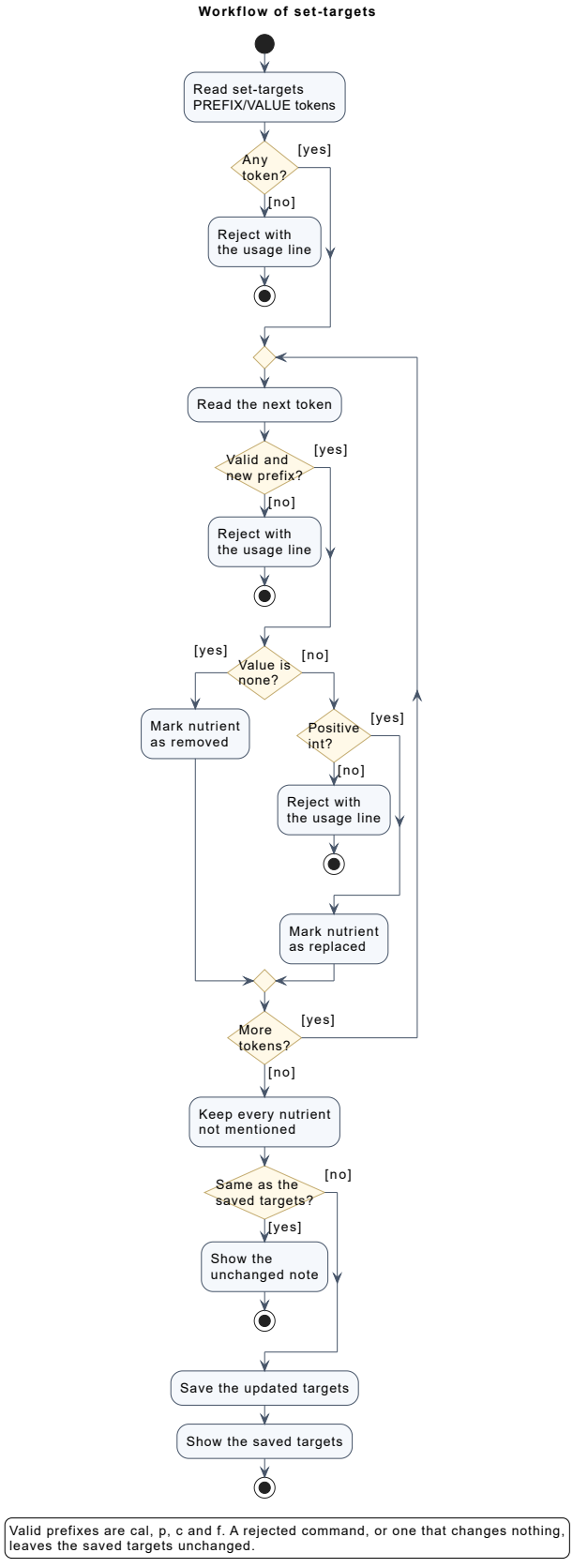
Selected classes and relationships for setting and viewing daily targets.

Only feature-relevant members are shown. Dashed arrows with an open head are dependencies; dashed arrows with a hollow triangle are interface realization; solid arrows are associations with role names and multiplicities. `Parser` creates the one `TargetSettings` and shares it with the commands.

`TargetChanges` records, for each nutrient the user mentioned, either a new value or `Optional.empty()` for “remove”. Nutrients that were not mentioned are absent from its map. This three-way distinction (keep, replace, remove) cannot be expressed with `NutritionTargets` alone, where `null` already means “no target”.

Setting targets

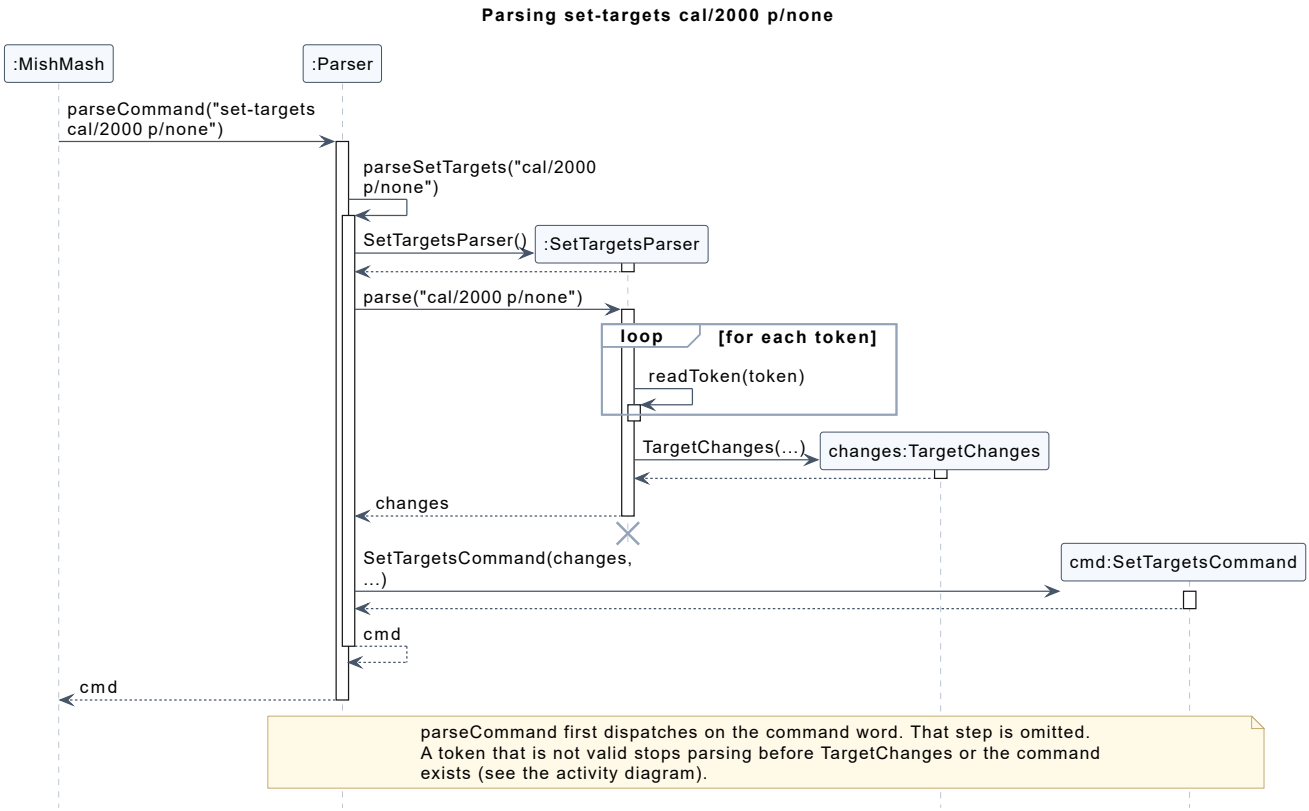
The activity diagram shows how `set-targets` treats each token, including every way it can be rejected.



Activity diagram of the set-targets workflow.

Diamonds are decisions, with their guards in square brackets. A rejected command, or one that would change nothing, leaves the saved targets unchanged.

The next two sequence diagrams follow `set-targets cal/2000 p/none`, first parsing and then execution.

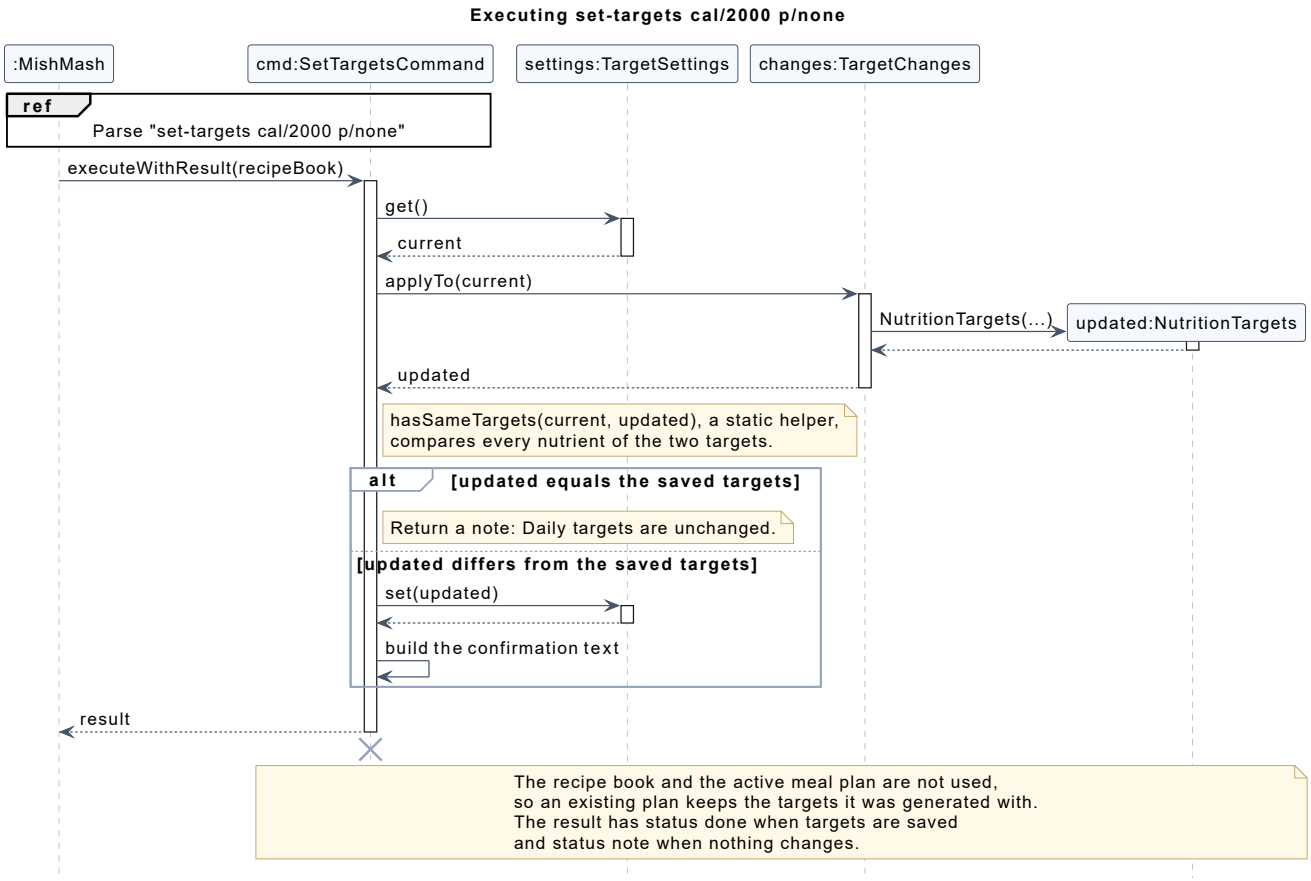


Sequence for parsing set-targets.

Parsing succeeds completely before any command exists, so a bad token never changes the saved targets. The X marks `SetTargetsParser`, which is no longer used once it has returned the changes.

`SetTargetsParser` reads each whitespace-separated token as `PREFIX/VALUE`. Prefixes are case-sensitive, like every other prefix, and `none` is accepted in any letter case, as for `food/none`. A value must consist of digits only and fit in an `int`; zero is rejected. The parser validates all tokens before creating a `TargetChanges`, and `Parser.parseSetTargets` wraps any `IllegalArgumentException` in a `MishMashException` that carries the set-targets format as its usage. A command object is created only if every token is valid, so a bad command can never partly change the saved targets.

Once parsing succeeds, execution reads the saved targets, applies the validated changes, and stores the replacement value unless it equals what is already saved.

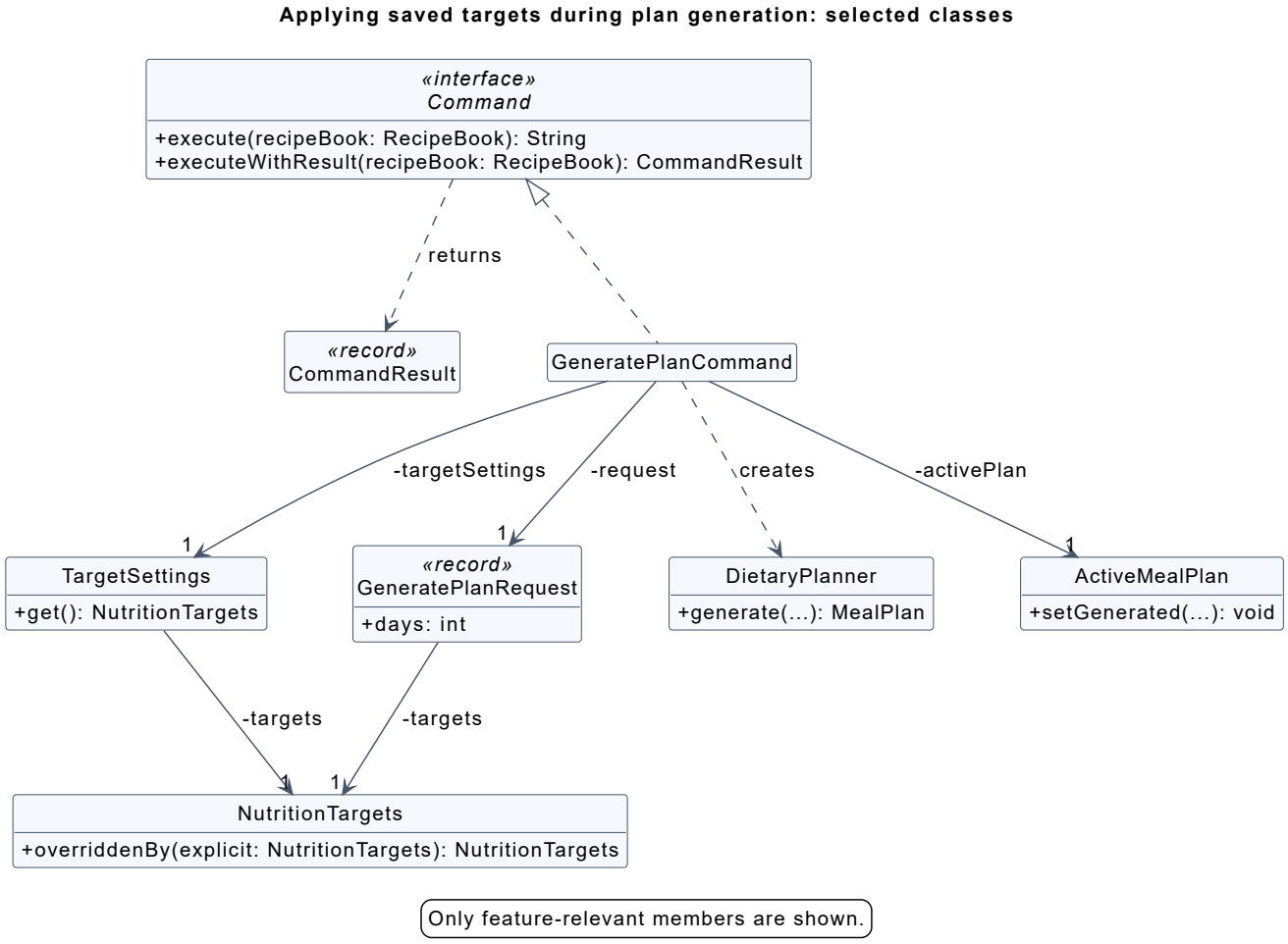


Sequence for executing set-targets.

The execution diagram refers to the parsing diagram with a `ref` frame. Its `alt` frame separates the case where nothing would change, which returns a note without storing anything, from the case that stores the new targets.

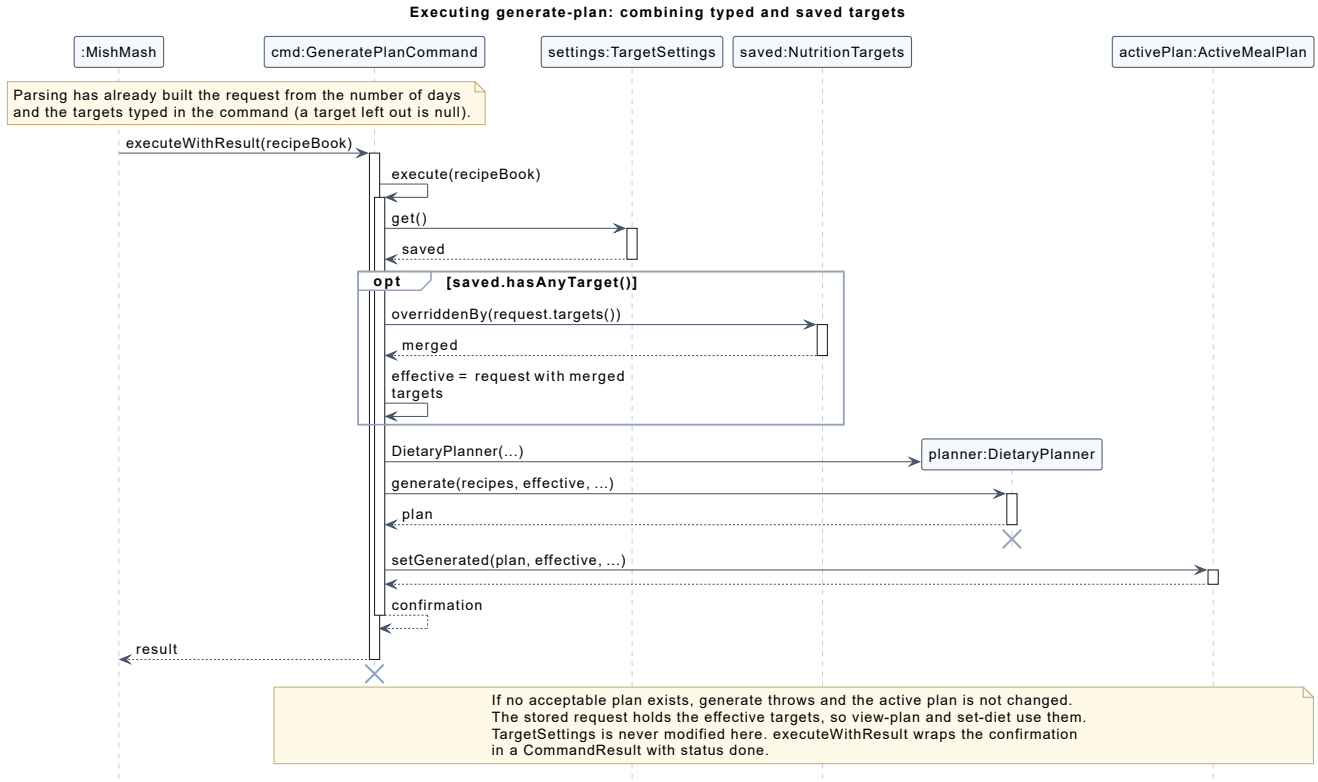
`SetTargetsCommand.executeWithResult` reads the saved targets, asks `TargetChanges.applyTo` for the updated targets, compares them with the saved ones, and stores them. `applyTo` builds a new `NutritionTargets` and does not modify the old one. If the updated targets equal the saved ones, it stores nothing and returns a `NOTE`, `Daily targets are unchanged. Nothing was changed.`, as `ClearDietCommand` does. Otherwise the reply lists every target now saved, or says `All daily targets cleared.` when none remain, and reminds the user that an existing plan is unchanged.

Using the targets in `generate-plan`



Classes that apply saved targets during plan generation.

Each association is drawn once, as a line with its role name and multiplicity, not also as an attribute.



How generate-plan combines typed and saved targets.

The `opt` frame runs only when something is saved. The command creates a `DietaryPlanner` for each execution and does not use it afterwards. If no acceptable plan exists, `generate` throws and `ActiveMealPlan.setGenerated` is never called, so the previous plan stays active.

`GeneratePlanCommand` reads `TargetSettings` when it executes, not when it is created, so a `set-targets` entered after parsing but before execution is respected, exactly as for dietary preferences. If nothing is saved, the request is used unchanged, so behaviour without `set-targets` is identical to before. Otherwise `NutritionTargets.overriddenBy` merges the typed targets over the saved ones and the command builds an effective `GeneratePlanRequest`.

The effective request, not the typed one, is passed to `DietaryPlanner` and stored with `ActiveMealPlan.setGenerated`. This matters in two places that read the stored request later: `view-plan` warns about daily totals that exceed the stored targets, and `set-diet` regenerates conflicting days using the stored targets. If the stored request kept only the typed targets, both would ignore the targets that actually shaped the plan.

When a saved target filled a nutrient that the command left out, the confirmation gains a `Daily targets used:` line and a failure message gains a `Daily targets applied:` line, both listing the merged targets. `GeneratePlanCommand.usesSavedTarget` decides this by checking whether any nutrient has a saved target but no typed one. Without these lines, a user who forgot an earlier `set-targets` would see a failure caused by a target they never typed in the command. When nothing is saved, or every saved nutrient is also typed, the messages are exactly as before.

Invalid input and recovery

Input or state	Where it is decided	Result
No arguments	`SetTargetsParser.parse`	`Supply at least one daily target to set or clear.` followed by the set-targets format.
Unknown, malformed, or wrong-case prefix (`x/5`, `2000`, `CAL/5`, `d/3`)	`SetTargetsParser.readToken`	`Unrecognised parameter: TOKEN` and the format.
Repeated prefix	`SetTargetsParser.readToken`	`Parameter cal/ was given more than once.` and the format.
Missing value (`cal/`)	`SetTargetsParser.parseValue`	`Missing value after cal/` and the format.
Zero, negative, decimal, text, signed, or too large value	`SetTargetsParser.parseValue`	An error naming the nutrient, for example `Calories target (cal/) must be a positive whole number from 1 to 2147483647, or 'none'.`, and the format.
Arguments after `view-targets`	`Parser.parseViewTargets`	`view-targets does not accept arguments.` and the view-targets format.
Saved target that no combination of recipes can meet	`GeneratePlanCommand.execute`	The generator's error plus `Daily targets applied: ... (including your saved targets). Change those with 'set-targets'.`; the previous plan is kept.

In every case the session continues, and an error leaves the saved targets unchanged.

Design choices and alternatives

Decision	Alternative	Reason for the current design
One `set-targets` command for calories and macronutrients	A separate command for calories and another for macronutrients	Both stories change the same object, and a single command lets the user set all four values in one line. Two commands would need two parsers and duplicate the validation for no additional capability.
Keep the saved targets in a session holder, `TargetSettings`	Store them in `ActiveMealPlan` or in the recipe book	Targets must survive `generate-plan` replacing the plan, and `ActiveMealPlan` is cleared by `set-diet` in some cases. `TargetSettings` copies the proven `DietarySettings` pattern.

Decision	Alternative	Reason for the current design
Update only the mentioned nutrients	Replace all four targets at once, as `set-diet` replaces dietary preferences	The user can set calories today and protein later without retyping. Removing a target is explicit, with `none`.
`none` removes a target	A separate `clear-targets` command	One command and the same word the user already knows from `food/none`. A separate command would be worth adding if clearing everything in one step were needed often.
Merge saved and typed targets when `generate-plan` executes, per nutrient	Merge when parsing, or let any typed target discard all saved ones	Execution-time merging respects the latest saved targets and keeps parsing free of session data. Per-nutrient merging lets `generate-plan d/3 cal/1800` change one number and still use the saved protein target.
Store the effective request in `ActiveMealPlan`	Store only what was typed	`view-plan` and `set-diet` then use the targets that actually produced the plan.
Name the merged targets in the `generate-plan` messages when a saved target contributed	Stay silent, or print the targets on every plan	A user who forgot a saved target can see where a number came from, and output is unchanged when nothing is saved. Printing on every plan would add noise for users who type all their targets.
No per-command way to ignore saved targets	Accept `cal/none` in `generate-plan`	Ignoring one needs `GeneratePlanParser` and `GeneratePlanRequest` to represent "explicitly no target", which changes shared code and its tests. `set-targets cal/none` already gives the same result. It could be added if users needed it.
Changing targets leaves the current plan alone	Regenerate or clear the plan	Plans are snapshots, as after `edit-recipe` and `delete-recipe`, and regenerating would change meals the user may be relying on.
`check-shortfall` is unchanged	Fall back on saved targets when none are typed	Its contract is to check the explicit targets given, and existing behaviour and tests stay valid. It would be a small change to add later.

Decision	Alternative	Reason for the current design
Persist targets with other user data	Separate target file	A complete snapshot avoids restoring targets and plans from different sessions.

Verification and maintenance

`TargetChangesTest` checks setting, keeping, and removing nutrients, that the original targets are never modified, the merge in `NutritionTargets.overriddenBy`, the summary text, and `TargetSettings`. `SetTargetsParserTest` checks every accepted form (any order, extra spaces, `none` in any case, leading zeros, the largest `int`) and every rejection above. `SetTargetsCommandTest` checks the confirmation texts, partial updates, removal, the untouched plan and recipes, `view-targets`, and the interaction with `generate-plan`: saved targets used, typed targets winning per nutrient, saved targets read at execution, removal, a typed target replacing an impossible saved one, a failed generation keeping the previous plan, and the `Daily targets used:` and `Daily targets applied:` lines appearing only when a saved target contributed. `ParserTargetsTest` checks that the parser creates the commands, shares one set of targets between them, reports errors with the set-targets usage only, and lists both commands in the usage and help text.

For manual verification, follow Set and View Daily Targets in the manual-testing appendix, or the User Guide's Setting Daily Targets example. Editable sources and rendering instructions are in `docs/diagrams/neil0618/targets`. When the implementation changes, check method names, message order, and branches against the code, then regenerate and inspect every SVG.

Generating a meal plan

Section author: GauthamNandi

`GeneratePlanCommand` combines command-specific targets with saved `TargetSettings`, filters recipes through the current dietary profile, and delegates selection to `MealPlanGenerator`. Each day contains three distinct recipes; recipes can recur on other days. With no targets, selection is random. With targets, the generator checks combinations against a 15% tolerance for every supplied nutrient. It checks every combination when there are at most 1,000,000 (about 180 recipes), otherwise up to 1,000,000 random combinations, and keeps at most 5,000 matches. Sampling bounds runtime but can miss a valid combination in a large recipe book.

`ActiveMealPlan` stores the successful plan and its generation request. Replacement generation requires confirmation when a plan exists. Success invalidates previous groceries; failure preserves the prior plan.

Viewing a meal plan

Section author: Rixte

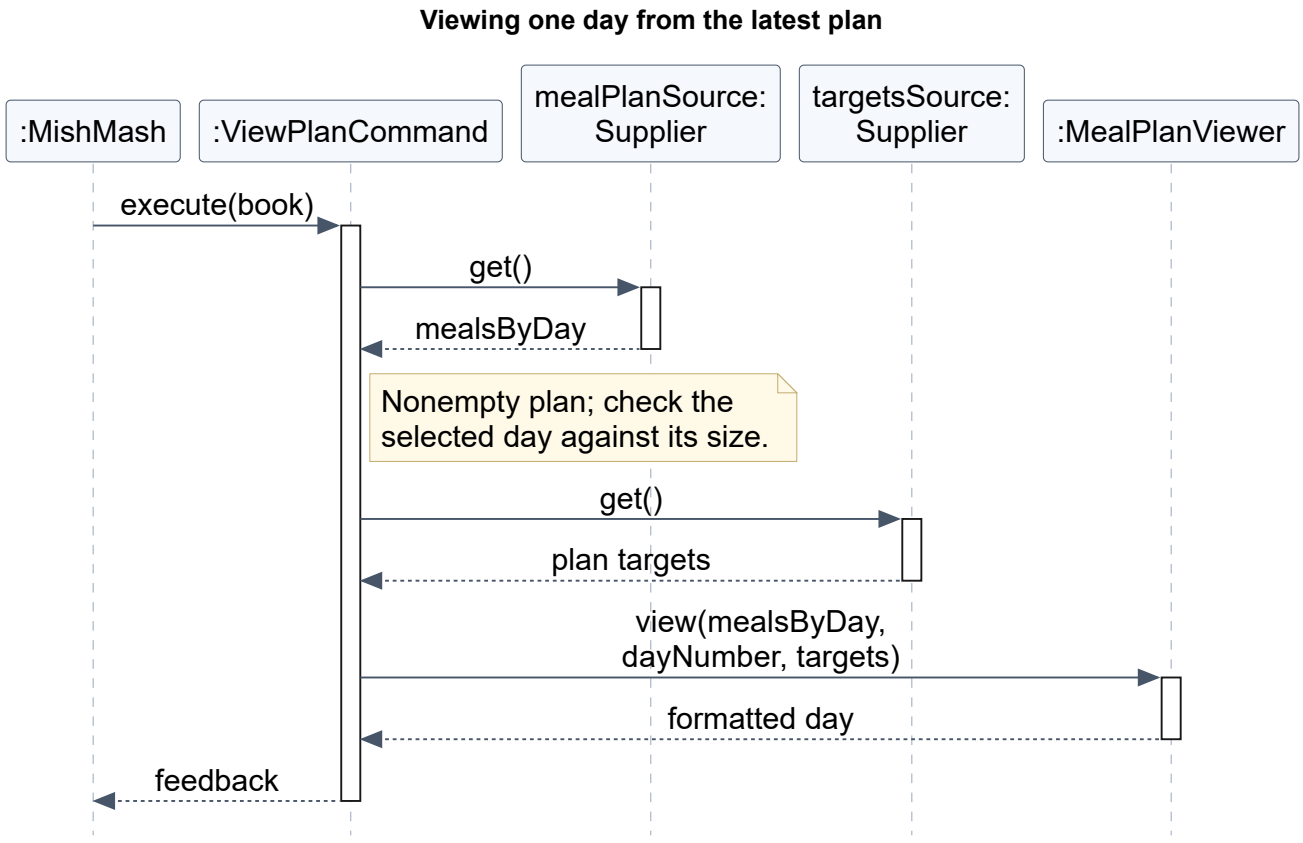
`view-plan [d/DAY_NUMBER]` displays all planned days or one selected day, without generating meals. See the User Guide for examples.

Responsibilities

Component	Responsibility
`Parser.parseViewPlan`	Validate the optional positive day and inject the plan and target suppliers.
`ViewPlanCommand`	Read the latest meals, check day bounds, and select the viewer overload.
`MealPlanViewer`	Return formatted meals, daily totals, and any target warnings.

The normal parser connects `mealPlanSource` to `ActiveMealPlan.get()` and `MealPlan.toMealSummaries()`. `targetsSource` reads the retained generation request, falling back to `NutritionTargets.none()`. Suppliers are invoked during execution, so a parsed command sees later regeneration or reset. Injected test sources can differ from the parser's holder; the command does not store a direct `ActiveMealPlan` reference.

Execution



Sequence diagram of viewing a selected day using current data

Successful `view-plan d/1` after parsing, with a nonempty plan. Supplier labels represent the injected Java functions. Their holder lookups and viewer formatting are condensed; viewer construction is omitted. With no day, the command calls `view(mealsByDay, targets)`. PlantUML source

1. `execute` calls `mealPlanSource.get()` and creates `MealPlanViewer`.
2. An empty day list returns the generate-first guidance without reading targets.

3. For a selected day, the command checks its upper bound against the current plan.
4. Only then does it read targets and format the selected day; without `d/`, it formats all days.

Nutrition summation and excess warnings are described in Combined daily nutrition totals and Detecting exceeded nutrition targets. Viewing returns text and changes neither planned data nor recipes.

Errors and boundaries

Condition	Result
Malformed, repeated, non-positive, or overflowing `d/` argument	Reject during parsing with view-plan usage.
No active plan or zero days	Return `No active meal plan. Use 'generate-plan' first.`, even if a day was supplied.
Selected day exceeds the current plan size	Raise `MishMashException` before reading targets.
Valid day contains no meals	Display the day with zero totals.
Supplier returns null, or displayed data contains null elements	Throw `IllegalStateException` or `NullPointerException`; `MishMash.run` logs and rethrows the failure, ending the session.

Design choice and verification

Deferred suppliers keep viewing independent of generation and make command reuse and tests straightforward. Capturing a plan during parsing would display stale data after regeneration. Keeping formatting in `MealPlanViewer` lets commands return text without owning terminal output.

`ViewPlanCommandTest`, `ViewPlanIntegrationTest`, `ViewPlanDiagnosticsTest`, `MealPlanViewerTest`, and the viewing cases in `ParserTest` cover selected days, missing plans, current-state lookup, large totals, and unchanged groceries and recipes. Manually generate two days, run `view-plan d/2`, then reset and run the view again; the second view should report missing data.

Adjusting planned meal portions

Section author: oilybikechain

`adjust-portion d/DAY m/MEAL portions/COUNT` changes one recipe occurrence without editing the recipe book. `AdjustPortionParser` accepts the three required prefixes in any order, rejects duplicates and unknown fields, and creates an immutable `PortionAdjustment`. Bounds against the current plan are deferred until `AdjustPortionCommand.execute`, so parsing does not read or mutate session state.

`PlannedMeal` owns a `Recipe` reference and a positive integer portion count. It supplies scaled nutrition as `long` values and new scaled `GroceryItem` values using `BigDecimal`. `DayPlan` retains its original recipe-oriented constructor and `getMeals()` compatibility API, while `getPlannedMeals()` exposes portion-aware data. New generated recipes are wrapped as one-portion planned meals.

The update path is copy-on-write: `MealPlan.withAdjustedPortion` replaces only the selected `DayPlan`, and `DayPlan.withAdjustedPortion` replaces only the selected `PlannedMeal`. Exact addition rejects nutrition overflow before publication. `ActiveMealPlan.update` then preserves the original

`GeneratePlanRequest` and clears the grocery snapshot. Validation finishes before field assignment, so a failed adjustment preserves the plan, request, and groceries. Requesting the current count returns the original objects and does not clear groceries.

`MealPlan.toMealSummaries` supplies the portion count and scaled nutrition to `MealPlanViewer`. Counts above one appear as `[N portions]`; one-portion output stays backward compatible. `CheckShortfallCommand` receives scaled daily totals through `DayPlan`, and `GroceryListGenerator.generateForPlan` aggregates scaled ingredients.

`PlannedMealTest`, `AdjustPortionParserTest`, `AdjustPortionCommandTest`, and `AdjustPortionIntegrationTest` cover scaling, validation, atomic state changes, viewing, shortfalls, and grocery generation. Existing model, viewer, and grocery tests provide regression coverage.

Customising planned meal ingredients

Section author: oilybikechain

`customize-meal d/DAY m/MEAL i/INGREDIENT:QUANTITYUNIT [...]` replaces the effective one-portion ingredients of one occurrence. It does not edit `RecipeBook`, the source `Recipe`, nutrition, or other occurrences. `IngredientParser` centralises the established recipe ingredient grammar so recipe and customisation commands cannot drift apart.

`CustomizeMealParser` accepts `d/` and `m/` once and repeated `i/` fields in any order. It creates an immutable `IngredientCustomization`; active-plan bounds remain execution-time checks. `CustomizeMealCommand` constructs the entire replacement plan before publishing it through `ActiveMealPlan.update`, which preserves the original generation request and invalidates stale groceries only after success.

`PlannedMeal` stores an optional copied override and exposes effective ingredients independently of its source recipe. `withIngredients` treats numerically equal `BigDecimal` quantities as equal, returns the same object for a no-op, and removes the override when values match the source recipe again. Portion changes retain overrides, and grocery generation multiplies effective quantities by the portion count. Nutrition continues to use the recipe values because the application has no ingredient-to-nutrition database.

Dietary category checks continue to use the source recipe's declared `FoodInfo`; MishMash cannot infer whether an arbitrary replacement name belongs to a named-diet or avoided category. Ingredient-name exclusions use the effective planned-meal ingredients and are rejected before state changes. When category restrictions are active, an accepted meal that still has custom ingredients returns `WARNING` and asks the user to verify the replacement. Restoring the source ingredients returns `DONE` because the resulting meal has no unclassified override. Later `set-diet` compatibility checks still inspect the source declaration together with effective ingredient exclusions.

Ingredient equality follows grocery aggregation: order, exact names, and exact units are significant, while numerically equal `BigDecimal` values may use different scales. Only an exact ordered match is a no-op that preserves the plan and grocery snapshot and returns `NOTE`; recasing or reordering creates an override.

`CustomizeMealParserTest`, `IngredientCustomizationTest`, `CustomizeMealCommandTest`, and `CustomizeMealIntegrationTest` cover syntax, defensive copying, atomic failure, dietary conflicts, source recipe isolation, portion composition, display markers, and grocery quantities. Existing recipe parsing tests protect the shared `IngredientParser` behaviour.

Replacing a planned meal

Section author: neil0618

`replace-meal d/DAY m/MEAL r/RECIPE_INDEX` swaps one planned meal for a recipe chosen from the recipe book. `ReplaceMealParser` mirrors `AdjustPortionParser`: it accepts the three required prefixes in any order, rejects duplicates, unknown prefixes, and non-positive or overflowing numbers, and creates an immutable `MealReplacement`. Bounds against the current plan and recipe book are deferred until `ReplaceMealCommand.execute`, so parsing does not read or mutate session state.

`ReplaceMealCommand` receives the shared `ActiveMealPlan` and `DietarySettings` and checks, in order: an active plan, the day, the meal, and the recipe index (using the displayed one-based position, since recipes can share names). Choosing the recipe already in that slot at one portion with no customisation returns a `NOTE` and changes nothing; the comparison is by reference, so a recipe replaced by `edit-recipe` after generation counts as different and refreshes the plan's snapshot. Otherwise `DietaryPreferences.isAllowed(recipe)` must pass, so a manual choice cannot bypass the profile saved by `set-diet`; unclassified recipes are rejected while restrictions are active.

The update path is copy-on-write like the other plan edits: `MealPlan.withReplacedMeal` replaces only the selected `DayPlan`, and `DayPlan.withReplacedMeal` replaces only the selected `PlannedMeal` with `PlannedMeal.onePortionOf(recipe)`. This deliberately drops the old meal's portion count and ingredient customisation, which belonged to the old recipe. Exact addition in `DayPlan` rejects nutrition overflow before publication and the command reports it as a `MishMashException`. `ActiveMealPlan.update` then keeps the original `GeneratePlanRequest` and clears the grocery snapshot, so a failed replacement preserves the plan, request, and groceries.

`ReplaceMealParserTest`, `ReplaceMealModelTest`, `ReplaceMealCommandTest`, and `ReplaceMealIntegrationTest` cover syntax, position validation, dietary conflicts, the no-op case, atomic failure, overflow, viewing, and grocery generation.

Resetting a meal plan

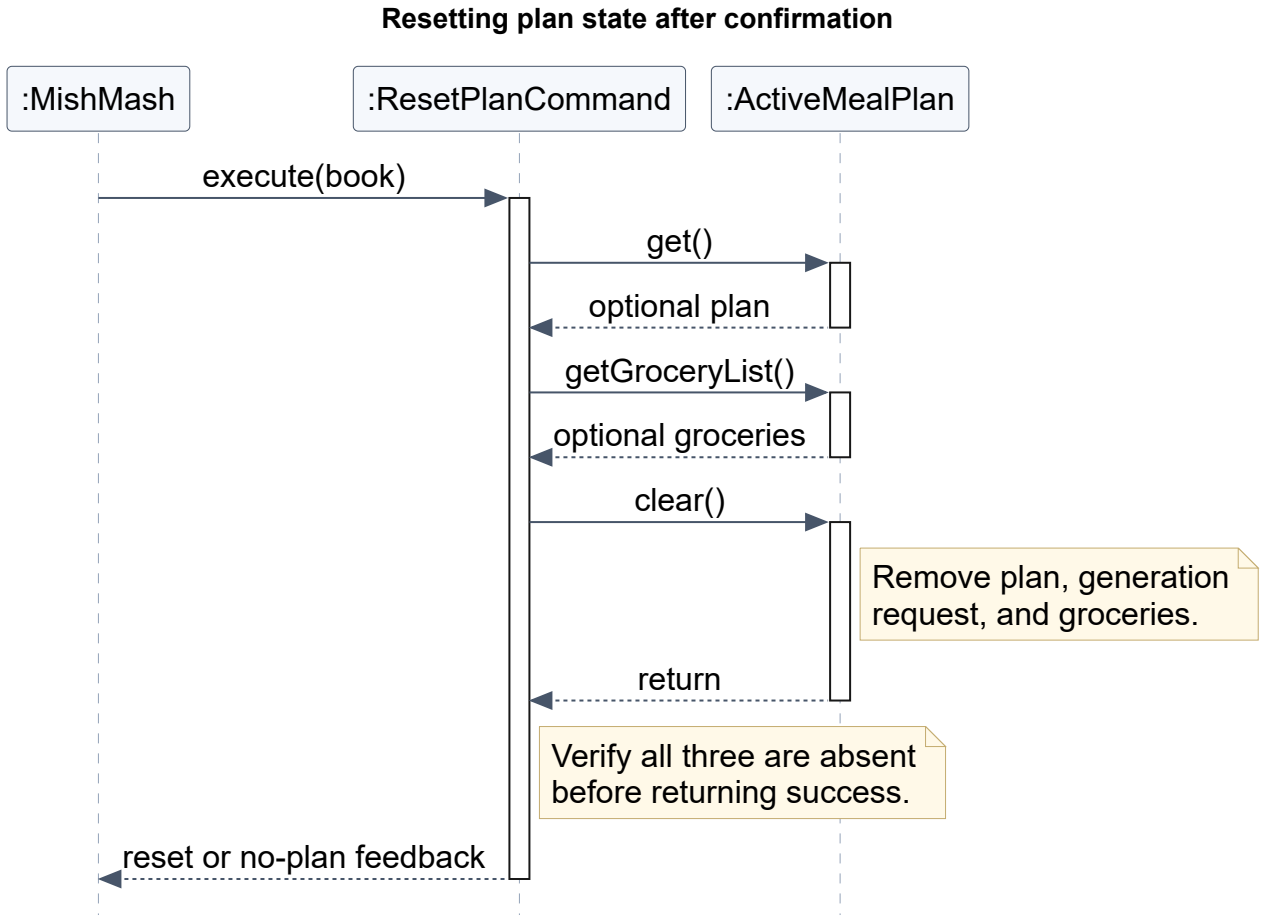
Section author: Rixte

`reset-plan` clears the current planning state while keeping the user's other session information. It takes no arguments. The User Guide describes the confirmation and messages.

Responsibilities

Component	Responsibility
`Parser.parseResetPlan`	Reject arguments and create a command using the shared holder.
`ResetPlanCommand`	Request confirmation, clear planning state, and verify the result.
`ActiveMealPlan`	Hold and clear the plan, its generation request, and derived groceries together.

Execution



Sequence diagram of clearing the active plan after agreement

Successful execution after the shared confirmation. Postcondition getter calls and logging are condensed in the note. PlantUML source

1. The command reads whether a plan or groceries were present when execution starts.
2. `ActiveMealPlan.clear()` removes the plan, generation request, and grocery snapshot.
3. The command checks that all three are absent before returning reset or no-plan feedback.

The command holds only `ActiveMealPlan`. Recipes, dietary preferences, personal targets saved through `set-targets`, and recorded meals are held elsewhere and remain available. Targets attached to the removed generation request are cleared; saved personal targets still apply to future generation. Previously returned immutable snapshots remain valid.

Errors and boundaries

Condition	Result
Arguments supplied	Reject with reset-specific usage before requesting confirmation.
No planning state	Clear remains safe; return `No active meal plan to reset.`

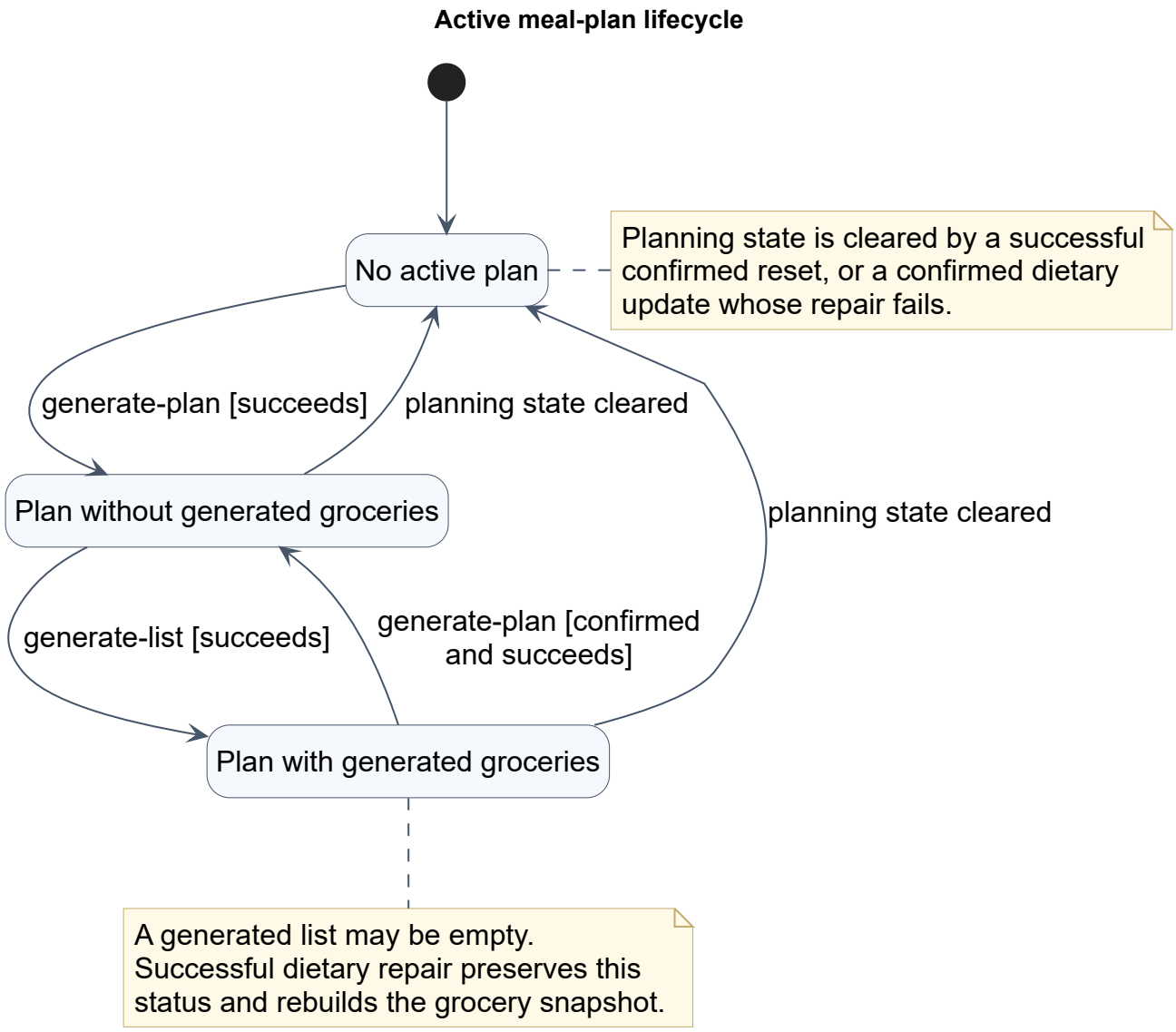
Condition	Result
Groceries remain without a plan, including an empty snapshot	Clear the snapshot as well.
`clear()` throws, or any planning state remains afterwards	Raise a recoverable `MishMashException`; keep the original cause when one exists.

The postcondition is checked even with Java assertions disabled. A failed clear does not promise rollback of planning state, but the command never changes unrelated session data.

Design choice and verification

Reusing `ActiveMealPlan.clear()` keeps clearing in one place. Replacing the holder would leave the parser and existing commands referring to the old object. Clearing individual fields in the command would expose model internals and risk leaving stale groceries or targets.

Plan-status lifecycle



State-machine diagram of active plans and generated grocery snapshots

This state-machine diagram shows the ordinary command-line lifecycle of the shared `ActiveMealPlan` holder. A generated grocery list can be empty; its presence distinguishes the bottom state from a plan whose groceries have not been generated. PlantUML source

Successful replacement generation discards groceries, so a plan with groceries returns to the plan-only state. A successful confirmed reset clears all planning state. A confirmed dietary update that cannot repair a conflicting plan also clears planning state, while retaining the newly saved dietary preferences.

The diagram omits transitions that keep the same status: viewing, invalid input, cancellation, failed generation, regenerating an existing grocery list, compatible dietary updates, and successful dietary repair. Repair preserves whether groceries existed and rebuilds them when present. Replacing a plan without groceries likewise keeps plan-only status. It omits legacy or injected holders with inconsistent state; the reset command's defensive checks are described in the boundaries table above.

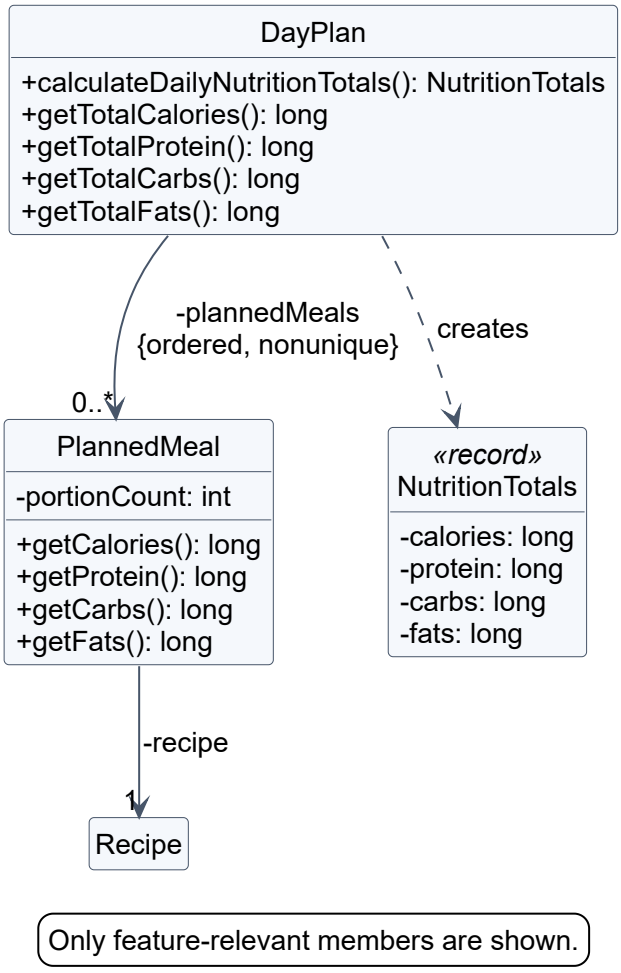
`ResetPlanCommandTest`, `ParserResetPlanTest`, and `ResetPlanDiagnosticsTest` cover validation, repeated resets, current-state lookup, and recoverable failures. `ResetPlanIntegrationTest` and `MishMashResetPlanTest` check retained recipes, preferences, saved targets, recorded meals, and regeneration. Manually reset a targeted plan with groceries, confirm both views report missing data, and generate again using the saved settings.

Combined daily nutrition totals

Section author: oilybikechain

The structural view focuses on the combined totals API and the recipe values it sums.

Combined daily nutrition totals



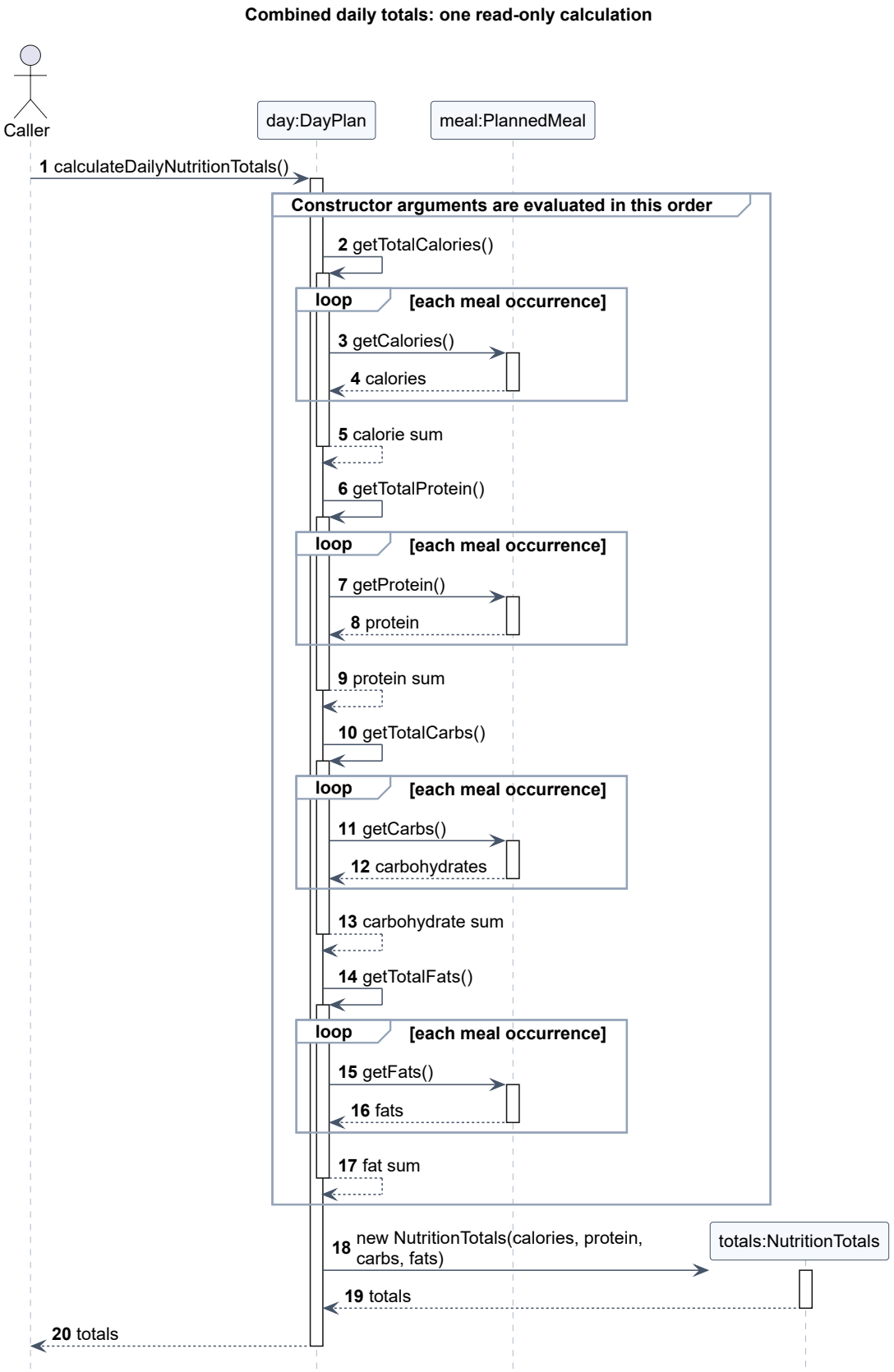
Nutrition class diagram showing daily totals, targets, generation requests, and viewing suppliers.

Nutrition structure: `DayPlan` retains its ordered meals and creates an immutable totals value. Only members used by this calculation are shown.

`DayPlan.calculateDailyNutritionTotals()` returns one immutable `NutritionTotals` record containing `calories()`, `protein()`, `carbs()`, and `fats()`. Calories are measured in kcal; the macronutrients are measured in grams. It delegates to the existing individual total getters, giving callers a single result without changing the plan, recipes, or the individual-getter API.

Every meal occurrence contributes its full nutrition, including repeated recipes. Each day is calculated independently; these are not totals across the entire plan. An empty day returns four zeros. All four record components use `long`, so sums of valid integer recipe values can exceed `Integer.MAX_VALUE` without wrapping.

For example, a day containing the User Guide's Protein Oatmeal, Chicken and Rice, and Salmon Salad has `new NutritionTotals(1650, 127, 150, 55)`. The record represents calculated data; it does not apply target tolerance, validate a nutrition goal, or introduce a new console command. The existing plan viewer computes its display totals from `MealSummary` values separately.



Sequence diagram of four daily nutrition sums followed by construction of NutritionTotals.

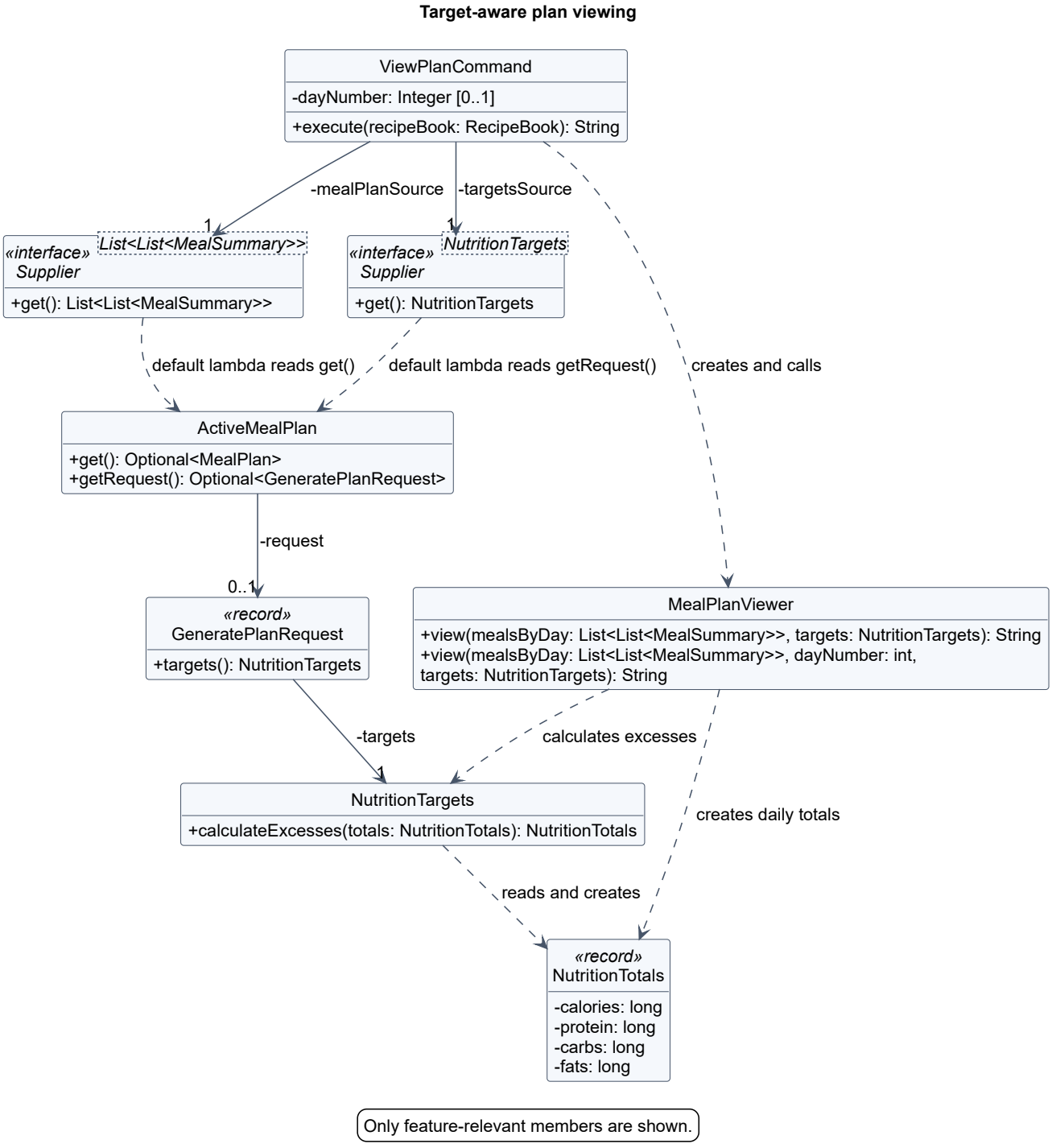
Combined-total calculation: constructor arguments are evaluated in calories, protein, carbohydrates, and fats order. Each existing getter traverses every meal occurrence using a `Long` sum. Stream internals are condensed, and the caller represents an API client rather than the plan viewer.

`DayPlanTest` covers ordinary meals, one meal, zero nutrition, an empty day, repeated recipes, independent days and repeated calls, unchanged meals, and totals above the integer range.

Detecting exceeded nutrition targets

Section author: oilybikechain

The target-aware viewing structure is kept separate from the combined-totals model so both class diagrams remain legible at guide width. The supplier interfaces are bindings of Java's existing `Supplier` type; the dashed dependencies show the default lambdas reading current shared state.



Class diagram of target-aware plan viewing.

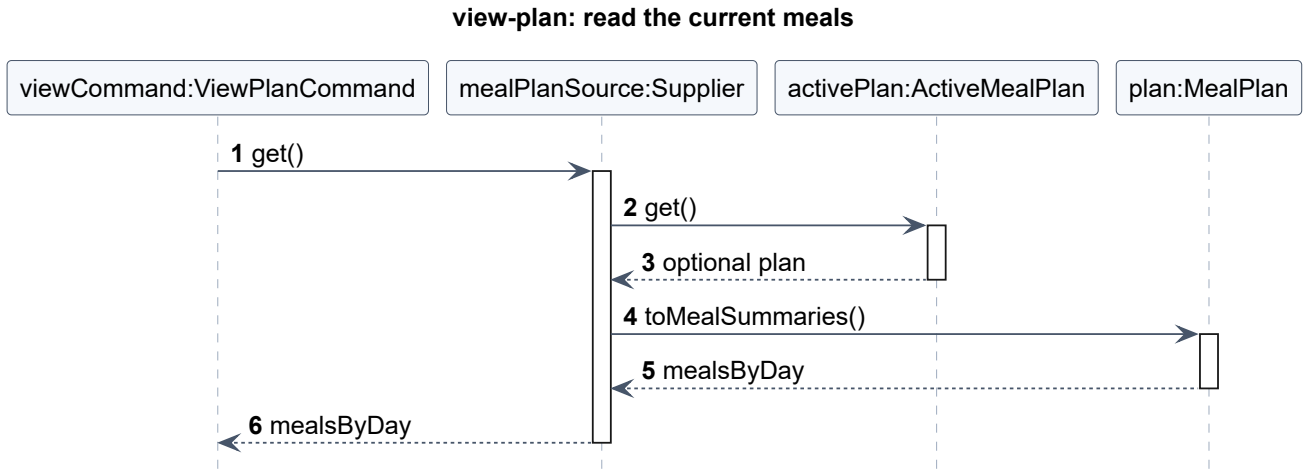
`ActiveMealPlan` already retains the original `GeneratePlanRequest` alongside a generated plan for dietary repair. Exceedance warnings reuse this request's immutable `NutritionTargets` rather than storing another copy in `MealPlan`. Successful generation replaces both meals and targets through `setGenerated`; failed generation preserves them. Dietary repair retains the request, and clearing a plan clears its request. Legacy plans published with `set(MealPlan)` have no request and therefore no targets for warnings.

`NutritionTargets.calculateExcesses(NutritionTotals)` returns each nutrient's positive excess over its exact target, or zero if the target is omitted or not exceeded. It uses `long` arithmetic, matching daily totals, so sums beyond the integer range remain valid. This comparison does not use `TOLERANCE`: generation still accepts days within ±15%, while viewing warns for any total strictly above the supplied target.

`Parser` supplies the current plan's meals and request targets to `ViewPlanCommand` at execution time. Commands parsed before generation or replacement therefore display the latest meals and targets. Legacy supplier-based parser and command constructors default to no targets. The new command constructor accepts meal and target suppliers plus a nullable day number (null selects every day). The suppliers must return corresponding meals and targets; omitted targets are represented by `NutritionTargets.none()`.

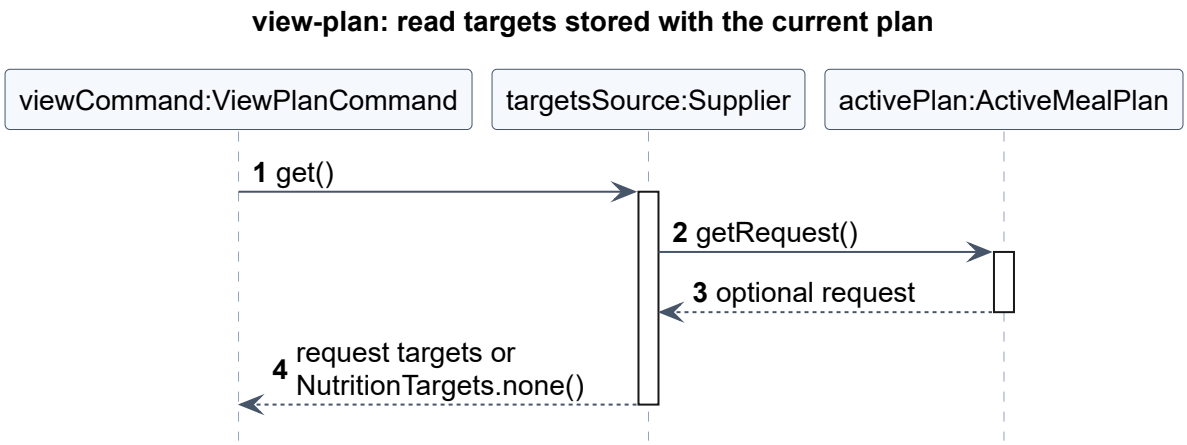
`MealPlanViewer` provides target-aware overloads for all days and one selected day; its original overloads delegate with no targets. It sums each day's meals, requests excesses, and places warnings before the day's closing separator in calories, protein, carbohydrates, and fats order. Warnings include the excess, actual total, and target, using kcal for calories and g for the macronutrients. They do not change application state. `check-shortfall` continues to use its explicitly supplied targets independently of this display.

State lookup and warning calculation are shown separately so their messages remain readable. The first sequence follows a valid selected day: the default meal supplier adapts the current active plan, then the target supplier reads the request stored with that plan.



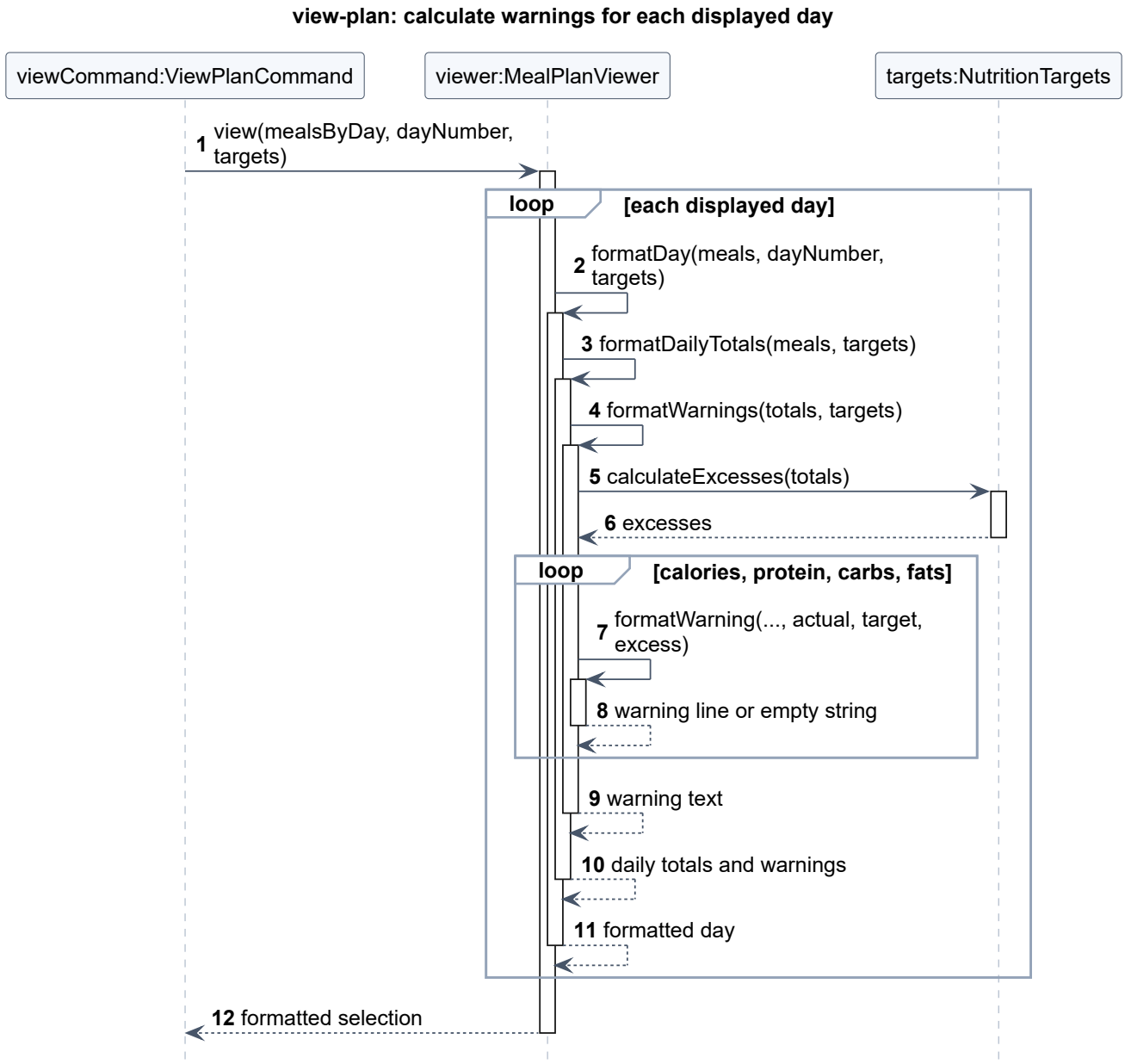
Sequence diagram of execution-time meal and target lookup.

The corresponding targets are read through a separate supplier after the selected day is known to be valid.



Sequence diagram of execution-time target lookup.

With no active plan, execution returns before reading the target supplier. An out-of-range selected day is also rejected before targets are read. For a valid selection, the viewer calculates and formats warnings only for the displayed days.



Sequence diagram of calculating exact excess warnings for displayed days.

Tests cover exact comparison boundaries, omitted targets, simultaneous excesses, large totals, exact output, and day selection. Integration tests verify the generation tolerance, replacement after success or failure, legacy plans without requests, dietary repair and clearing, and independence from `check-shortfall`. Run `gradlew.bat test checkstyleMain checkstyleTest` with Java 25.

Checking nutrition shortfalls

Section author: Rixte

`check-shortfall` reports exact daily deficits against one or more explicit targets:

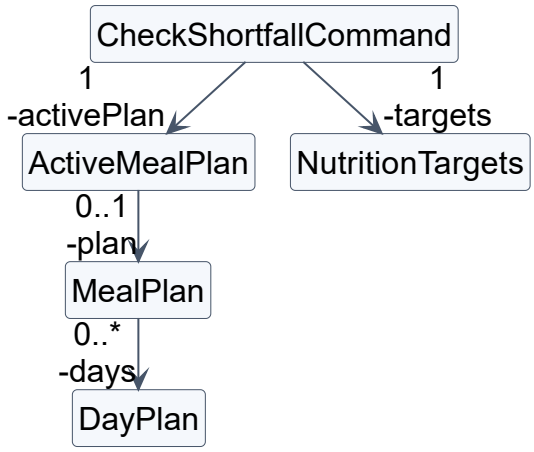
```
check-shortfall [cal/DAILY_CALORIES] [p/DAILY_PROTEIN] [c/DAILY_CARBS] [f/DAILY_FATS]
```

For example, `check-shortfall cal/2000 p/100` checks only calories and protein. It checks planned meals, rather than recorded meals, and does not save targets or rebalance the plan. The User Guide gives examples and expected output.

Responsibilities

Component	Responsibility
`Parser.parseCheckShortfall`	Coordinate parsing and attach command usage to validation errors.
`CheckShortfallParser`	Validate unique target tokens and create `NutritionTargets`.
`CheckShortfallCommand`	Read the current plan and report each day's deficits.
`DayPlan`	Supply the four existing nutrition totals as `long` values.

Shortfall state and targets



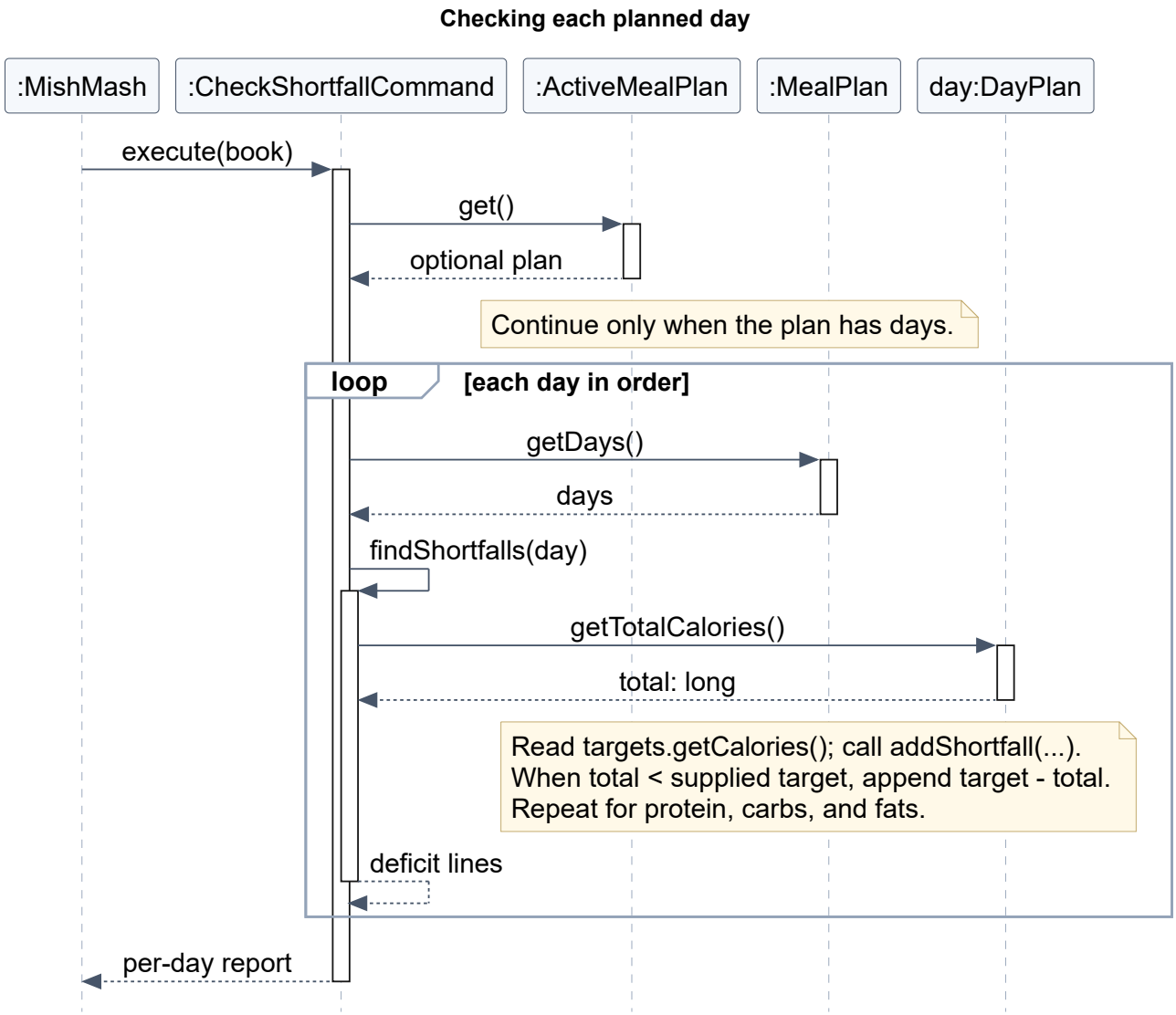
Class diagram of the shortfall command's state and targets

Selected stored references, with private role names and target multiplicities. The ordered day list can hold repeated references. Parsing dependencies, recipe internals, and the command interface are omitted. PlantUML source

The command retains the shared `ActiveMealPlan` and request-specific `NutritionTargets`. The normal parser shares that holder with generation. Caller-provided viewing suppliers do not replace this data source. The `RecipeBook` execution parameter is unused.

Parsing and daily comparison

`CheckShortfallParser` accepts each of `cal/`, `p/`, `c/`, and `f/` once, in any order. Values must be digit strings representing 1 to `Integer.MAX_VALUE`; omitted nutrients remain null. `Parser` wraps validation failures in `MishMashException`, preserving their cause and attaching shortfall-specific usage.



Sequence diagram of daily shortfall comparison

Successful execution for a plan containing days. One day's calorie lookup illustrates the repeated comparison. Plan-size checks, target getters, `addShortfall`, report assembly, and logging are condensed in the note. PlantUML source

1. `execute` reads the current plan; no plan or zero days returns generate-first guidance.
2. For each day, `findShortfalls` reads its calorie, protein, carbohydrate, and fat totals.
3. `addShortfall` appends `target - total` only when a target is supplied and the total is smaller.
4. Output lists each day's deficits, or states that the supplied targets have no shortfalls.

Comparisons use `long` totals and subtraction, so several large recipe values cannot wrap into false deficits. Days are checked independently; a surplus on one cannot hide another's deficit. Equality is sufficient, and omitted nutrients are ignored.

Errors and boundaries

Condition	Result
No targets, unknown or repeated prefix, missing value, extra token, or space after a prefix	Reject before accessing the plan, with shortfall usage.
Zero, signed, negative, decimal, unit-suffixed, nonnumeric, or overflowing value	Reject; only positive whole numbers in range are accepted.
No active plan or zero days	Return `No active meal plan. Use 'generate-plan' first.`
A day contains no meals	Treat totals as zero; report each supplied target as its full deficit.
Target omitted or total at/above target	Add no deficit for that nutrient.

Every check preserves the plan, generation settings, grocery snapshot, recipe book, and saved personal targets.

Design choice and verification

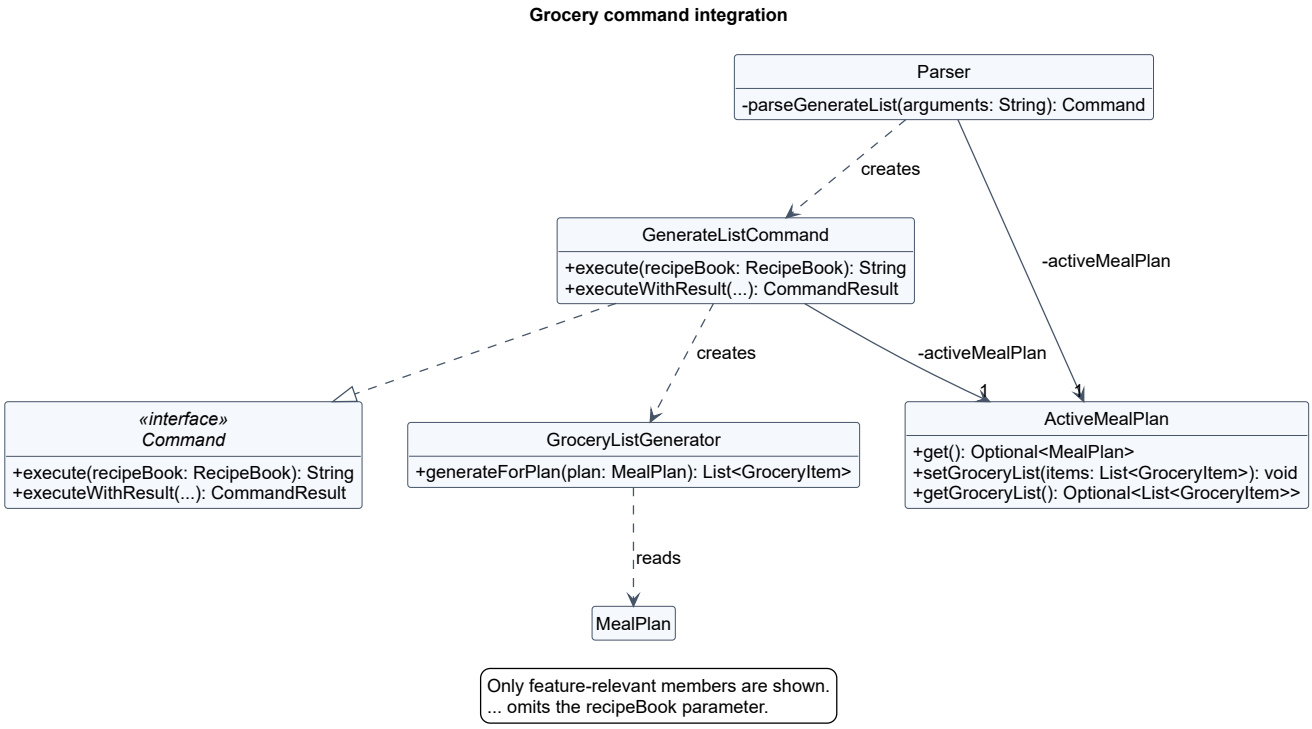
Exact comparison explains the gap; the generator's +/-15% tolerance only decides whether a candidate is acceptable. A 1900 kcal day can therefore satisfy generation for 2000 kcal and still show a 100 kcal shortfall. Explicit targets allow a different comparison without changing saved settings. Automatic rebalancing would need selection rules and grocery updates.

`CheckShortfallParserTest` and `ParserCheckShortfallTest` cover target orders, limits, wrapped errors, and unchanged state after rejection. `CheckShortfallCommandTest`, `CheckShortfallIntegrationTest`, and `CheckShortfallDiagnosticsTest` cover daily boundaries, large totals, repeated execution, and snapshot preservation. Manually check a plan against higher and lower targets, then use both viewing commands to confirm its data is unchanged.

Generating a grocery list

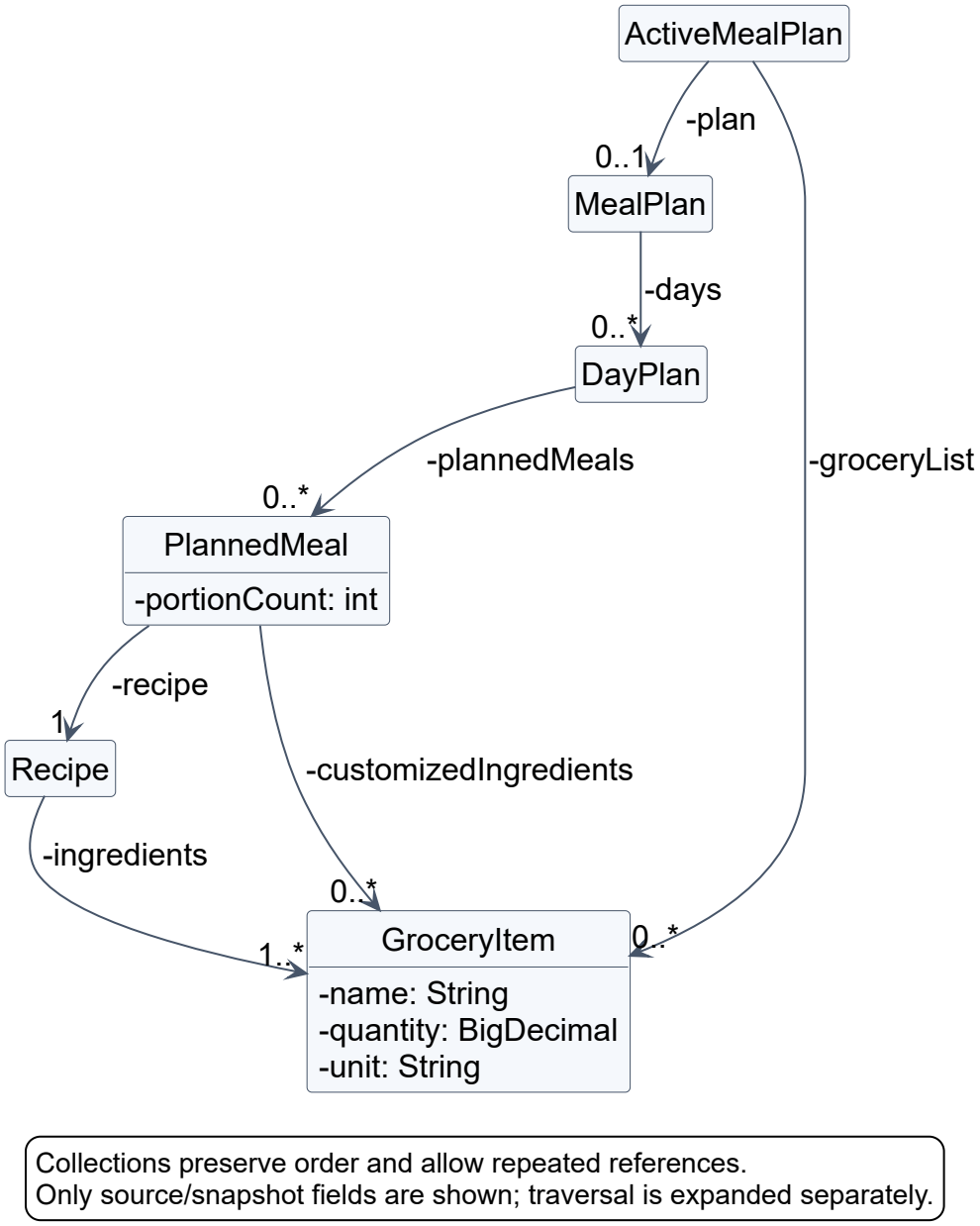
Section author: oilybikechain

The grocery structure is divided by responsibility: command integration, source and snapshot state, and aggregation values.



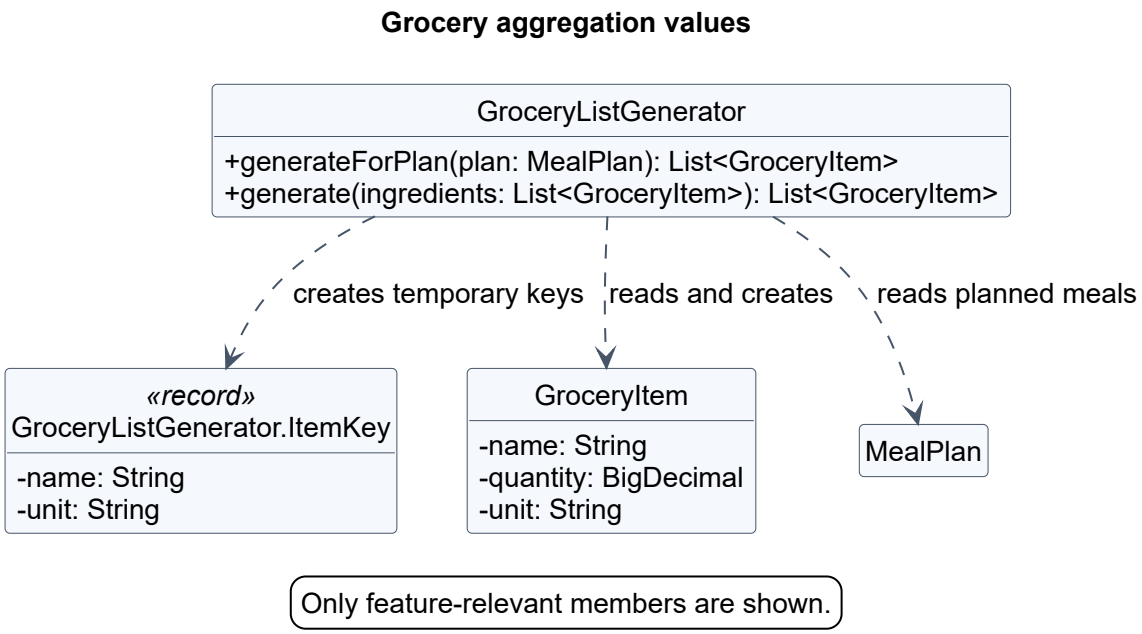
Class diagram of grocery command integration.

Grocery source and snapshot model



Class diagram of the grocery source and snapshot model.

Stored references and portions only; traversal methods are expanded in the collection sequence below. Collection ends preserve order and allow repeated references. PlantUML source.



Class diagram of grocery aggregation values.

The parser and command retain the same active-plan holder. Plain navigable associations allow shared model objects; they do not claim exclusive ownership. The optional grocery snapshot distinguishes an absent snapshot from a present empty list. The temporary aggregation key is a dependency rather than a stored generator field.

Ingredient aggregation

`GroceryItem` holds a trimmed, non-blank ingredient name and unit plus a nonnegative `BigDecimal` quantity. Recipe input separately requires strictly positive ingredient quantities; zero is supported by the grocery model and aggregation API. `toString()` uses plain decimal notation and removes insignificant trailing zeros, for example `Oil: 0.3ml`.

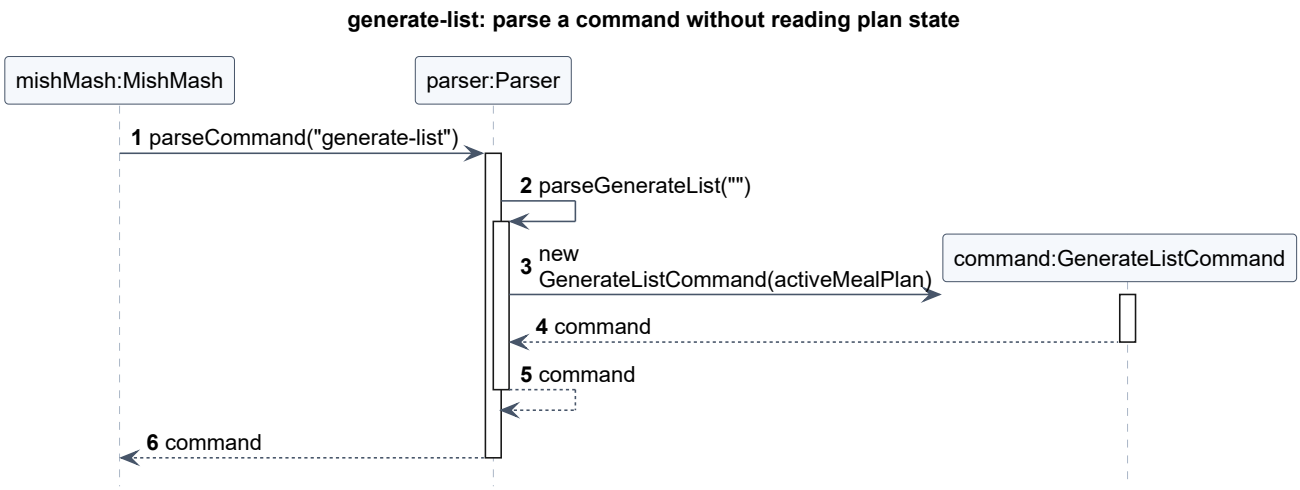
`GroceryListGenerator.generate(List<GroceryItem>)` uses a `LinkedHashMap` keyed by an `ItemKey` record containing the ingredient name and unit. Each occurrence is merged with `BigDecimal::add`. Exact decimal addition avoids floating-point rounding, and insertion order preserves the first appearance of each key. `Rice:200g` and `Rice:100g` combine; `Rice:1kg`, `rice:100g`, and `Rice:1cups` remain separate. No unit conversion or spelling normalisation occurs. When groceries are generated from a plan, `PlannedMeal` first scales each ingredient by its exact `BigDecimal` portion count.

The method rejects a null list or item, returns an empty list for empty input, and publishes an unmodifiable result with `List.copyOf`. Inputs remain unchanged, and successive calls do not reuse previous totals. `GroceryListGeneratorTest` checks exact matching, incompatible units, different names, decimal precision, zero quantities, ordering, immutable output, repeated calls, and null inputs.

Command parsing and shared state

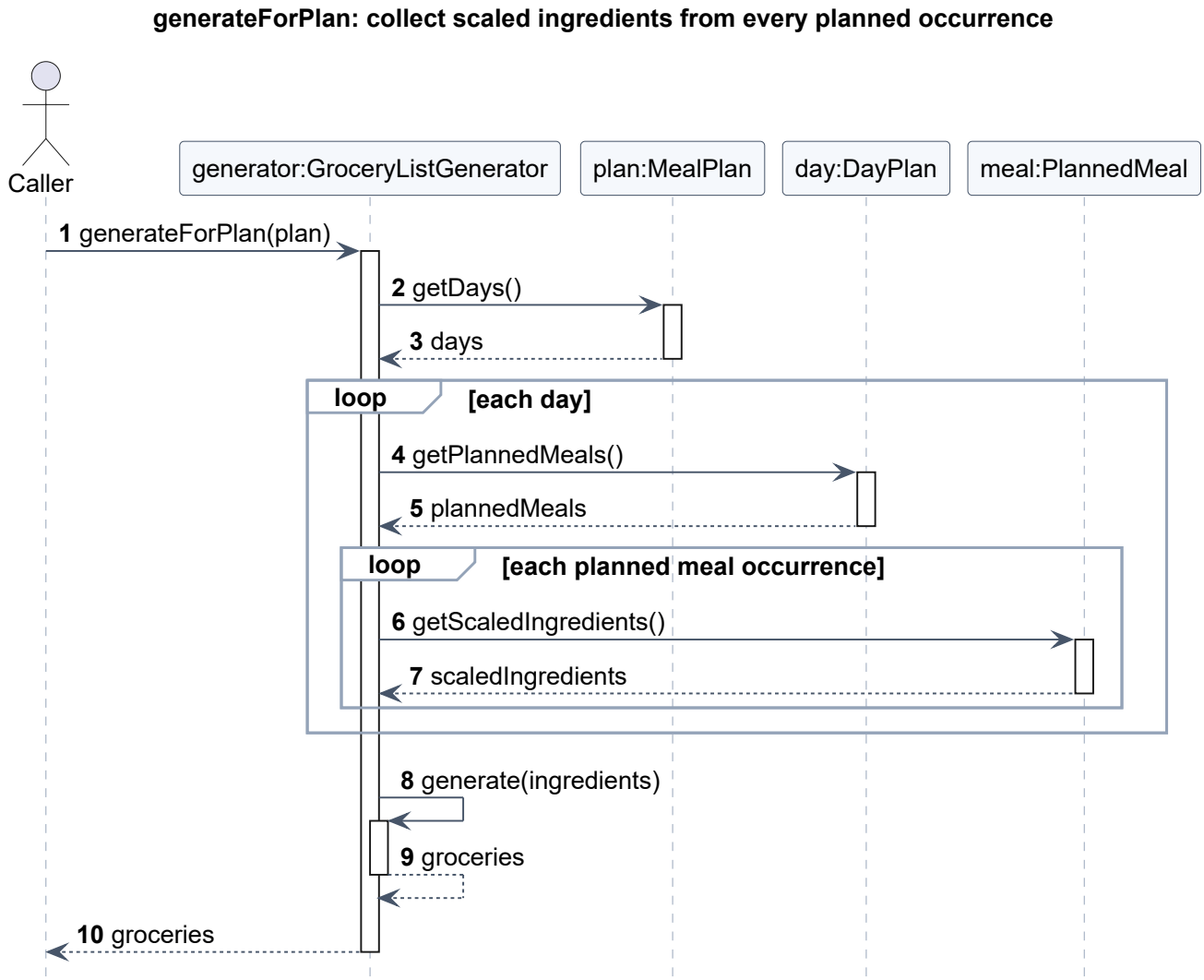
`Parser()` shares one `ActiveMealPlan` between meal-plan generation, grocery generation, and viewing. For deterministic tests or an externally managed plan, use `Parser(MealPlanGenerator, ActiveMealPlan)`. The supplier-based constructors retain their caller-provided viewing sources; their grocery viewers do not automatically read the internally generated grocery snapshot.

The parser accepts `generate-list` without arguments and returns `GenerateListCommand` without reading or modifying the plan.



Sequence diagram of parsing generate-list without reading plan state.

During execution, the command traverses `MealPlan.getDays()`, `DayPlan.getPlannedMeals()`, and each planned meal's scaled ingredients. Every meal occurrence contributes its portion-adjusted quantities. `GroceryListGenerator` combines exact name/unit matches with `BigDecimal` arithmetic, preserving first-appearance order and keeping incompatible units separate.

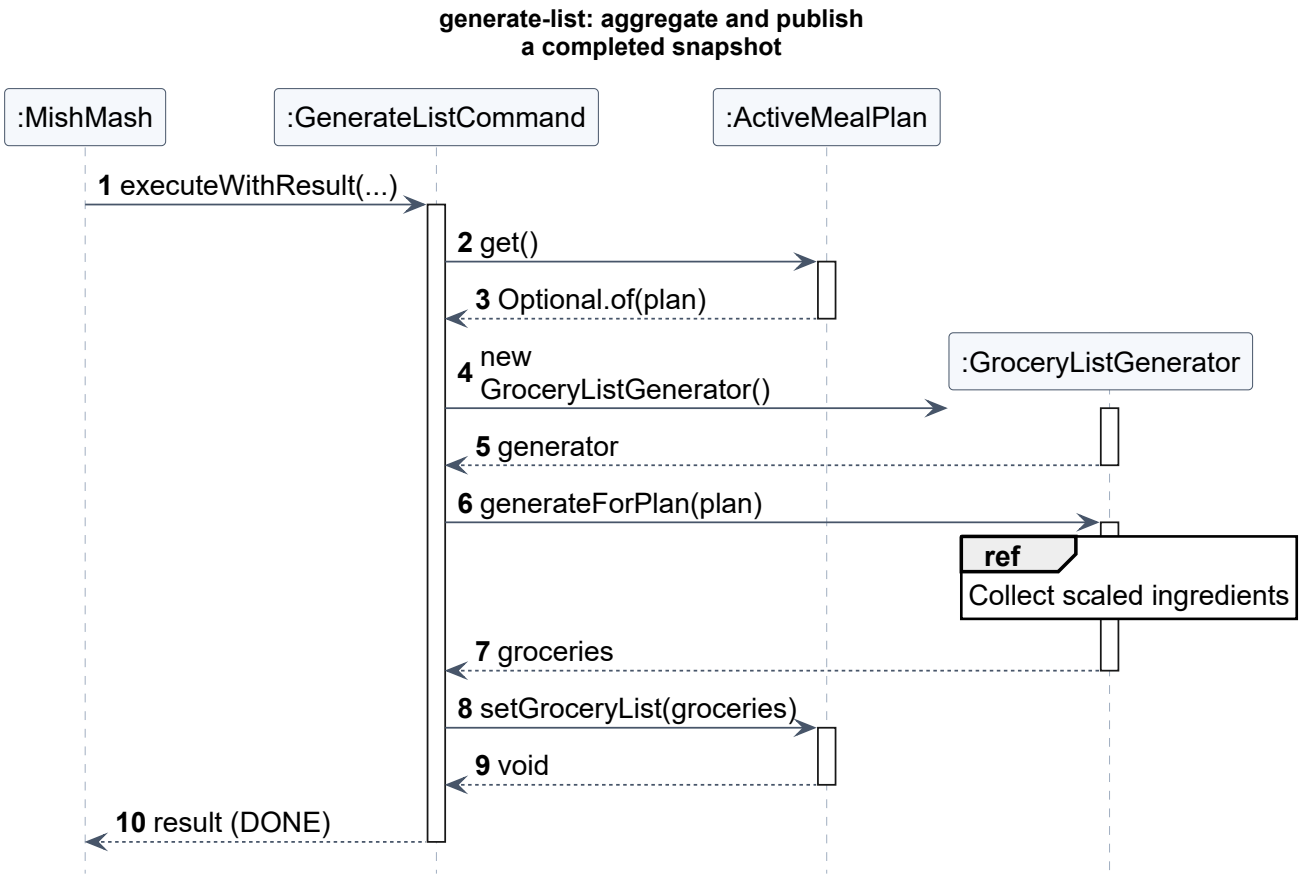


Sequence diagram of collecting every ingredient occurrence.

`ActiveMealPlan.setGroceryList` copies the completed result into a read-only snapshot. `getGroceryList` returns an optional: absence means no list has been generated for the current plan, whereas a present empty list is valid. `ViewListCommand` reads this snapshot during execution and delegates formatting to `GroceryListViewer`; viewing never regenerates data.

Successfully storing a replacement plan clears its old groceries. Failed plan generation or invalid command syntax leaves the previous state intact. The complete state is autosaved for later sessions.

The parse step rejects arguments with `generate-list does not accept arguments.` and `Format: generate-list`; it does not inspect the active plan. The command reads the latest plan when executed, so a command parsed before generation can use a subsequently generated plan. Without a plan it returns `No active meal plan. Use 'generate-plan' first.` without publishing a snapshot. Successful generation returns `Grocery list generated. Use 'view-list' to view it.` Only the finished result is stored, preserving any previous snapshot if generation fails.



Sequence diagram of grocery aggregation and snapshot publication.

The execution sequence follows `executeWithResult`; `...` omits its `recipeBook` parameter. The collection reference expands in the preceding `generateForPlan` sequence. Only a completed grocery snapshot is published, and the command returns a `DONE` result. Autosaving follows the command result, as shown under Persistence and recovery. The PlantUML source keeps the publication sequence separate from ingredient traversal so its labels remain readable at normal PDF width. Ingredient-key lookup, decimal merging, and result-item construction happen inside `GroceryListGenerator.generate` and are described in the ingredient-aggregation section above.

`GenerateListCommandTest`, `ParserGenerateListTest`, and `ActiveMealPlanTest` cover missing plans, execution-time plan lookup, every recipe occurrence, repeated generation, syntax rejection, copying, and snapshot replacement. `MishMashGeneratePlanTest` exercises the generation-to-viewing connection

through the console. Grocery-list formatting is supplied by the existing viewer rather than this aggregation contribution.

Grocery parsing and viewing use `FINE` logs, successful generation logs counts at `INFO`, and rejected generation requests use `WARNING`. These loggers inherit the existing application configuration; no additional handlers are installed. New diagnostics exclude ingredient names and raw input.

Run `gradlew.bat test checkstyleMain checkstyleTest` on Windows with Java 25 to verify the integration.

Maintaining the grocery and nutrition diagrams

The class and sequence diagrams above have editable sources and regeneration instructions in `docs/diagrams/oilybikechain`. Generated SVGs are checked in under `docs/images/oilybikechain`. Use Java 25 and PlantUML 1.2026.9 with the existing shared style and Smetana class layout. Check method names, retained references, multiplicities, supplier indirection, message order, and failure paths against current code, then syntax-check, render, and inspect every image. Keep sources and SVGs synchronised. The final integration review records owner evidence and both grocery readability decisions in the exported PDF. The contribution history identifies the relevant commits; teammates' viewing, generation, formatting, and dietary components are shown as integration context.

Viewing a grocery list

Section author: Rixte

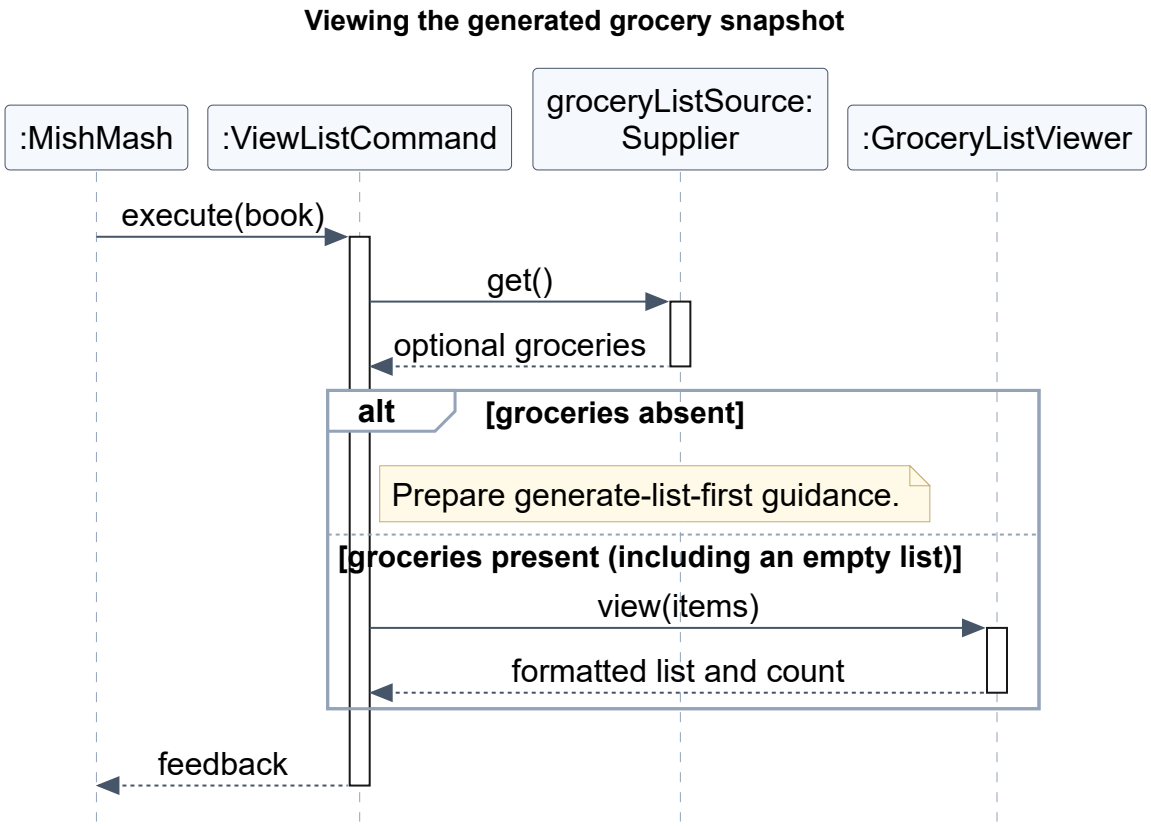
`view-list` displays the grocery snapshot already created by `generate-list`. It accepts no arguments and never aggregates ingredients itself. See the User Guide for the output format.

Responsibilities

Component	Responsibility
`Parser.parseViewList`	Reject arguments and inject the grocery supplier.
`ViewListCommand`	Read the current optional snapshot and distinguish absence from an empty list.
`GroceryListViewer`	Format items in supplied order, with `[ ]` markers and a total count.

The normal parser supplies `activeMealPlan::getGroceryList`. The command retains the supplier rather than a copied list, so later generation or reset is visible on execution. Grocery aggregation and publication are covered in Grocery command integration.

Execution



Sequence diagram of reading and displaying generated groceries

Both missing and generated snapshots are shown. A present empty list follows the formatting branch; `items` is the list extracted from the present `Optional`. Supplier holder access and item formatting are condensed; viewer construction is omitted. The common return represents either guidance or the formatted list. PlantUML source

1. `execute` calls `groceryListSource.get()` to read the latest snapshot.
2. An absent `Optional` returns `No grocery list generated. Use 'generate-list' first.`
3. A present list is passed to `GroceryListViewer.view`, including a list with zero items.

Errors and boundaries

Condition	Result
Arguments supplied	Reject during parsing with view-list usage.
No generated snapshot	Show generate-list-first guidance without creating groceries.
Generated empty list	Display the normal heading and `Total items to purchase: 0`.
Successful `generate-plan` or `reset-plan`	The normal holder removes the old snapshot; later views show missing-list guidance. A successful `set-diet` repair regenerates an existing snapshot instead.

Condition	Result
Null supplier result or null item	Throw `IllegalStateException` or `NullPointerException`; `MishMash.run` logs and rethrows the failure, ending the session.

The viewer preserves item order, quantities, and units. Recipe edits or deletions do not rewrite an existing snapshot; users regenerate the plan and groceries to incorporate them.

Design choice and verification

`Optional<List<GroceryItem>>` distinguishes “not generated” from “generated with no items”. Using an empty list for both would lose that distinction. Keeping generation separate also lets users inspect the saved snapshot without silently replacing it.

`ViewListCommandTest`, `ViewListIntegrationTest`, `ViewListDiagnosticsTest`, `GroceryListViewerTest`, and `ParserTest` check missing and empty snapshots, execution-time lookup, formatting, invalid arguments, and preservation of recipes. Manually generate groceries and view them twice. Then successfully run `generate-plan` or confirm `reset-plan`; verify the next `view-list` asks for `generate-list`. To check the different `set-diet` behaviour, generate groceries before changing the diet and verify that a successful repair updates the existing grocery snapshot rather than showing missing-list guidance.

Recording meals and tracking nutrition

Section author: GauthamNandi

Purpose and scope

This V2.0 enhancement supports two user stories:

- **As a returning user, I want to record meals that I have eaten so that I can keep track of my actual food intake.**
- **As a user tracking nutrition, I want to calculate the total calories and macronutrients for each meal so that I know the nutritional content of what I am eating.**

It adds these commands:

```
record-meal INDEX [date/YYYY-MM-DD] [s/SERVINGS]
view-meals [date/YYYY-MM-DD]
edit-meal MEAL_NUMBER [date/YYYY-MM-DD] [s/SERVINGS]
delete-meal MEAL_NUMBER
```

The date defaults to today and the servings default to 1. `MEAL_NUMBER` is a meal’s position in the whole log, as shown by `view-meals`.

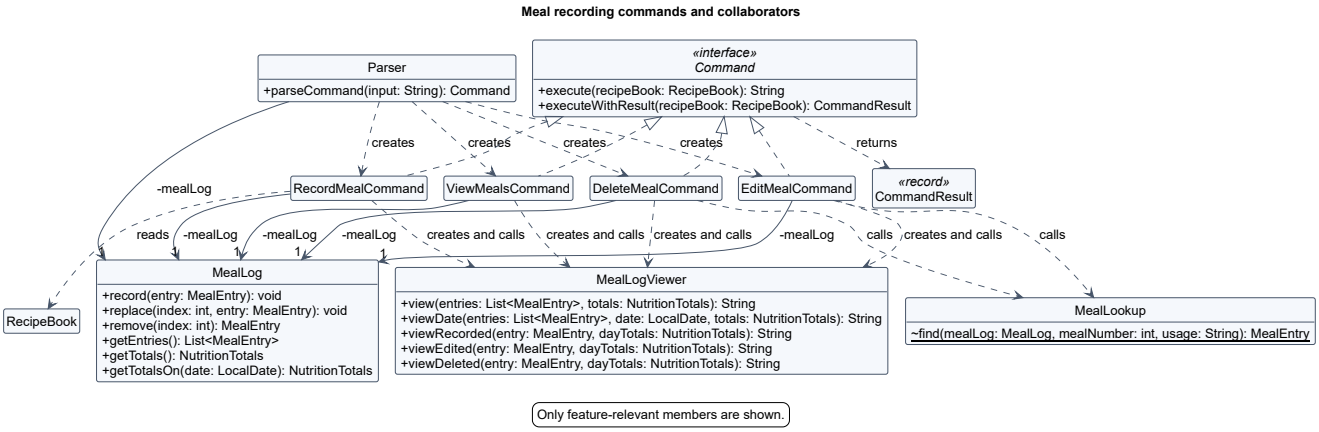
`INDEX` is the position of a recipe in the `list-recipes` output, counting from 1, as for `delete-recipe`. `view-meals` lists every recorded meal with its calories (kcal), protein, carbs and fats (g), followed by the totals for all recorded meals. Unlike `check-shortfall`, which examines the *planned* meals of the active plan, this feature records meals the user *has eaten*. Recorded meals are retained as independent recipe snapshots across sessions.

A meal is a whole number of servings of a recipe, and a recipe stores the nutrition the user entered with `add-recipe`; nutrition is not derived from ingredients. The nutritional content of one meal is therefore its recipe's values multiplied by its servings. The feature calculates the totals across all recorded meals and shows them together with each meal's stored nutrition values.

Architecture-level view

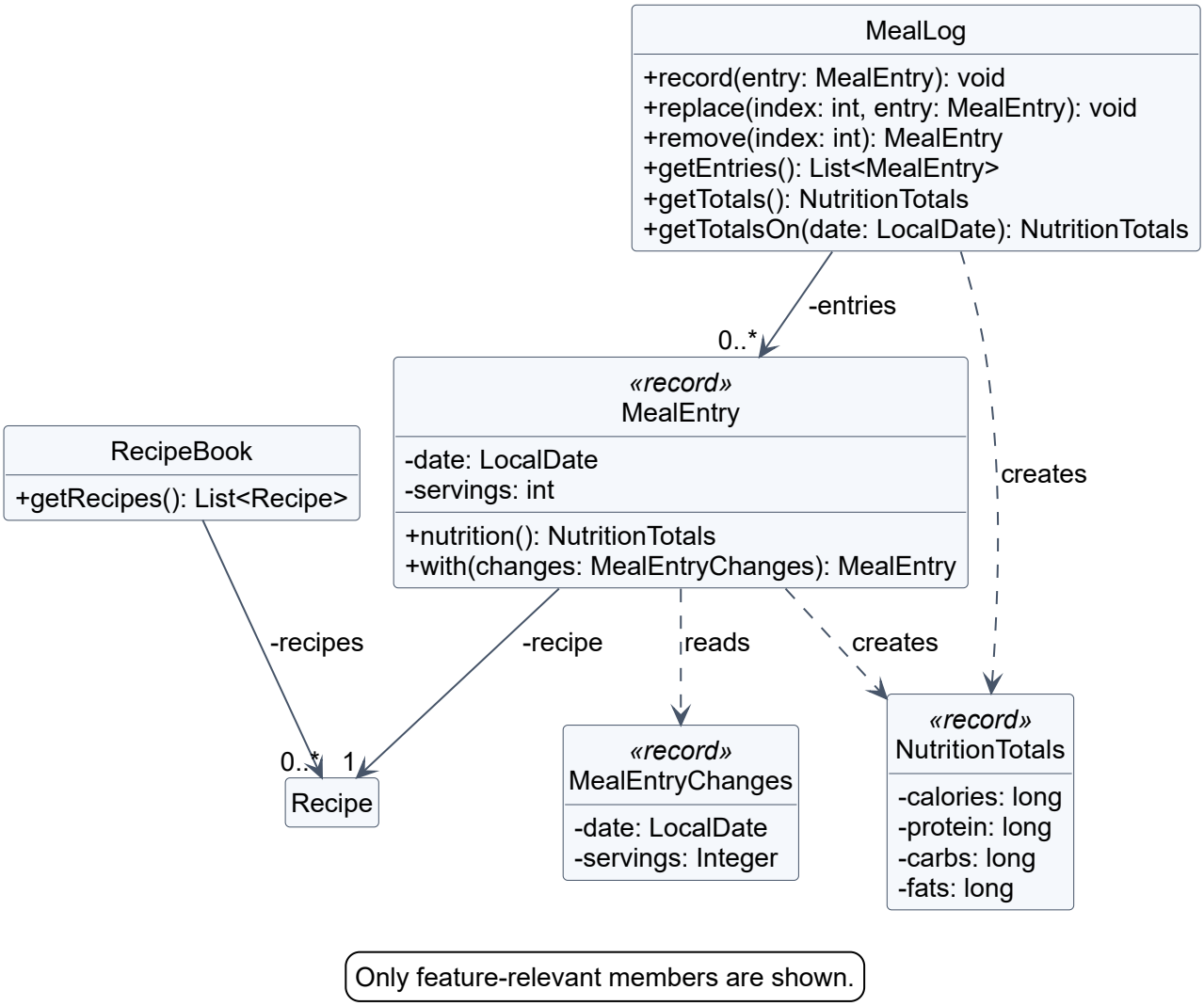
The feature follows the existing command workflow. `MishMash` reads input, asks `Parser` for a `Command`, executes it, and prints the returned text. Parsing validates syntax and creates the action; execution reads or changes session data. A new model class, `MealLog`, holds the recorded meals. Like `ActiveMealPlan`, it is session data shared between commands, but it is separate from the plan: the plan holds meals the user intends to eat, and the log holds meals the user has eaten.

The class diagrams separate command collaborators from stored model relationships.



Meal-recording commands and collaborators.

Meal recording model



Meal-recording model relationships.

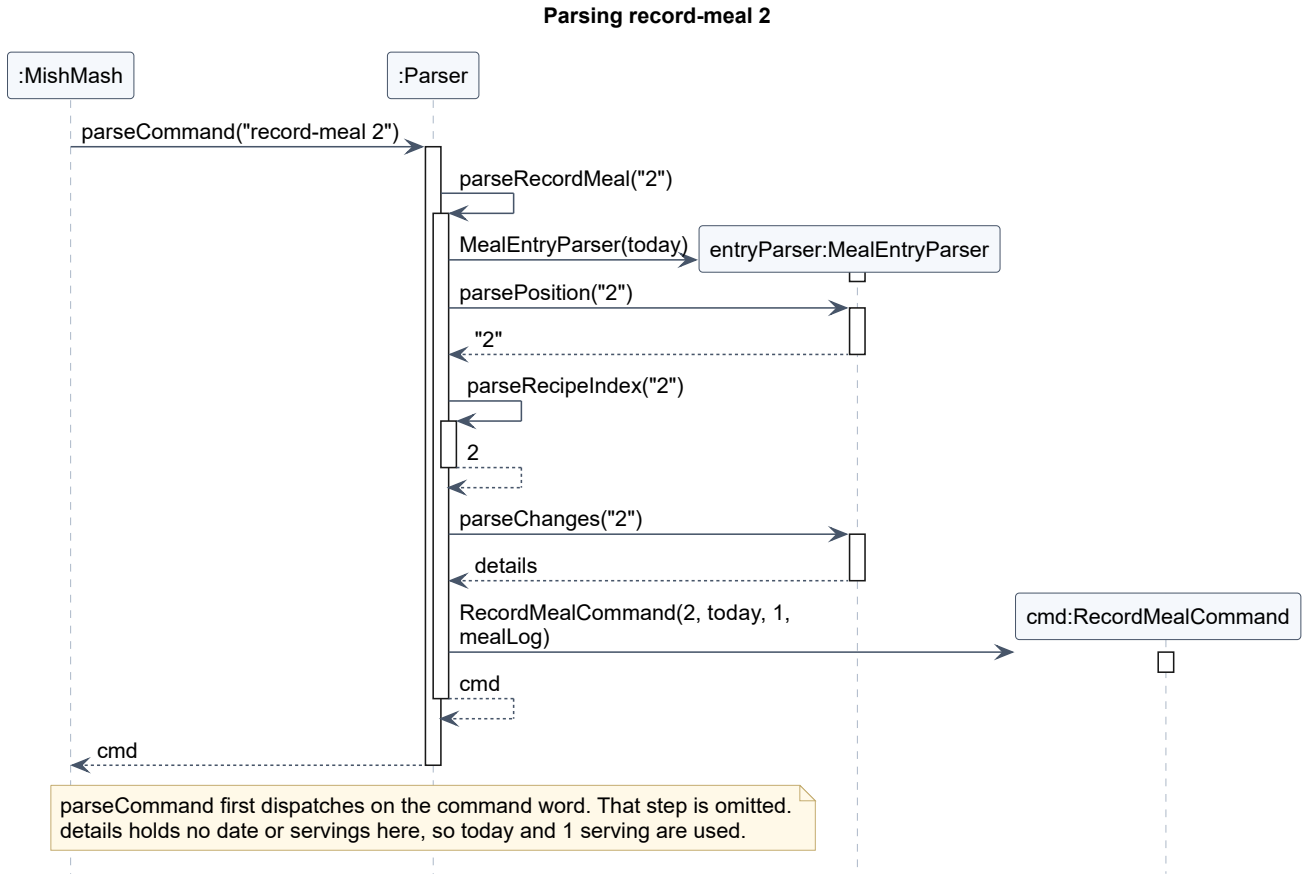
Only members relevant to this enhancement are shown. Solid arrows are navigable associations with private role names and multiplicities; dashed arrows with an open head show dependencies, and dashed arrows with a hollow triangle show interface realization. `NutritionTotals` *is the existing value record that* `DayPlan` *also uses for daily totals, and* `MealLog` *reuses it; its values are* `Long` *so that adding many recipes cannot overflow.*

- `MealEntry` is an immutable record of a recipe, a date and a serving count, and its `nutrition()` multiplies the recipe's values by the servings. `MealEntryChanges` holds the optional typed date and servings, and `MealEntryParser` parses them for `record-meal` and `edit-meal`. It receives today's date as an argument, so tests can check future dates without depending on the system clock.
- `MealLog` stores the entries in the order they were recorded. It offers `record(...)`, `replace(...)`, `remove(...)`, read-only `getEntries()`, `getTotalsOn(date)` for one date, and `getTotals()`, which sums calories, protein, carbs and fats into a `NutritionTotals`, the same record `DayPlan` returns for a day's meals. The commands never add up nutrition values themselves.

- `RecordMealCommand`, `ViewMealsCommand`, `EditMealCommand` and `DeleteMealCommand` implement `Command`. Each retains the shared `MealLog`. `MealLookup` gives the last two the same checks on a meal number.
- `MealLogViewer` (ui) turns the meals and totals into text, in the same way `MealPlanViewer` formats a meal plan. Keeping formatting out of the model and commands lets them be tested without comparing layout text.
- `Parser` owns the `MealLog` and passes it to the commands it builds, because `Command.execute` only receives the `RecipeBook`. Every `Parser` constructor that is not given a log creates a fresh one, so isolated test sessions start with no recorded meals; production restores the saved log. `Parser(MealLog)` and `Parser(MealLog, MealPlanGenerator, ActiveMealPlan)` accept one, which lets tests supply and inspect the log. The command classes receive the log through their own constructors.

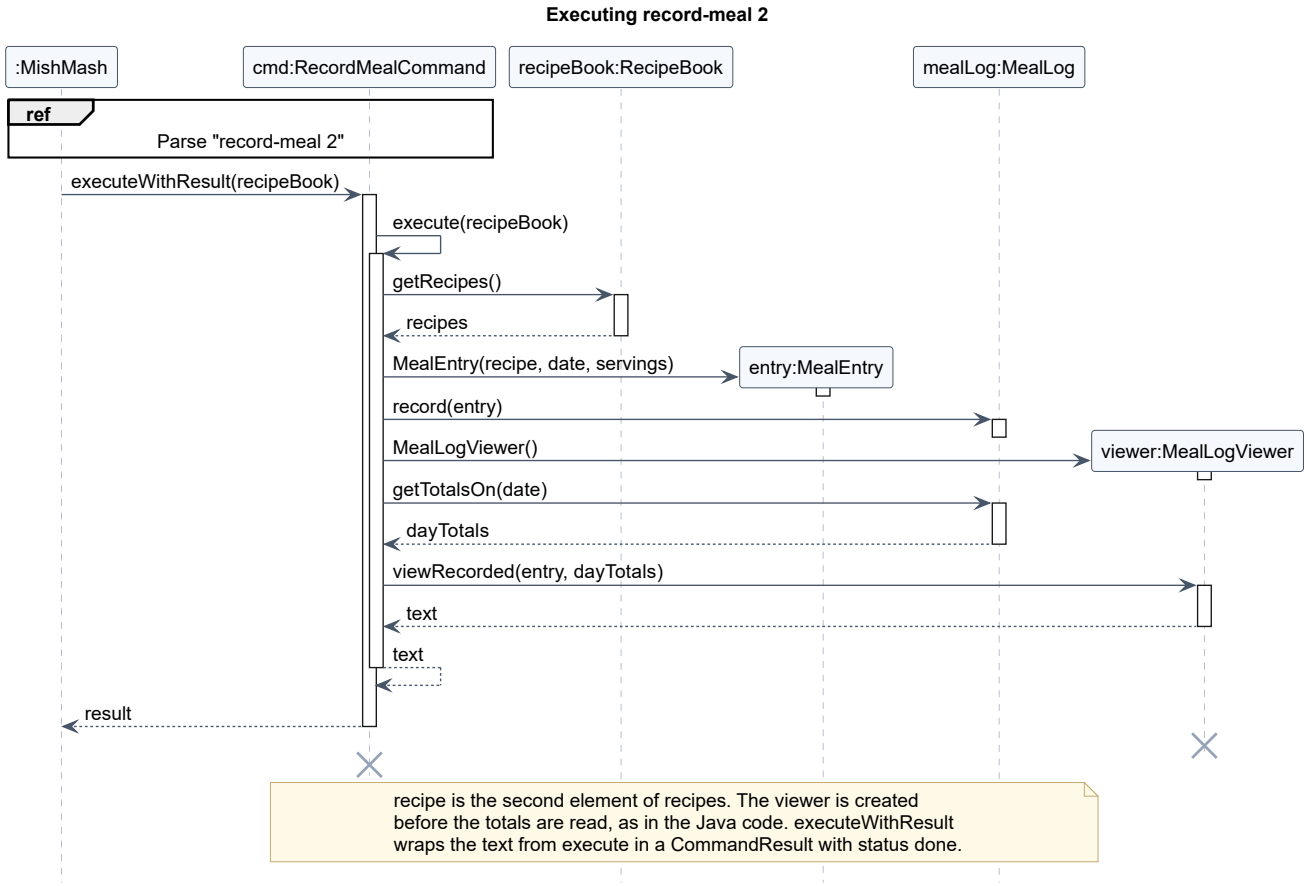
Recording a meal

Separate sequence diagrams show parsing and execution at a readable size. The parsing diagram follows the successful parsing of `record-meal 2`; command dispatch is condensed into the parser lifeline.



Parsing a valid record-meal command.

The execution diagram starts with the command returned by parsing and follows its successful execution.



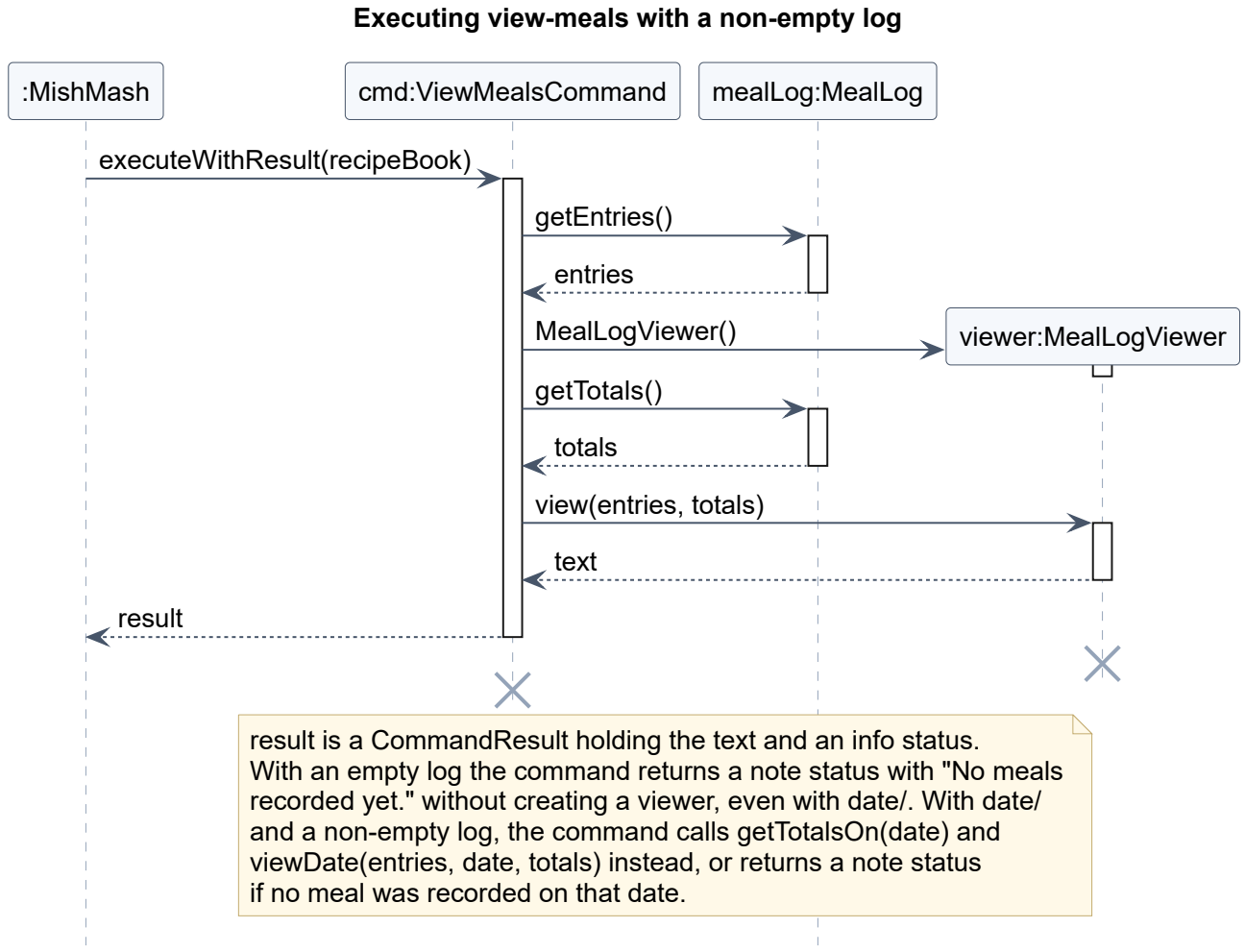
Recording and displaying the selected meal.

Parsing returns a command without reading application data, and the execution diagram refers to it with a `ref` frame. The inherited confirmation step is omitted because the command returns an empty `Optional`. Invalid indexes are described below rather than combined with either successful interaction. The x marks objects that are no longer used once the command has returned.

`Parser` uses `MealEntryParser` to separate the index from the optional `date/` and `s/` fields, then checks that the index is a single positive whole number. A missing, non-numeric, signed, zero or extra argument is rejected with the `record-meal` usage line; nothing is read or changed. The command is created only after validation succeeds.

During execution the command converts the index to a zero-based position, as `DeleteRecipeCommand` does, and looks the recipe up in the `RecipeBook` passed to `execute`. If the book is empty or the index exceeds the number of recipes, it throws a `MishMashException` and nothing is recorded. Otherwise it calls `MealLog.record(entry)` with a new `MealEntry`, creates a `MealLogViewer`, reads `MealLog.getTotalsOn(date)`, and asks `MealLogViewer.viewRecorded` to format the recorded meal's calories and macros with that date's totals, so the user sees the effect immediately. The viewer is created before the totals are read because Java evaluates the object a method is called on before that method's arguments. Recording a meal needs no confirmation: it is quick to repeat, and it changes only the log.

Viewing recorded meals and totals



Sequence for viewing recorded meals.

Parsing mirrors `record-meal` and is omitted. The sequence follows a non-empty log; the empty-log result is described below.

`view-meals` accepts one optional `date/` field; anything else is rejected with its usage line. A date keeps each meal's number from the whole log, so a number shown for one date works with `edit-meal` and `delete-meal`. `ViewMealsCommand` reads the entries from the `MealLog`. An empty log is a valid state, not an error, so the command returns `No meals recorded yet.` Otherwise it creates a `MealLogViewer`, asks the log for the totals and delegates formatting to the viewer. Each meal appears with its own calories and macros, followed by one totals line. If the log is not empty but has no meals on a requested date, it returns `No meals recorded on DATE.` instead; an empty log returns `No meals recorded yet.` even when `date/` is supplied. Viewing never changes the log.

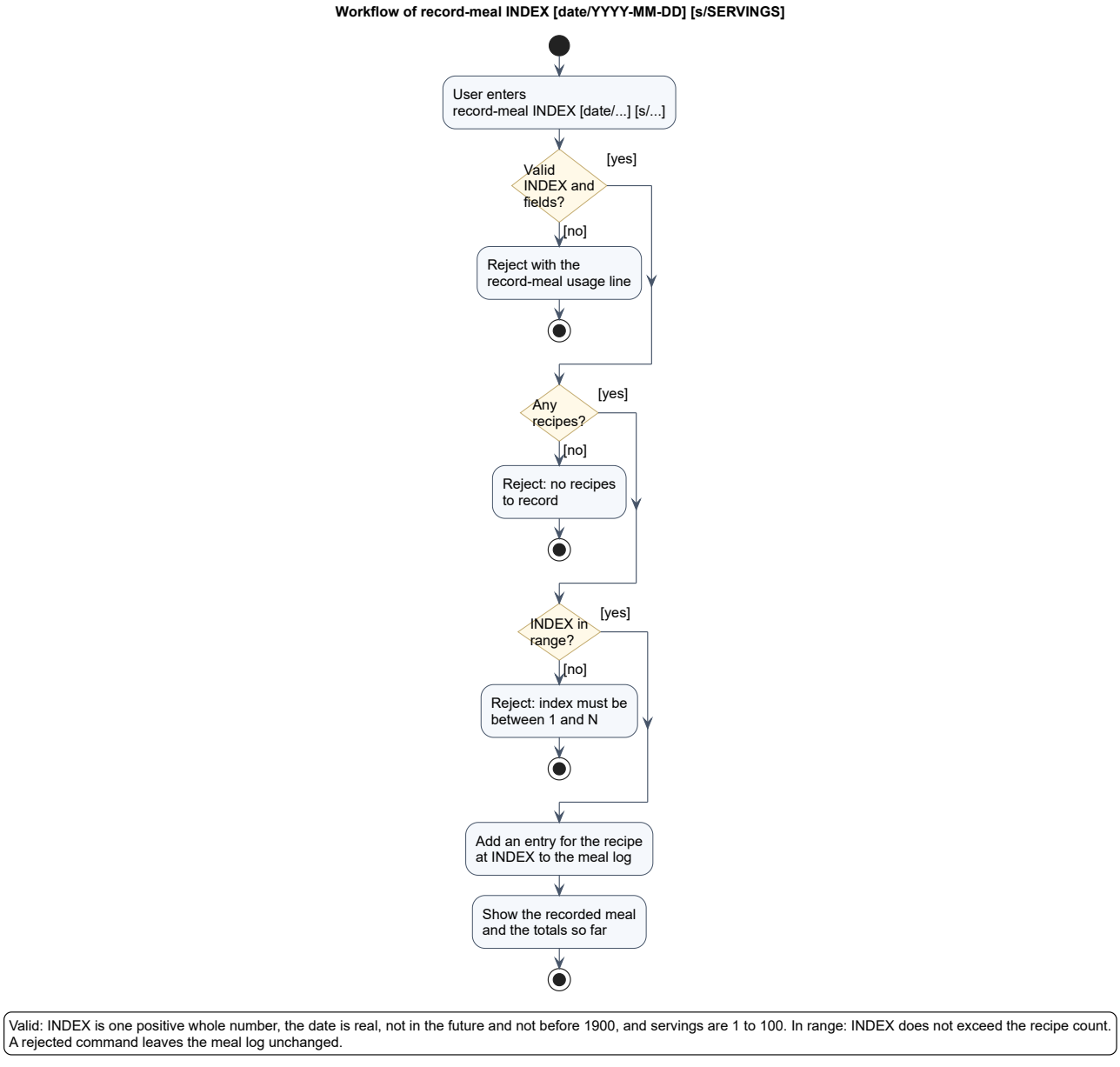
Correcting recorded meals

`edit-meal` and `delete-meal` find a meal by the number `view-meals` shows, using `MealLookup` so that both report a bad number the same way. `EditMealCommand` builds a new `MealEntry` with `MealEntry.with(changes)` and calls `MealLog.replace`, so the meal keeps its place and no other number changes. `DeleteMealCommand` overrides `confirmationPrompt`, which also rejects a number that does not exist before the user is asked anything, and then calls `MealLog.remove`; later meals move up

by one. Both show the totals for the affected date, and totals are recomputed from the log, so a correction cannot leave them out of step. Editing needs no confirmation because it is easy to redo, whereas a deleted entry would have to be recorded again.

Invalid input and recovery

The recoverable errors reuse the existing path: `Parser` or the command throws `MishMashException`, `MishMash.executeCommand` catches it, displays the message, and prompts for another command. The activity diagram shows where `record-meal` can be rejected; every rejection happens before the log changes.



Activity diagram of the record-meal workflow.

Diamonds are decisions, with their guards in square brackets. The first decision is made by `Parser`, the other two by `RecordMealCommand`.

Boundary or invalid condition	Behaviour
`record-meal` with no argument, a non-numeric, signed, zero or decimal index, or an extra token	Reject with the command format; no data is read or changed.
Index larger than the number of recipes	Reject with `Recipe index must be between 1 and N.`; nothing is recorded.
`record-meal` while the recipe book is empty	Reject with a message that there are no recipes to record; nothing is recorded.
`date/` that is not `YYYY-MM-DD` or is not a real date	Reject with the command format; nothing changes.
`date/` after today in `record-meal` or `edit-meal`	Reject, because a recorded meal has already been eaten. `view-meals` accepts any real date.
`s/` outside 1 to 100	Reject with the command format; the limit catches typos and keeps totals within `long`.
`view-meals` with anything other than one `date/`	Reject with the command format.
`view-meals date/...` with no meals on that date	Return `No meals recorded on DATE.`
`edit-meal` with no field, or either correction command with a missing or invalid number	Reject with the command format; the log is unchanged.
`edit-meal` or `delete-meal` with a number beyond the log	Reject with `Meal number must be between 1 and N.`; `delete-meal` reports this before asking for confirmation.
`view-meals` before any meal is recorded	Return `No meals recorded yet.`
Recipe deleted, edited with `edit-recipe`, or otherwise changed after a meal was recorded	The recorded meal keeps its original calories and macros.
Same recipe recorded twice	Two entries are kept, and both count toward the totals.
Recipe with 0 g of protein, carbs and fats	The meal is recorded and contributes zeros for those nutrients.

Rationale and alternatives considered

Decision	Alternative considered	Reason for the current choice
Store the `Recipe` itself in the log	Store the recipe's index	`Recipe` is immutable, so the log keeps the nutrition as it was when the meal was eaten. Indexes shift whenever `delete-recipe` runs, so an old entry could silently point to a different recipe.

Decision	Alternative considered	Reason for the current choice
`MealLog.getTotals()` sums the meals on every call	Keep running totals updated by `record`	The meal list is the single source of truth, so totals cannot drift out of step with it. Sessions are small, so the repeated sum costs nothing noticeable. Running totals would need extra care if removing meals is added later.
`Parser` owns the `MealLog` and passes it to the commands	Widen `Command.execute` to take a shared context object	`Parser` already owns and shares `ActiveMealPlan`, and this leaves the `Command` interface and every existing command unchanged. Constructors that take a `MealLog` let tests supply their own, as the `ActiveMealPlan` constructors do. A context object becomes worthwhile if many more data holders are added.
Pick a meal by recipe index	Type nutrition values in directly	Reusing stored recipes avoids duplicating the input and validation of `add-recipe` and keeps unchecked numbers out of the log. Free-text entry would help when eating out, so it is a possible extension.
Return formatted text through `Command.execute`, using a small viewer	Print from the command or the model	The viewer matches `MealPlanViewer` and keeps layout out of the logic, so the model and commands can be tested without comparing layout text.

Limitations and possible extensions

- The log is autosaved and restored across application sessions, including dates and servings.
- A later version could compare a date's totals with the targets given to `check-shortfall`.
- `edit-meal` cannot change which recipe was eaten; deleting the entry and recording the right recipe does the same job.
- Servings are whole numbers, matching the whole-recipe portion counts of `adjust-portion`.

Verification

The feature is covered by `MealLogTest`, `MealEntryTest`, `MealEntryParserTest`, `MealCorrectionCommandsTest`, `MealLogViewerTest`, `RecordMealCommandTest`, `ViewMealsCommandTest`, `ParserRecordMealTest`, and console-workflow tests in `MishMashTest`. They check the totals (including values beyond the `int` range), the exact output layout, every invalid index in the table above, the empty-log and empty-book cases, that failed commands leave the log unchanged, and that the log is separate from the active plan. The manual checks are under Record and View Meals; the use cases are UC11, UC12, UC19 and UC20.

Diagram maintenance

The six editable diagram sources are under `docs/diagrams/record-meal`. Rendered SVGs are checked in under `docs/images/record-meal` so the diagrams display without a plugin. The source README gives the regeneration commands. Check every depicted call, branch and message against the code after any change, and update the sources and images together.

Command help

Section author: peilingggg

`HelpCommand` returns the supported command names, purposes, and formats to the application loop. Syntax constants are shared with parsers where available, reducing duplication when a command changes. The command does not read console input or change application data. Invalid commands are handled by the loop, which displays an error and allows another command. See the User Guide help instructions.

Confirming plan changes

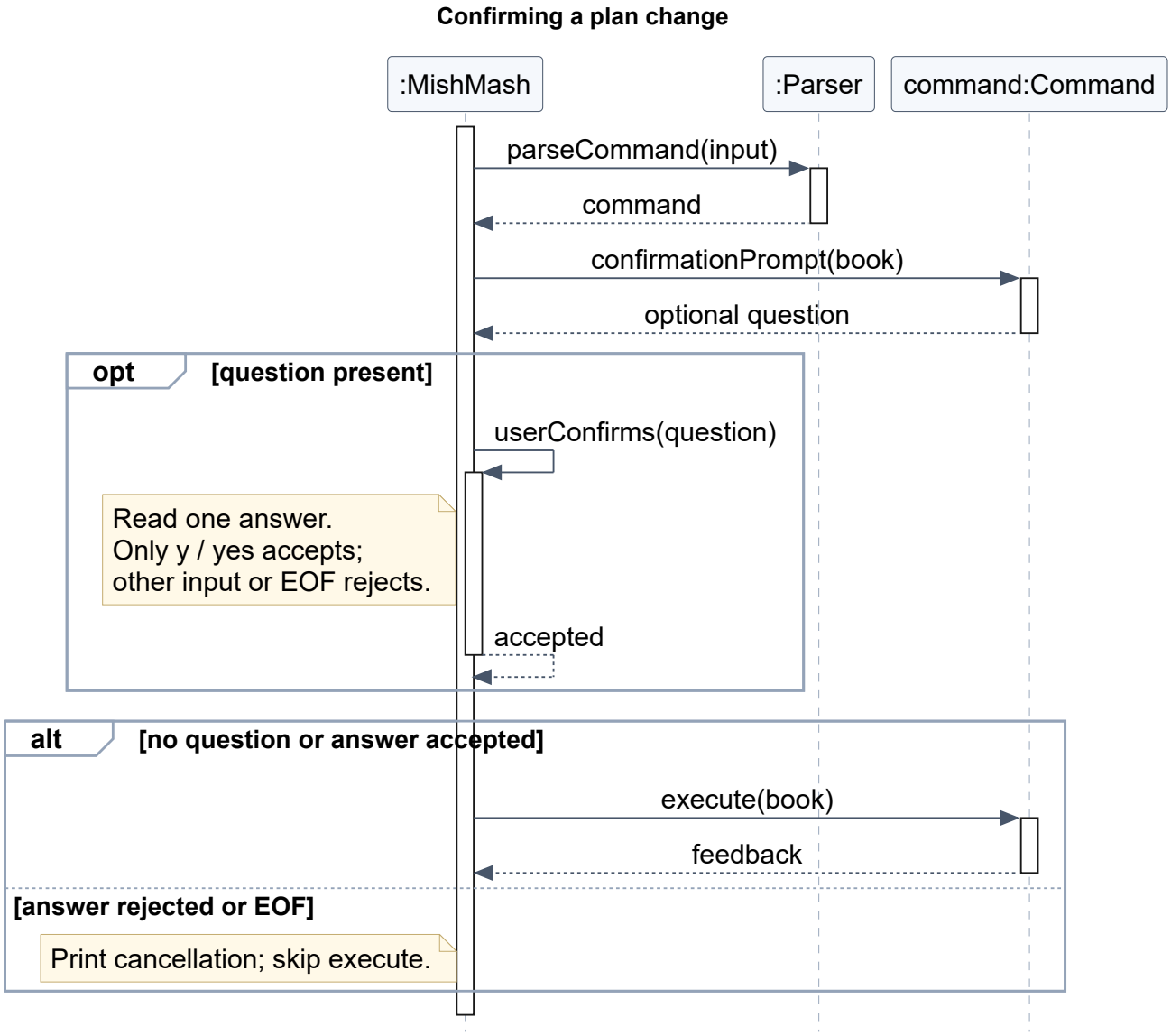
Section author: Rixte

Users confirm before an existing plan is cleared or replaced. This implements #61. See the User Guide for prompts and examples.

Responsibilities

Component	Responsibility
`Parser`	Validate syntax and create a command before an answer can be requested.
`Command.confirmationPrompt`	Return an optional question based on current state, without changing it.
`MishMash`	Read the answer, skip cancelled commands, and display feedback or errors.

Confirmation flow



Sequence diagram of shared plan-change confirmation

One shared flow covers acceptance, cancellation, and commands needing no answer. Scanner operations, printing, and command internals are condensed. PlantUML source

1. `MishMash.executeCommand` parses the input, then calls `confirmationPrompt(recipeBook)`.
2. An empty `Optional` allows execution immediately. Otherwise `userConfirms` reads one line.
3. Only `y` or `yes`, ignoring case and surrounding spaces, permits `execute`. Other answers and EOF print `Cancelled. Nothing was changed.` and skip execution.

Command	When a question is needed
`ResetPlanCommand`	A plan, generation request, or grocery snapshot exists, including an empty snapshot.
`GeneratePlanCommand`	A plan already exists.

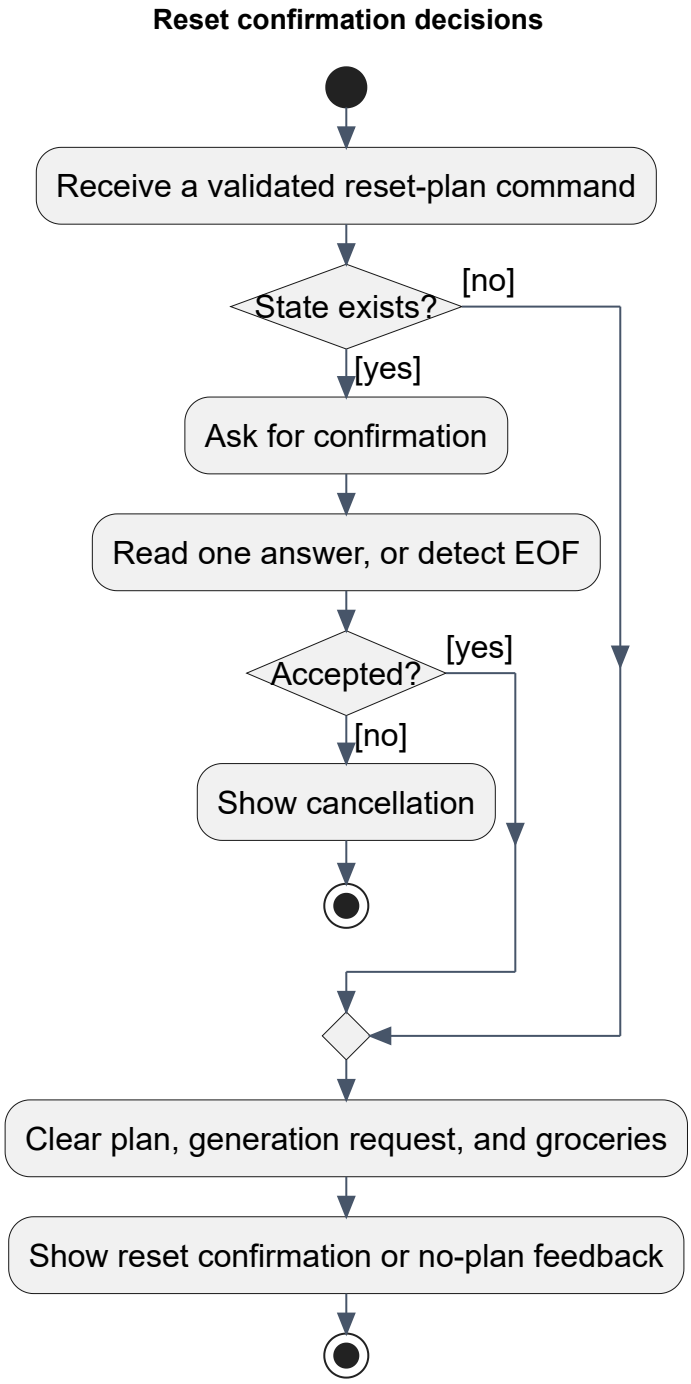
Command	When a question is needed
`SetDietCommand`	An existing plan conflicts with the proposed preferences.

Errors and boundaries

Invalid syntax is reported before confirmation. A command typed as an answer is consumed as that answer; users must enter it again at the command prompt. Prompt lookup reads the latest holder state and never generates replacements or changes preferences.

Agreement permits execution; it does not guarantee success. Failed replacement generation preserves the previous plan and groceries. Dietary repair saves the new preferences even when repair fails and clears the incompatible plan; its question explains that outcome. Cancelling preserves the old preferences, plan, and groceries.

Reset activity



Activity diagram of reset confirmation, cancellation, and clearing

This activity diagram starts after `reset-plan` arguments have been validated. Diamond decision nodes and mutually exclusive guards show when confirmation is needed and when execution is skipped. Answers are trimmed and compared without regard to case; EOF follows the rejection branch. Cancellation preserves all session data. The execution path assumes `clear()` and its postcondition checks succeed; recoverable failures are described in the reset section instead of repeating them here. Recipes, preferences, saved targets, and recorded meals are not modified. PlantUML source

Design choice and verification

The existing confirmation hook keeps console input in `MishMash` and commands independently testable. Putting a scanner inside every command would duplicate interaction logic. Direct callers of `execute` must perform any required confirmation themselves. Feasibility checks run after agreement, avoiding speculative generation and cached candidates during prompting.

`PlanConfirmationTest` covers accepted and rejected answers, EOF, invalid syntax, execution-time state lookup, preservation after cancellation, and confirmed failures. For a manual check, generate a plan and groceries, cancel `reset-plan`, and verify both views are unchanged. Repeat with `yes` and verify both report missing data.

Defensive checks and generation diagnostics

Section author: oilybikechain

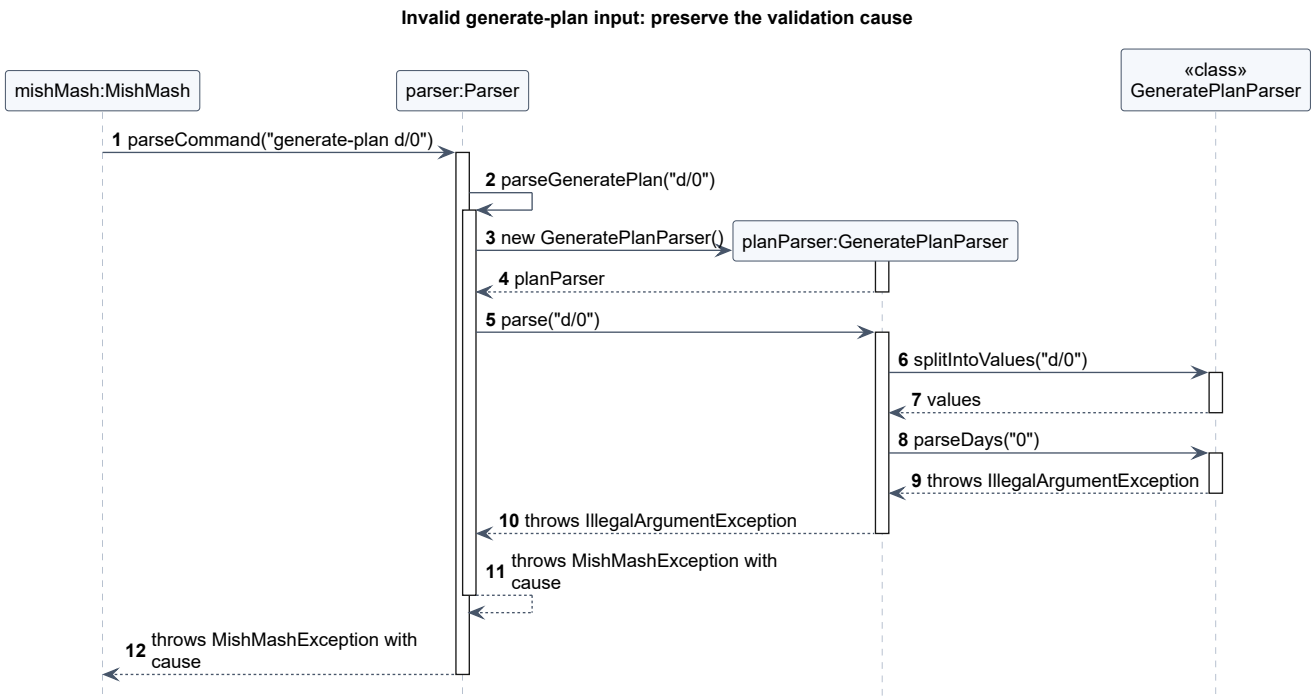
`GeneratePlanCommand` retains an `IllegalArgumentException` as the cause when wrapping a generation failure in `MishMashException`. Parser validation for `generate-plan`, `delete-recipe`, and `list-recipes` likewise preserves the cause while adding the relevant command usage. Users receive the existing recoverable error and can retry; developers can inspect the original exception through `getCause()` or the application's rejection log. This does not add new command syntax.

After a failure, `GeneratePlanCommand` also asks `PlanFailureAdvisor` (logic) to explain it. If the dietary preferences removed any recipes, the advisor reports how many remain. It then compares each target with the sum of the three lowest and three highest values available, which proves cheaply that a target cannot be reached. If every target passes that check, the targets conflict, so it suggests relaxing one. For recipe books large enough to be sampled it also suggests retrying, because `MealPlanGenerator.checksEveryCombination` is false and a valid combination could have been missed. The advice runs only on failure, so the generator's selection rules do not change. The error shows the `generate-plan` usage rather than every command, which keeps the advice visible. A more advanced alternative is to report the closest combination found, but that would have to keep extra state during generation for a similar suggestion.

`MealPlanGenerator` uses assertions to check two internal invariants: the returned number of days matches the request, and candidate construction starts with enough recipes. Normal argument validation still handles user mistakes; assertions diagnose programming errors. Tests enable assertions through the existing Gradle configuration. Enable them in a direct application run with `java -ea -jar build/libs/MishMash.jar` when investigating invariants.

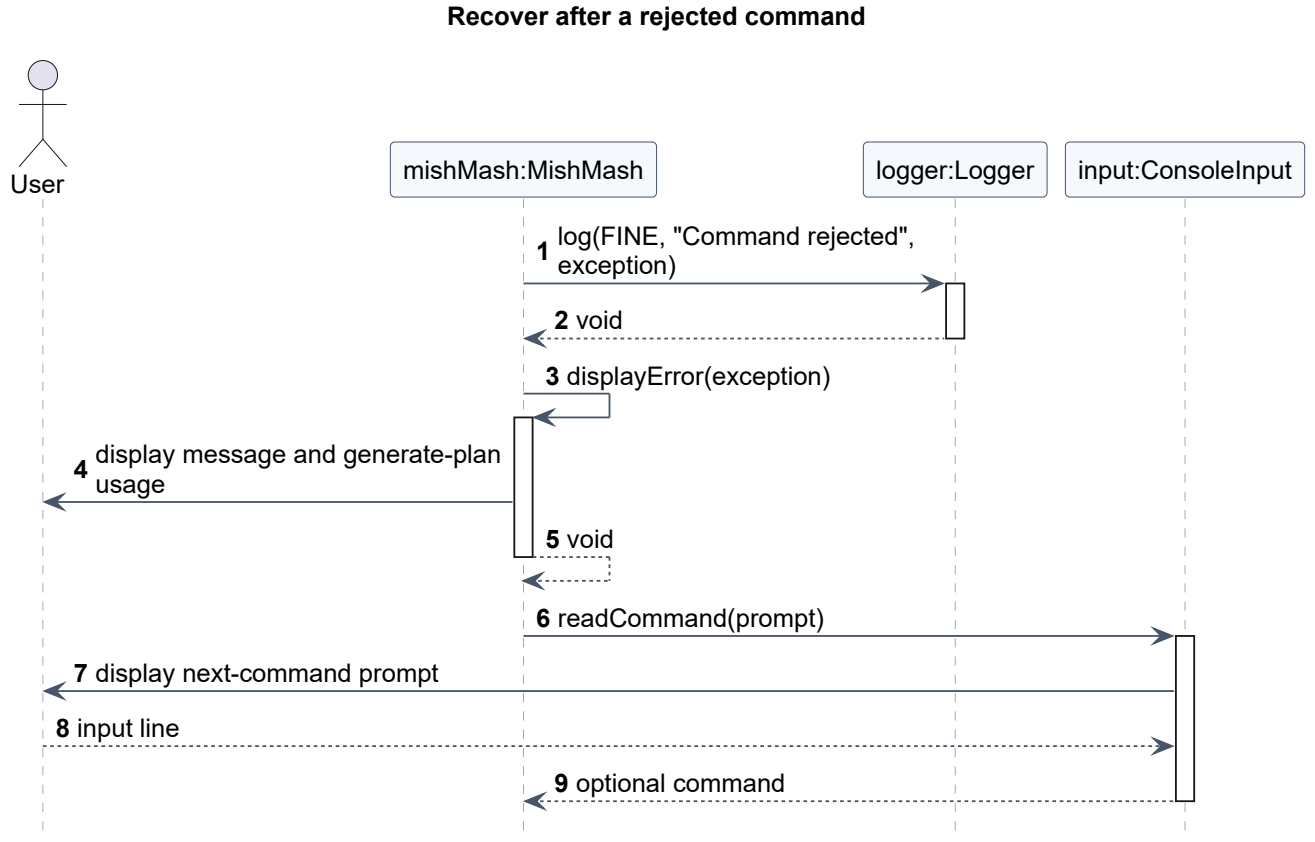
Generation logs at `FINE` record the requested day count, recipe count, whether targets were supplied, and how many candidate combinations matched. They use the existing logger configuration and do not record recipe contents. `ParserTest.parseCommand_invalidArguments_preservesCauseAndUsage` checks the cause and usage for all three parsing paths; `GeneratePlanCommandTest` verifies the original generation cause is retained. These changes strengthen diagnostics around teammates' commands rather than implementing those commands.

The validation and application-recovery interactions are separate so their exception returns and activation bars remain clear.



Sequence diagram of invalid generate-plan input preserving its validation cause.

`generate-plan d/0` fails while parsing, so no command is constructed or executed. The outer exception retains the validation cause. `MishMash` then logs the rejection, displays the recoverable error, and prompts for another command.



Sequence diagram of logging a rejected command and continuing.

Documentation, logging, testing, dev-ops

Documentation

The project documentation is maintained in Markdown under `docs`. The main documents are `DeveloperGuide.md` and `UserGuide.md`

Keep command formats, examples, prerequisites, and expected results consistent with the code. Document planned behaviour separately from available behaviour.

For RepoSense attribution and corrections, see Contributing to MishMash.

Logging

The application uses `java.util.logging`. Diagnostics are written to rotating files using the pattern `mishmash.%g.log`, with an approximate limit of 1,000,000 bytes per file and three files retained.

Levels include `FINE` for command diagnostics, `INFO` for lifecycle events and successful grocery-generation counts, `WARNING` for selected rejected operations or logging problems, and `SEVERE` for unexpected application failures. If the log file cannot be opened because of an I/O error, the application continues with its existing logging configuration.

Testing

The project uses JUnit Jupiter. Tests are under `src/test/java/mishmash` and mirror the production packages. They cover validation, commands, models, generation, display formatting, and full console interactions.

JLine's automated UI tests also belong in this JUnit suite. `ui/JLineConsoleInputTest` exercises real in-memory editing and uses a scripted reader for interrupt/EOF handling. `ui/CommandCompleterTest` covers command and prefix suggestions, while `ui/ConsoleInputsTest` covers adapter selection and fallback. `JLineSessionTest` is an integration test: it runs application commands, confirmations, and meal corrections through the real JLine input adapter. These tests do not require native console access. Actual terminal display, native-provider initialisation, and platform key sequences still need the manual terminal checklist.

Run verification from the repository root:

```
.\gradlew.bat check jacocoTestReport
```

On Linux or macOS:

```
./gradlew check jacocoTestReport
```

Use Java 25. JUnit reports appear at `build/reports/tests/test/index.html`, and Checkstyle reports are under `build/reports/checkstyle`. JaCoCo HTML coverage is at `build/reports/jacoco/test/html/index.html`. The `check` task includes tests, Checkstyle, and a coverage gate requiring 100% line and branch coverage for `ResetPlanCommand`, `GeneratePlanCommand`, and `SetDietCommand`; this is not a project-wide 100% requirement. `jacocoTestReport` generates the report separately. Tests enable assertions, and packaged-JAR storage tests use temporary working directories.

The add-recipe invalid-input dataset is `src/test/resources/add-recipe-invalid.tsv`.

Configuration

The current application has no user configuration file. Users supply recipe values, plan length, and optional daily nutrition targets through commands.

The supported plan length is 1 to 7 days. Each generated day contains exactly three different stored recipe entries. Generation uses a fixed target tolerance of 15%; users cannot change that tolerance through a command. In contrast, `check-shortfall` reports any amount strictly below the explicit targets supplied to that check.

DevOps

GitHub Actions runs `./gradlew check` for pushes and pull requests on Ubuntu, macOS, and Windows using Java 25. Check the resulting reports before merging changes.

The Gradle build packages the application as `build/libs/MishMash.jar`. Release documentation should use that filename and describe the features included in the actual release.

Appendix A: Product Scope

Target user profile: Users who prepare meals in advance, want to organise recipes and review calorie or macronutrient totals, and are comfortable entering text commands. Users provide their own recipes and nutrition information.

Value proposition: MishMash connects recipe management, meal selection, and grocery preparation in one workflow. It generates plans from user-entered recipes and combines ingredient quantities for the selected meals.

Current boundaries: Data is retained locally across sessions. Nutrition is entered by the user rather than calculated from an ingredient database, including after occurrence-specific ingredient customisation. Plans support positive whole-recipe portion counts, not fractional servings. Ingredient names and units are matched exactly; automatic unit conversion and grocery-item ticking are not supported.

Appendix B: User Stories

Priorities follow the AB3 convention: *** essential, ** useful, * optional. Grey rows are the v1.0 MVP; green rows are the v2.0 MVP. Text labels retain that meaning when colour is unavailable. Owners identify the feature contributors; the v2.0b and v2.1 owners are based on the authors of the merged feature PRs in the team repository. Uncoloured rows cover implemented v2.0b and v2.1 stories. Release labels follow the team milestones.

Release	Priority	As a …	I want to …	So that I can …	Owner	Current support
v1.0 MVP	***	Meal prep user	Add a recipe	Use my own dishes in meal plans	Pei Ling	`add-recipe`

Release	Priority	As a …	I want to …	So that I can …	Owner	Current support
v1.0 MVP	***	User managing recipes	Delete a recipe	Remove dishes I no longer need	Neil	`delete-recipe`
v1.0 MVP	***	User managing recipes	List recipes and inspect one recipe	Choose a dish and check its details	Neil	`list-recipes [INDEX]`
v1.0 MVP	***	Meal prep user	Generate a random meal plan	Reduce manual meal selection	Gautham	`generate-plan`
v1.0 MVP	***	Shopper	Generate a grocery list	Obtain combined ingredient quantities for my plan	Justin	`generate-list`
v1.0 MVP	***	Shopper	View the grocery list	See what to purchase	Samantha	`view-list`
v1.0 MVP	***	Meal prep user	View my meal plan	Review the meals selected for each day	Samantha	`view-plan`
v2.0 MVP	***	User with dietary requirements	Set dietary preferences and restrictions	Generate meals compatible with my declared requirements	Pei Ling	`set-diet`
v2.0 MVP	***	New user	Receive assistance from a help command	Find available commands and their formats	Pei Ling	`help`
v2.0 MVP	***	User tracking nutrition	Set a daily calorie target	Plan toward my desired calorie intake	Neil	`set-targets cal/…`
v2.0 MVP	***	User tracking nutrition	Set daily macronutrient targets	Plan toward my protein, carbohydrate, and fat goals	Neil	`set-targets p/… c/… f/…`
v2.0 MVP	***	User tracking intake	Record meals eaten	Keep track of my actual food intake	Gautham	`record-meal`
v2.0 MVP	***	User tracking nutrition	Calculate nutrition totals for each meal	Review each recorded meal's calories and macronutrients	Gautham	`view-meals`

Release	Priority	As a …	I want to …	So that I can …	Owner	Current support
v2.0 MVP	***	User tracking nutrition	Calculate daily nutrition totals	Review the combined nutrition of each planned day	Justin	`view-plan` for planned days; `view-meals date/YYYY-MM-DD` for recorded daily totals
v2.0 MVP	***	User tracking nutrition	Detect meal plans that exceed nutrition targets	Identify days above my exact generation targets	Justin	`view-plan` warnings
v2.0 MVP	***	User tracking nutrition	Detect meal plans that fall short of nutrition targets	Identify deficits against the targets I supply	Samantha	`check-shortfall`
v2.0 MVP	***	User managing recipes	Edit an existing recipe	Keep its details consistent with how I prepare it	Samantha	`edit-recipe`
v2.0b	**	Returning user	Reset my active plan and groceries	Start again while keeping recipes, settings, and intake records	Samantha	`reset-plan`
v2.0b	**	Meal prep user	Customise one planned meal's ingredients	Adapt one occurrence without editing its source recipe	Justin	`customize-meal`; nutrition is not recalculated from ingredients
v2.0b	**	Returning user	Confirm plan removal or replacement	Avoid accidentally losing my plan	Samantha	confirmation before reset, replacement generation, and incompatible dietary changes
v2.0b	**	User with dietary requirements	View my dietary settings	Check which restrictions apply	Pei Ling	`view-diet`
v2.1	**	Returning user	Retain my data between sessions	Continue planning after restarting	Justin	local JSON autosave, restoration, backup recovery, and process locking
v2.1	**	Meal prep user	Replace an individual planned meal	Adapt the plan to what I want to eat	Neil	`replace-meal`; one portion, source ingredients, and current dietary checks
v2.1	**	Meal prep user	Adjust a planned meal's portion	Scale nutrition and groceries to my intended quantity	Justin	`adjust-portion`; positive whole-recipe portions only
v2.1	**	Meal prep user	Understand unsuccessful plan generation	Know what to change before retrying	Gautham	generation failure advice for eligible recipes and nutrition targets

Release	Priority	As a …	I want to …	So that I can …	Owner	Current support
v2.1	**	User tracking intake	Review recorded meals by date	Understand intake for a selected day	Gautham	dated records, servings, and `view-meals date/YYYY-MM-DD`
v2.1	**	User tracking intake	Correct or remove recorded meals	Keep intake records accurate	Gautham	`edit-meal` and confirmed `delete-meal`
v2.1	**	User with dietary requirements	Clear my dietary restrictions	Generate without restrictions	Justin	`clear-diet`
v2.1	**	CLI user	Edit commands and use completion and history	Enter long commands more easily	Samantha	JLine editing, Tab completion, in-memory history, and plain-input fallback

Use cases

The actor is a user who has started a session and entered a name. MSS means Main Success Scenario.

Shared rules

- Invalid syntax or values produce command-specific error/usage feedback, preserve existing data, and allow another command.
- Viewing and checking commands do not mutate application data.
- Confirmation accepts only `y` or `yes`, ignoring case and surrounding whitespace. Any other answer, a blank answer, or EOF cancels without applying the pending change. EOF from piped or redirected input also ends the session. In interactive editing, Ctrl+D at an empty confirmation cancels the action and returns to the command prompt.
- Each MSS ends after its final step. Each extension ends the use case after its stated response unless it explicitly resumes an MSS step. The session then accepts another command, except after exit/EOF.

UC01 Add a recipe

MSS

1. The user enters `add-recipe` with a name, nutrition values, and at least one ingredient.
2. MishMash validates the input.
3. MishMash adds the recipe and displays its details.

Extensions

- 2a. A field is missing, blank, repeated, or invalid: reject the addition.

UC02 List recipes

MSS

1. The user enters `list-recipes`, optionally selecting an index.
2. MishMash displays the numbered names and calories of all recipes, or the full details of the selected recipe.

Extensions

- 1a. The index syntax is invalid: reject the command.
- 2a. The book is empty: MishMash displays guidance to add a recipe.
- 2b. The selected index does not exist: MishMash displays an index-range error.

UC03 Delete a recipe

MSS

1. The user enters `delete-recipe INDEX`.
2. MishMash checks the index and asks for confirmation, naming the selected recipe.
3. The user enters `y` or `yes`.
4. MishMash deletes the entry and displays its name and the remaining recipe count.

Extensions

- 1a. The index syntax is invalid: MishMash displays an error.
- 2a. The book is empty or the index does not exist: reject before confirmation.
- 3a. Confirmation is declined: cancel deletion.

UC04 Generate a meal plan

MSS

1. The user enters `generate-plan d/DAYS`, optionally including daily nutrition targets.
2. MishMash validates the requested days and targets.
3. If a plan is already active, MishMash asks whether to replace it and clear its groceries; the user confirms.
4. MishMash generates exactly three different recipe entries per day, meeting every applied target within 15%.
5. MishMash makes the new plan active, clears the previous groceries, and displays a confirmation.

Extensions

- 2a. Days are outside 1 to 7, a target is not a positive integer, or arguments are malformed: MishMash rejects the command.
- 3a. Confirmation is declined: cancel replacement.
- 4a. Fewer than three recipe entries are available, or no acceptable plan is found: MishMash displays an error, the likely cause, and what to try next. The previous plan and groceries remain available.

Recipes may be repeated on different days. With no targets, there are no nutrition constraints on the selection. Targets saved with `set-targets` (UC13) apply to every nutrient that the command leaves out, and a target typed in the command replaces the saved one for that plan.

UC05 View a meal plan

MSS

1. The user enters `view-plan`, optionally selecting `d/DAY_NUMBER`.
2. MishMash displays all days or the selected day, including each meal's nutrition and daily totals.

Extensions

- 1a. The day argument is malformed: reject the command.
- 2a. No plan is active: MishMash asks the user to generate one first.
- 2b. The selected day exceeds the active plan's length: MishMash displays a range error.

UC06 Generate a grocery list

MSS

1. The user enters `generate-list` without arguments.
2. MishMash combines ingredients from every meal occurrence in the active plan.
3. MishMash stores the completed groceries and confirms that they can be viewed using `view-list`.

Extensions

- 1a. Arguments are supplied: MishMash rejects the command without changing the stored list.
- 2a. There is no active plan: MishMash asks the user to generate one first.

Only exact ingredient-name and unit matches are combined. Different units or letter case remain separate entries.

UC07 View a grocery list

MSS

1. The user enters `view-list` without arguments.
2. MishMash displays the stored entries, their quantities and units, and the total number of entries.

Extensions

- 1a. Arguments are supplied: MishMash displays an error.
- 2a. No grocery list has been generated: MishMash asks the user to generate one first.

Viewing does not regenerate groceries. The displayed `[ ]` markers cannot be ticked through a command.

UC08 View help or exit

- Entering `help` displays command purposes and formats. Additional arguments produce an error.

- Entering `bye` displays the goodbye message and ends the session. The latest successfully saved user data is retained.

UC09 Check planned nutrition shortfalls

MSS

1. The user enters `check-shortfall` with at least one daily nutrition target.
2. MishMash validates the target arguments and obtains the active plan.
3. MishMash compares each day's planned totals with the supplied targets.
4. MishMash displays each day's deficient nutrients and amounts, or a no-shortfalls message for that day.

Extensions

- 2a. Targets are missing, repeated, unknown, or invalid: reject the check.
- 2b. No plan exists or the plan has no days: ask the user to generate a plan.

Omitted nutrients and excesses are not checked. All plan, recipe, and grocery data remains unchanged. A shortfall is any amount below a target, even within generation's tolerance.

UC10 Detect exceeded nutrition targets

MSS

1. The user enters `view-plan`, optionally selecting a day.
2. MishMash calculates each displayed day's nutrition totals and compares them with the saved generation targets.
3. For each exceeded target, MishMash displays a warning with the excess, actual total, and target below that day's totals.

Extensions

- 2a. A target is omitted, equal to the total, or above it: skip that nutrient and resume step 3 for the remaining nutrients.
- 2b. No generation request is retained, or the plan was generated without targets: only the normal plan display is shown.
- 2c. No plan exists or a selected day is invalid: the existing `view-plan` messages apply.

The comparison is strict and does not apply the generator's 15% tolerance. Viewing leaves the meals, targets, and groceries unchanged. A successful replacement uses its new targets; failed generation preserves the previous warnings. Dietary repair recalculates warnings using the repaired meals and the retained generation targets. Clearing the plan removes the warnings.

UC11 Record a meal eaten

MSS

1. The user enters `record-meal INDEX`, using an index shown by `list-recipes`, optionally with a date and servings.
2. MishMash validates the input and finds the recipe at that position.

3. MishMash adds the recipe, date and servings to the recorded meals. The date defaults to today and the servings default to 1.
4. MishMash displays the meal and the totals of all recorded meals.

Extensions

- 2a. The index is missing or is not a single positive whole number, or the date or servings are invalid: reject without recording.
- 2b. There are no recipes, or the index is larger than the number of recipes: reject without recording.

No confirmation is requested. Recording changes only the recorded meals; recipes, the meal plan, and groceries are unchanged. Recording the same recipe again adds another entry.

UC12 View recorded meals

MSS

1. The user enters `view-meals`, optionally with a date.
2. MishMash reads the recorded meals and sums their nutrition, for that date if one was given.
3. MishMash displays each meal in the order recorded with its calories, macronutrients, servings and date, followed by the totals.

Extensions

- 1a. The user adds anything other than one valid date: reject the command.
- 2a. No meal has been recorded: display `No meals recorded yet.`
- 2b. No meal was recorded on the given date: display `No meals recorded on DATE.`

Without a date, the totals cover every retained recorded meal.

UC13 Set daily targets

MSS

1. The user enters `set-targets` with one or more of `cal/`, `p/`, `c/`, and `f/`, each followed by a positive whole number or `none`.
2. MishMash validates every argument.
3. MishMash replaces the target of each nutrient given a number, removes the target of each nutrient given `none`, and keeps the targets of the other nutrients.
4. MishMash displays the targets now saved, or `All daily targets cleared.` if none remain.

Extensions

- 2a. No argument is given, a prefix is unknown or repeated, a value is missing, or a value is neither `none` nor a positive whole number up to 2147483647: reject the update.
- 3a. The result equals the targets already saved: display `Daily targets are unchanged. Nothing was changed.` and save nothing.

The command does not change the recipe book or an existing meal plan. The saved targets are used by later `generate-plan` commands (UC04) and remain available after restarting from the same working directory.

UC14 View daily targets

MSS

1. The user enters `view-targets`.
2. MishMash displays the saved target of each nutrient, with `not set` for a nutrient that has none.

Extensions

- 1a. The user adds an argument: reject the command.
- 2a. No target is saved: display `No daily targets saved. Use 'set-targets' to set them.`

UC15 Set dietary preferences

MSS

1. The user enters `set-diet` with at least one supported prefixed field.
2. MishMash validates the complete replacement profile.
3. If any planned day conflicts, MishMash asks for confirmation. After agreement, it preserves compatible planned days and replaces conflicting days using the original targets.
4. MishMash rebuilds existing groceries after repair, saves the profile, and displays the outcome.

Extensions

- 2a. Syntax or values are invalid: reject the command.
- 3a. No active plan exists: save the profile for future generation and display the outcome.
- 3b. The complete plan is compatible: save the profile, preserve the plan and groceries, and display the outcome.
- 3c. The user declines confirmation: preserve the profile, plan, and groceries.
- 3d. Replacement fails or the original request is unavailable: save the profile, clear the plan and groceries, and explain how to generate again.

UC16 View dietary preferences

MSS

1. The user enters `view-diet` without arguments.
2. MishMash reads the current session profile and displays the diet and both exclusion lists.

Extensions

- 1a. Arguments are supplied: reject the command.
- 2a. No restrictions are active: display `none` for the diet and both lists.

UC17 Clear dietary preferences

MSS

1. The user enters `clear-diet` without arguments.
2. MishMash resets the complete dietary profile to unrestricted and displays confirmation.
3. MishMash preserves recipes, the active plan and its customisations, groceries, saved targets, and recorded meals.

Extensions

- 1a. Arguments are supplied: reject the command.
- 2a. The profile is already unrestricted: report that nothing changed and preserve all data.

UC18 Edit a recipe

MSS

1. The user lists recipes and chooses an index.
2. The user enters `edit-recipe INDEX` with at least one replacement field.
3. MishMash validates the input, replaces the recipe at that index, and displays its updated details.

Extensions

- 2a. The index or a supplied field is invalid, or no edits are supplied: reject the edit.

Existing meal plans and recorded meals keep their previous recipe snapshots.

UC19 Correct a recorded meal

Precondition: At least one meal has been recorded.

MSS

1. The user enters `edit-meal MEAL_NUMBER` with a new date, new servings or both, using a number shown by `view-meals`.
2. MishMash validates the input and replaces the meal, keeping its number.
3. MishMash displays the corrected meal and the totals for its date.

Extensions

- 1a. The number or a supplied field is invalid, or no field is supplied: reject the correction.
- 2a. The number is larger than the number of recorded meals: reject the correction.

UC20 Delete a recorded meal

Precondition: At least one meal has been recorded.

MSS

1. The user enters `delete-meal MEAL_NUMBER`, using a number shown by `view-meals`.
2. MishMash describes the meal and asks for confirmation.
3. The user confirms.
4. MishMash removes the meal, moves later meals up by one, and displays the deleted meal and the totals remaining for its date.

Extensions

- 1a. The number is missing, invalid or larger than the number of recorded meals: reject before confirmation.
- 3a. The user does not confirm: cancel deletion.

UC21 Reset the active plan

MSS

1. The user enters `reset-plan`.
2. MishMash asks for confirmation when plan state exists.
3. The user confirms with `y` or `yes`.
4. MishMash clears the plan, retained generation request, and groceries, preserving recipes, dietary settings, saved targets, and recorded meals.

Extensions

- 1a. Arguments are supplied: reject the command.
- 2a. No plan state exists: report that there is no active plan to reset.
- 3a. Confirmation is declined: cancel reset.

UC22 Adjust a planned meal's portions

MSS

1. The user enters `adjust-portion d/DAY m/MEAL portions/COUNT`.
2. MishMash validates the positions and positive whole-recipe portion count.
3. MishMash updates only that occurrence, scales its nutrition and ingredient quantities, and clears existing groceries.

Extensions

- 2a. No plan exists: ask the user to generate one first.
- 2b. Syntax, positions, count, or resulting totals are invalid: reject the change.
- 3a. The count is unchanged: report a no-op and preserve groceries.

The source recipe, other occurrences, saved targets, dietary profile, and recorded meals are unchanged. Run `generate-list` to obtain updated groceries.

UC23 Customise a planned meal's ingredients

MSS

1. The user enters `customize-meal d/DAY m/MEAL i/INGREDIENT:QUANTITYUNIT`, repeating `i/` for the complete replacement list.
2. MishMash validates the selected occurrence, positive quantities, and current dietary compatibility.
3. MishMash replaces that occurrence's one-portion ingredients and clears existing groceries.

Extensions

- 2a. No plan exists: ask the user to generate one first.
- 2b. Arguments, positions, quantities, or dietary compatibility are invalid: reject the change.
- 3a. The effective ingredients are unchanged: report a no-op and preserve groceries.
- 3b. Ingredients match the source recipe again: remove the customisation, then complete step 3.
- 3c. Named-diet or avoided-category rules are active: complete step 3 with a warning that replacement names are not reclassified; the occurrence keeps its source food categories.

Nutrition and portion count remain unchanged. Grocery generation scales the replacement quantities by the portion count. The source recipe and other occurrences remain unchanged.

UC24 Replace a planned meal

MSS

1. The user enters `replace-meal d/DAY m/MEAL r/RECIPE_INDEX`.
2. MishMash validates the positions, recipe index, current dietary compatibility, and resulting totals.
3. MishMash replaces only that occurrence with the chosen recipe at one portion and its source ingredients, then clears existing groceries.

Extensions

- 2a. No plan exists: ask the user to generate one first.
- 2b. Arguments, positions, recipe selection, compatibility, or totals are invalid: reject the change.
- 3a. The occurrence already uses that exact recipe at one portion without customisation: report a no-op and preserve groceries.

Other occurrences, source recipes, recorded meals, and settings remain unchanged. Replacement does not enforce generation’s distinct-recipe rule or target tolerance; displayed totals and warnings reflect the changed plan.

Appendix C: Non-Functional Requirements

- The application requires Java 25 and provides a command-line interface.
- The project targets Windows, macOS, and Linux, with verification configured on all three operating systems.
- Ordinary invalid command input must allow the session to continue and preserve existing data.
- Decimal ingredient quantities must retain their numeric precision.
- Generating meals and groceries must leave the stored recipe entries unchanged.
- The normal application workflow does not require a network connection or an external database.
- Snapshot loading must reject files larger than 10 MiB without partially publishing application state.
- Concurrent production instances using the same data directory must be prevented by an exclusive process lock.

These are compatibility, integrity, and resource constraints. No measured recipe-capacity or response-time guarantee is specified; the generator’s limits of 1,000,000 checked combinations and 5,000 kept matches are algorithm limits, not a performance benchmark.

Appendix D: Glossary

Term	Meaning
Recipe	A named entry with calorie and macronutrient values and required ingredients.
Recipe book	The ordered recipes available in the current session.
Recipe index	A recipe's current displayed position, starting at 1.
Meal plan	Recipes assigned to one or more numbered days.
Active plan	The single plan currently available for viewing and grocery generation.
Macronutrient	Protein, carbohydrate, or fat, measured in grams.
Daily target	An optional desired daily total for calories or a macronutrient.
Grocery list	The combined ingredient quantities generated from the active plan.
Session	One running instance of the application. Successfully saved application data remains available across sessions.

Appendix E: Instructions for Manual Testing

Each subsection starts in a fresh temporary working directory with the sample recipes where required. Follow steps within a subsection in order.

For every rejected command, verify unchanged data and a successful retry. Answer `yes` to replacement/reset/dietary confirmations unless testing cancellation; this includes generation attempts expected to fail. First generation and malformed commands need no answer. See Reset and Confirm Plan Changes for cancellation checks.

Launch

1. Install Java 25. To test this checkout, run `./gradlew shadowJar` (Windows: `.\gradlew.bat shadowJar`) from the repository root, then copy `build/libs/MishMash.jar` into an empty temporary folder and open a terminal there. To test a published release instead, download its JAR from the team's Releases page and limit checks to the features listed for that release.
2. Run `java -ea -jar "MishMash.jar"`, replacing `MishMash.jar` with the downloaded file's actual name if it differs. The `-ea` option enables the application's internal assertion checks during manual testing.
3. Confirm that the banner and name prompt appear. Enter a name and check that a command prompt follows.

Sample Recipes

Enter each command on one line:

```
add-recipe n/Chicken Rice cal/600 p/40 c/70 f/15 i/Chicken:200g i/Rice:150g
add-recipe n/Tofu Rice cal/500 p/25 c/60 f/12 i/Tofu:200g i/Rice:100g
add-recipe n/Oats cal/400 p/15 c/50 f/10 i/Oats:80g i/Milk:200ml
```

With exactly these three entries, each generated day contains all three. Meal order may vary. Daily totals are 1500 kcal, 80g protein, 180g carbs, and 37g fats.

Add a Recipe

Input variation on the first sample recipe	Expected result
Unchanged sample	One addition with nutrition and ingredients.
Omit both `i/` fields	Error requiring an ingredient; no addition.
`cal/abc` or `cal/2147483648`	Integer/range error.
`cal/0`	Accepted; zero nutrition is valid.
`i/Chicken:0g`	Rejected; quantity must be positive.
`i/Chicken:200.5g`	Exact decimal retained.
A second `n/` field	Rejected; repeated `i/` remains valid.

List Recipes

After loading the samples:

- Enter `list-recipes`. Expect three numbered entries in insertion order, showing names and calories.
- Enter `list-recipes 2`. Expect Tofu Rice's full nutrition and ingredients.
- Enter `list-recipes 4`. Expect an out-of-range error.
- In a fresh empty session, enter `list-recipes`. Expect guidance to add a recipe.

Delete a Recipe

After loading the samples:

- Enter `delete-recipe 2`. Expect a confirmation naming Tofu Rice. Answer `n`; verify that nothing changed.
- Repeat and answer `yes`. Expect deletion, a remaining count of 2, and renumbering of the later recipe.
- Enter `delete-recipe 0` or an index larger than the current book size. Expect an error without deletion.

Generate a Meal Plan

Use a fresh session containing exactly the three sample recipes:

- Enter `generate-plan d/3`. Expect confirmation of a three-day plan. Check with `view-plan` that each day has exactly three meals.
- Enter `generate-plan d/2 cal/1500 p/80 c/180 f/37`. Expect success. Each day's totals match the sample totals exactly.
- Enter `generate-plan d/8`. Expect rejection because the day limit is 1 to 7.
- Enter `generate-plan d/1 cal/1`. Expect failure to find acceptable meals. Verify that the previous plan remains available.
- In a separate session containing fewer than three recipes, try `generate-plan d/1`. Expect a minimum-recipe error.

View a Meal Plan

Generate a two-day plan first.

Command	Expected result
`view-plan`	Both days, meal nutrition, and daily totals.
`view-plan d/1`	Day 1 only.
`view-plan d/3`	Day-range error.
`view-plan` in a separate empty session	Generate-first guidance.

Combined Daily Nutrition Totals

Using exactly the three sample recipes, run `generate-plan d/2` and `view-plan`. Both days must show `Daily Totals: 1500 kcal | Protein: 80g | Carbs: 180g | Fats: 37g` regardless of meal order. Run `view-plan d/2`: its totals must match the second day's previous display, rather than adding the two days together. Repeating the command must not change the plan or totals.

The combined `NutritionTotals` API's empty-day, repeated-meal, and integer-overflow cases are tested by `DayPlanTest`; ordinary generation always supplies three distinct recipes per day.

Nutrition Target Warnings

In a fresh session with no dietary restrictions, enter:

```
add-recipe n/A cal/500 p/20 c/50 f/10 i/Rice:100g
add-recipe n/B cal/600 p/30 c/60 f/15 i/Rice:100g
add-recipe n/C cal/700 p/40 c/70 f/20 i/Rice:100g
generate-plan d/2 cal/1700 p/85 c/180 f/45
view-plan
view-plan d/2
```

Each day must total 1800 kcal, 90 g protein, 180 g carbohydrates, and 45 g fats, regardless of meal order. Expect exactly these warnings below each displayed day's totals:

```
Warning: Calories exceed target by 100 kcal (1800 kcal actual; 1700 kcal target).
Warning: Protein exceeds target by 5 g (90 g actual; 85 g target).
```

`view-plan d/2` must show only day 2. Then try `generate-plan d/1 cal/1`: generation must fail and `view-plan` must still display the previous two days and their warnings. Run `check-shortfall cal/1900` and verify that a subsequent `view-plan` retains those warnings. Finally run `generate-plan d/1` and `view-plan`: the replacement must show one day with no warnings because no targets were supplied.

Check Nutrition Shortfalls

1. In a fresh session, enter `check-shortfall cal/2000` before generating a plan. Expect `[NOTE] No active meal plan. Use 'generate-plan' first.` and another `mishmash >` prompt.
2. Load exactly the three sample recipes, enter `generate-plan d/2`, and then `generate-list`. Each day's planned totals are 1500 kcal, 80 g protein, 180 g carbohydrates, and 37 g fat.
3. Enter `check-shortfall cal/2000 p/100 c/200 f/60`. Both Day 1 and Day 2 should report deficits of 500 kcal, 20 g protein, 20 g carbohydrates, and 23 g fat. For each day, verify that the displayed total and target are included alongside the deficit.
4. Enter `check-shortfall cal/1500 p/80 c/100`. Both days should display `No shortfalls for the supplied targets.` Calories and protein exactly match, carbohydrates exceed the target, and fat is omitted.
5. Enter `check-shortfall cal/1501`. Expect a 1 kcal deficit for each day even though that difference is within the generation tolerance.
6. Try `check-shortfall`, `check-shortfall cal/0`, `check-shortfall p/1.5`, `check-shortfall cal/2000 cal/2100`, and `check-shortfall d/1`. Each should produce a recoverable error with the check-shortfall format. Retry a valid check afterwards.
7. Enter `view-plan` and `view-list`. The two-day plan and all five grocery entries should be unchanged. Their quantities should match the table in Generate a Grocery List. Repeat a check to verify that results are stable, and enter `help` to confirm the new command is listed.

The CLI cannot generate an empty day or an empty plan; those defensive cases, null dependencies, and integer-overflow boundaries are covered by the automated tests.

Generate a Grocery List

With exactly the sample recipes, generate a two-day plan and enter `generate-list`. Expect:

```
[DONE] Grocery list generated. Use 'view-list' to view it.
```

Then enter `view-list` and verify:

Ingredient	Quantity for two days
Chicken	400g
Rice	500g
Tofu	400g
Oats	160g

Ingredient	Quantity for two days
Milk	400ml

There should be five displayed entries. Their order follows the meals' first appearances and may vary. The generation confirmation itself does not display an item count.

- Run `generate-list` again. Expect unchanged quantities rather than doubled totals.
- Enter `generate-list extra`. Expect rejection without changing the stored list.
- In a fresh session without a plan, enter `generate-list`. Expect guidance to generate a plan first.

For aggregation edge cases, use the User Guide's ingredient combining example in a fresh session. Expect exactly `Rice: 0.6g`, `Rice: 2kg`, and `rice: 100g`, with no rounding, unit conversion, or case normalisation. Repeat `generate-list` and check that quantities stay the same. Then enter `generate-list extra` and `view-list`: syntax rejection must leave those entries unchanged. Generate a replacement plan and confirm that groceries must be regenerated before viewing.

Defensive Diagnostics and Recovery

With a plan and grocery list already generated, enter `generate-plan d/0`, `list-recipes 0`, and `delete-recipe 0`. Each must display its own command format, then allow another command. Verify with valid viewing commands that no recipes, meals, or groceries changed. Attempt `generate-plan d/1 cal/1` with the standard sample recipes and confirm that generation fails while the previous plan and grocery list remain available.

Exception causes and assertion invariants are developer diagnostics, not extra console output. Verify cause preservation with `ParserTest` and `GeneratePlanCommandTest` under Java 25 with assertions enabled. Inspect `FINE` generation records only when that level is enabled in the existing logging configuration; absence of a `FINE` record at a higher log level is expected.

View a Grocery List

Action	Expected result
`view-list` after grocery generation	Quantities, units, `[ ]` markers, and entry count.
`view-list extra`	Format error.
Generate a replacement plan, then `view-list`	Generate-list-first guidance.
Generate groceries, attempt `generate-plan d/1 cal/1`, then `view-list`	Failed generation preserves groceries.

Record and View Meals

Start empty, then follow the rows in order. Dates default to today and servings to 1.

Action	Expected result
`record-meal 1`, then `view-meals` before adding recipes	No-recipes error and no recorded meals.

Action	Expected result
Load sample recipes; `record-meal 1`	Chicken Rice; totals 600 kcal, 40g protein, 70g carbs, 15g fats.
`record-meal 2`, then `view-meals`	Two numbered entries with dates/servings; totals 1100 kcal, 65g protein, 130g carbs, 27g fats.
`record-meal 1` again	Third entry; totals 1700 kcal, 105g protein, 200g carbs, 42g fats.
Add `add-recipe n/Black Coffee cal/5 p/0 c/0 f/0 i/Coffee:1cup`; `record-meal 4`	Totals grow by 5 kcal only.
`record-meal 1 date/2026-03-02 s/2`; `view-meals date/2026-03-02`	Only that entry, doubled nutrition, original meal number, and date-specific total.
`edit-meal 1 s/2`	Updated confirmation; meal nutrition doubles.
`delete-meal 1`, answer `n`; repeat and answer `y`	Cancellation preserves log; confirmation deletes and renumbers remaining entries.
Try `record-meal`, `record-meal 0`, `record-meal 5`, `record-meal two`, `record-meal -1`, `record-meal 1 2`, and `view-meals now`	Recoverable errors; log unchanged.
`edit-recipe 2 cal/100`; `delete-recipe 1` and confirm	Recorded nutrition snapshots remain unchanged.
`generate-plan d/1`; view meals and plan	Recorded and planned meals remain independent.
`help`; exit and restart from the same directory	All four meal-log commands listed; remaining entries restore dates, servings, order, and recipe snapshots.

Set and View Daily Targets

Use exactly the sample recipes in a fresh directory; follow rows in order.

Action	Expected result
`view-targets`	No saved targets; guidance to use `set-targets`.
`set-targets cal/1500 p/80`; `set-targets f/37`; `view-targets`	Earlier values retained: 1500 kcal, 80g protein, 37g fats; carbs not set. Existing plan unchanged.
`generate-plan d/1`; `view-plan`	Applied-target summary; all three recipes, 1500 kcal/80g protein/37g fats, no warnings.
`generate-plan d/1 cal/1400`; view plan and targets	Applied calories 1400 and a 100 kcal excess warning; saved calories remain 1500.
`set-targets p/200`; `generate-plan d/1`	Failure with effective targets and saved-target guidance; previous plan preserved.
`generate-plan d/1 p/80`	Success using 1500 kcal, 80g protein, 37g fats.

Action	Expected result
`set-targets cal/none c/none f/none p/none`; view targets; generate again	All cleared; no saved targets and no applied-target summary.
Try bare `set-targets`; then try it with `cal/0`, `cal/-5`, `cal/abc`, `cal/1.5`, `cal/`, `x/5`, `2000`, or `cal/1 cal/2`; also try `view-targets now`	Recoverable error with matching usage.
`set-targets p/5 f/oops`; `view-targets`	Invalid update is atomic: protein was not saved.
`set-targets cal/NONE`; `set-targets CAL/5`; `Set-Targets cal/5`	Uppercase NONE accepted (a no-change note when no calorie target was saved); uppercase prefix and command rejected.
`set-targets cal/2000`; exit and restart; `view-targets`	2000 kcal restored.

Help and Recovery

- Enter `help`. Expect all supported command names, purposes, and formats.
- Enter `help extra`, a blank line, or an unknown command. Expect an error followed by another prompt.
- After an invalid command, enter a valid command and verify that it executes normally.

Adjust, Customise, and Replace Planned Meals

Use a fresh temporary working directory, load exactly the three sample recipes, and enter `generate-plan d/1`, `generate-list`, and `view-plan`. Note which recipe occupies Meal 1; random ordering is expected.

1. Enter `adjust-portion d/1 m/1 portions/2`. Check that Meal 1's nutrition doubles and the day's totals increase by one copy of that recipe's nutrition. Other meals and `list-recipes` details must stay unchanged. `view-list` must require grocery generation again.
2. Run `generate-list`. Quantities must equal the original one-day groceries plus one copy of Meal 1's source ingredients. Repeat the same portion command: expect a no-op and preservation of these groceries.
3. Enter `customize-meal d/1 m/1 i/TestRice:25g`. Nutrition and two portions must remain unchanged; the previous groceries must be cleared. Run `generate-list`: Meal 1 contributes `TestRice:50g` and no longer contributes its original ingredients. Ingredients shared with other meals must retain those other meals' contributions.
4. Repeat the identical customisation: expect a no-op with groceries preserved. Check `list-recipes` to confirm that the source recipe's ingredients are unchanged.
5. Enter `replace-meal d/1 m/1 r/1`. Expect Meal 1 to become Chicken Rice at one portion, with Chicken 200g and Rice 150g as its ingredients, without the customisation. The other two meals remain unchanged, even if one already uses Chicken Rice. Groceries must be cleared.
6. Run `generate-list`, repeat the same replacement, and verify a no-op that preserves groceries. Daily totals must equal the three displayed meals; replacement need not retain generation's target tolerance or recipe uniqueness.

7. Try `adjust-portion d/1 m/1 portions/0`, `adjust-portion d/1 m/1 portions/1.5`, `customize-meal d/1 m/1 i/TestRice:0g`, `replace-meal d/1 m/1 r/999`, and `replace-meal d/2 m/1 r/1`. Each must be rejected while the plan, groceries, recipes, and recorded meals stay unchanged.
8. In a separate empty session, try valid forms of all three commands before generating a plan. Expect generate-first guidance.
9. For dietary checks, use the classified recipes from Successful repair and grocery refresh. After applying vegetarian restrictions, try replacing a meal with Chicken A, or exclude `Forbidden` with `set-diet diet/vegetarian exclude/Forbidden` and then try `customize-meal d/1 m/1 i/Forbidden:1g`. Rejections must preserve the plan and groceries. Customising to an allowed new ingredient under vegetarian rules must warn that names are not reclassified, keep source food categories, and leave nutrition unchanged.

Reset and Confirm Plan Changes

Use a fresh temporary directory. Add these classified versions of the sample recipes (do not also add the unclassified versions):

```
add-recipe n/Chicken Rice cal/600 p/40 c/70 f/15 i/Chicken:200g i/Rice:150g food/poultry
add-recipe n/Tofu Rice cal/500 p/25 c/60 f/12 i/Tofu:200g i/Rice:100g food/soy,legume
add-recipe n/Oats cal/400 p/15 c/50 f/10 i/Oats:80g i/Milk:200ml food/oat,milk,lactose
set-targets cal/1500
set-diet exclude/Forbidden
record-meal 1
generate-plan d/1
generate-list
```

All three recipes are classified and contain no `Forbidden` ingredient, so they remain eligible under the ingredient exclusion. Each day's total is 1500 kcal, matching the saved target. Keep the general Sample Recipes unchanged for tests that require unclassified recipes.

1. Enter `generate-plan d/2` and answer `n`. Check that the original one-day plan and groceries remain. Repeat with a blank answer and an unrecognised answer; neither authorises replacement.
2. Enter `generate-plan d/2` and answer `yes`. Expect a two-day plan and cleared groceries. Generate groceries again.
3. Enter `reset-plan` and answer `n`. Check all views are unchanged. Enter `reset-plan extra`: expect syntax rejection without a confirmation.
4. Enter `reset-plan` and answer `y`. Both `view-plan` and `view-list` must report absent data. `list-recipes`, `view-diet`, `view-targets`, and `view-meals` must retain their previous values.
5. Enter `reset-plan` again. Expect a no-plan note without a confirmation.
6. Regenerate a plan and groceries, then exit with `bye`. Relaunch using piped or redirected input containing a name followed by `reset-plan`, with no confirmation answer. EOF must cancel the pending reset and end the session. Restart from the same working directory: the plan and groceries must still exist. Repeat for replacement generation, using `generate-plan d/2` as the final input command. This check uses stream EOF; interactive Ctrl+D at an empty confirmation instead cancels the action and returns to the command prompt.
7. Test dietary-change cancellation and failed repair separately with Set, View, and Clear Dietary Preferences.

Session Data

Retention across normal exits

Create recipes, a plan with adjusted portions and custom ingredients, groceries, recorded meals, preferences, and targets. Exit with `bye`, restart from the same working directory, and verify that all views restore their earlier values. Repeat using EOF instead of `bye`.

Missing, damaged, and locked files

Use disposable working directories only. Close MishMash before copying or editing snapshots. The default files, relative to the directory from which Java is launched, are:

File	Role
`data/mishmash.json`	Current validated snapshot.
`data/mishmash.backup.json`	Previous valid snapshot; recovery can therefore lose the most recent change.
`data/mishmash.lock`	Process lock; the operating-system lock, not the file's mere existence, prevents concurrent use.

Create a reproducible seed in an empty directory:

1. Launch MishMash, enter a name, and add the Chicken Rice sample recipe.
2. Add the Tofu Rice sample recipe, then exit with `bye`.
3. Check that the primary contains both recipes and the backup contains Chicken Rice alone. Copy this closed seed directory for each independent case below.

To corrupt a file, replace its entire contents with `{broken` in a text editor. Preserve copies of the valid seed files outside each case directory.

Case and action before launch	Expected result after launch
Leave both seed snapshots unchanged.	Both recipes restore without recovery feedback.
Rename the primary to `mishmash.saved.json`; leave the valid backup.	Warning about the missing primary; Chicken Rice restores from backup and a valid primary is recreated. The renamed file is not loaded.
Corrupt only the primary.	Recovery warning; Chicken Rice restores from backup. Damaged primary is preserved as `mishmash.corrupt-<timestamp>-<uuid>.json`.
Corrupt both primary and backup.	Warning and empty application state. Both originals remain under unique `mishmash.corrupt-...` and `mishmash.backup.corrupt-backup-...` JSON names.
In another empty directory, launch without either snapshot.	Normal empty state; no damage-recovery warning.

Case and action before launch	Expected result after launch
Keep one instance running on the seed; launch a second terminal in the same working directory and run the same JAR.	Second launch reports `MishMash is already using this data folder. Close the other instance first.` and stops before the command loop. The first instance stays usable.

After each recovery, use `list-recipes` to verify the exact contents, exit, and restart again to check stable restoration. After the lock case, close the first instance and confirm that a new instance starts normally; do not delete the lock file while a process holds it.

Filesystem-access failures must stop startup rather than quarantine valid data. Permission failures, schema upgrades, interrupted writes, and save-retry behaviour are covered by the temporary-directory and packaged-JAR storage tests.

Interactive Command Entry

Build the current JAR with Java 25 using `./gradlew shadowJar` (Windows: `.\gradlew.bat shadowJar`). Copy `build/libs/MishMash.jar` into a disposable empty working directory, open a terminal there, and launch it with `java -jar "MishMash.jar"`. Record the operating system and terminal used; repeat on each supported platform before release.

1. At the name prompt, type `check-sh` and press Tab. Check that it does not become `check-shortfall`. Clear the line and enter your name, then type `check-sh` at the command prompt and press Tab. It should complete the command. Type `ca`, press Tab, then type `2000`. Check that the line reads `check-shortfall cal/2000`. Enter it; the no-plan message should appear, with another command prompt.
2. Recall that command with Up, move through it with Left/Right and Home/End, and change `2000` to `2100` using Backspace/Delete. Check that the cursor and displayed text agree, then submit it. Start a new draft, press Up then Down, and check that Down restores the draft.
3. Type `reset-plan` without pressing Enter, then press Ctrl+C. Check that it is discarded and the session remains usable. Add a recipe using the User Guide example, enter `delete-recipe 1`, then press Ctrl+C at the confirmation. Check cancellation and verify the recipe remains with `list-recipes`.
4. Repeat `delete-recipe 1` and answer `no`. Use Up/Down at the next command prompt to check that commands are recalled, while your name and `no` are absent. Type `bye`, restart in the same directory, and check that the recipe is retained while the earlier session's command history is absent.
5. Enter `delete-recipe 1` and press Ctrl+D on the empty confirmation line. Check cancellation and the return to the command prompt. Verify the recipe remains with `list-recipes`, then press Ctrl+D at an empty command prompt to exit normally. Restart and check that the recipe is still retained.
6. Run from an IDE Run console or with redirected input. Verify that ordinary commands and confirmation answers still work with plain input. Test `TERM=dumb` where supported and check that it also uses plain input. JLine editing shortcuts are not expected in these fallback modes.

These checks complement the JUnit tests by exercising the real terminal and its native provider.

Confirmation answers in manual workflows

The shared instructions at the start of Manual Testing apply throughout; cancellation checks are in Reset and Confirm Plan Changes.

Edit a Recipe

1. With the sample recipes loaded, enter `edit-recipe 1 cal/700 p/55`, then `list-recipes 1`. Check that calories and protein change while the omitted fields remain unchanged.
2. Enter `edit-recipe 1`. Expect an error requiring at least one edit; the recipe stays unchanged.
3. If a plan or log already contains that recipe, check it still shows its earlier snapshot. Regenerate to use edited recipe values in a plan.

Set, View, and Clear Dietary Preferences

Use a fresh session without the general sample recipes for profile checks. Run the classified-recipe repair scenario below separately.

Action	Expected result
`view-diet`	Diet and both exclusion lists show `none`.
`set-diet diet/vegetarian avoid/peanut exclude/Olive Oil`, then `view-diet`	All three restrictions appear.
`set-diet diet/invalid` or `set-diet diet/keto`	Supported values and usage; previous profile remains.
`set-diet avoid/milk,dairy`, `set-diet exclude/Beans, beans`, or `view-diet extra`	Error; profile remains unchanged.
`clear-diet`, then `view-diet`	All restrictions show `none`.
Repeat `clear-diet`	Note that preferences are already clear.
`clear-diet diet/none`	Argument error with `Format: clear-diet`; profile stays unrestricted.
Save a profile, exit, restart from the same directory, and `view-diet`	Saved profile is restored.

For vegan filtering and replacement semantics, use the restricted-plan example in another fresh directory: verify two classified vegan days and Rice, Beans, and Tofu at 400g each. Add `add-recipe n/Lentils cal/600 p/10 c/20 f/5 i/Lentils:200g food/legume`, apply `set-diet diet/vegan avoid/soy`, and confirm. Expect Rice, Beans, and Lentils each day, 1800 kcal per day, and 400g of each ingredient. Then apply `set-diet exclude/Beans,Tofu` and confirm: the profile replaces the prior restrictions, but only Rice and Lentils qualify, so the plan and groceries clear.

In a separate session with vegan restrictions, add an unclassified recipe without `food/`; new generation must exclude it. Automated tests cover deterministic partial repair of mixed compatible/conflicting days.

Successful repair and grocery refresh

Start a fresh session and enter your name. Do not load the general sample recipes or save daily targets. Enter these commands one line at a time:

```
add-recipe n/Chicken A cal/400 p/30 c/0 f/10 i/Chicken:100g food/poultry
add-recipe n/Chicken B cal/400 p/30 c/0 f/10 i/Chicken:100g food/poultry
add-recipe n/Chicken C cal/400 p/30 c/0 f/10 i/Chicken:100g food/poultry
generate-plan d/2
generate-list
view-plan
view-list
```

Each day must contain all three chicken recipes, in any order. The grocery list contains `Chicken` with a total of 600g. Now add three recipes suitable for a vegetarian plan:

```
add-recipe n/Rice Bowl cal/400 p/10 c/70 f/5 i/Rice:100g food/none
add-recipe n/Potato Bowl cal/400 p/10 c/70 f/5 i/Potato:200g food/none
add-recipe n/Oat Bowl cal/400 p/10 c/70 f/5 i/Oats:50g food/oat
set-diet diet/vegetarian
n
view-diet
view-plan
view-list
```

Check that cancellation preserves the unrestricted profile, both chicken-plan days, and the 600g chicken grocery entry. Adding recipes alone must not change those snapshots. Apply the change again and agree:

```
set-diet diet/vegetarian
y
view-diet
view-plan
view-list
```

Expect confirmation that conflicting days were replaced and the grocery list was updated. Both days now contain Rice Bowl, Potato Bowl, and Oat Bowl, in any order. The refreshed groceries contain Rice 200g, Potato 400g, and Oats 100g, with no Chicken entry. Do not run `generate-list` again here: `view-list` must demonstrate that dietary repair refreshed the existing groceries automatically.

Now enter `clear-diet`, followed by `view-diet`, `view-plan`, and `view-list`. The profile must be unrestricted, while both vegetarian plan days and their grocery quantities remain exactly as displayed before clearing. This verifies that clearing affects future eligibility without silently regenerating or deleting existing snapshots.

Generating the original plan before adding the vegetarian recipes makes the conflict certain; having exactly three compatible recipes makes the repaired contents predictable despite random meal ordering.

Failed repair clears the incompatible plan and groceries

Continue from the successfully repaired plan above:

```
set-diet diet/vegetarian avoid/oat
y
view-diet
view-plan
view-list
```

Expect the new vegetarian profile with avoided category `oat` to be saved. Only two recipes now satisfy it, so repair must fail with an explanation that at least three compatible, classified recipes are required. Both the plan and groceries must be cleared; the view commands should report that no plan or generated grocery list is available. The recipe book must still contain all six recipes. This checks the failure path separately from the successful-repair path.